

QUANTUM COMPUTING FOR EVERYONE

A Practical Guide from Basics to Quantum Algorithms

TEACHER'S GUIDE

Teaching notes • Self-tests with answers •
Complete step-by-step solutions to every practice problem

QCFE

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Quantum Computing for Everyone: Teacher's Guide

The instructor's companion to the QCFE Student Edition

This guide was built with the help of artificial intelligence (AI), under the direction and review of the author.

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This guide and its companion Student Edition are provided free of charge for anyone who wishes to learn or teach. Voluntary donations in support of their continued development are warmly welcomed and appreciated.

How to Use This Guide

This Teacher's Guide mirrors the Student Edition chapter for chapter — Chapter 12 here serves Chapter 12 there — and gives you, for every chapter: **Teaching Notes** (what the chapter is really about, where students stumble, and how to run the activities), a **Self-Test** of quick questions with answers and explanations (use them as warm-ups, exit tickets, homework, or revision), and **complete, step-by-step solutions to every practice problem**, numbered exactly as in the Student Edition and tagged with the same [Basic]/[Standard]/[Challenge] levels.

Three suggestions from the author's own classroom. First, resist handing out solutions early: the Student Edition repeatedly tells students to attempt every problem on paper first, and the learning lives in that struggle — release solutions after honest attempts, not instead of them. Second, the worked solutions deliberately show *efficient* routes; when a student's different route reaches the same answer, celebrate it and have them present it. Third, the [Challenge] problems are genuinely harder and often open a door to the next chapter — they make excellent group work and classroom discussions rather than individual grading material.

The three review appendices of the Student Edition (Complex Numbers; Vectors and Matrices; Probability) have their solutions here as Appendices A–C. The reference appendices (Formula Sheet; Python Quick Start) have no problem sets. Where a solution relies on a formula, it names the Student Edition chapter that teaches it, so you can point a struggling student to exactly the right lesson.

A note on numerical answers: values are rounded to the displayed precision (usually three significant figures); a student's answer differing in the last digit from honest rounding deserves full credit. Every computed number in both volumes was machine-verified; the verification scripts ship in the LaTeX source package.

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CHAPTER ONE

Bits and Classical Computing

Teaching notes, self-test, and complete solutions

TEACHING NOTES

This chapter buys two things for the whole course: fluency in binary patterns (students must read 101 as “five” without thinking by Chapter 9) and the 2^n explosion, which returns as the state-space wall, the simulation argument, and the exponential enemy in error budgets. Drill conversions briskly — they are skills, not concepts — and spend saved time on the two ideas students actually misjudge: that XOR/parity thinking recurs throughout quantum error correction, and that exponential growth defeats hardware improvements (Problem 15 is the eye-opener; run it live). The parity lesson is a deliberate seed for Chapter 29: when you reach syndromes later, remind the class they invented them here in Problem 16. Common stumbles: mixing up “bits” and “patterns” (n vs 2^n), and reading binary right-to-left place values. The activity works well as a two-team race; keep it under ten minutes.

SELF-TEST — QUESTIONS WITH ANSWERS

Q1. How many patterns can 10 bits hold, roughly and exactly?

A. Exactly $2^{10} = 1024$; roughly 10^3 . This “ $2^{10} \approx 10^3$ ” bridge is used for every big estimate in the book.

Q2. Convert 1101 to decimal.

A. $8 + 4 + 0 + 1 = 13$.

Q3. What single question does the XOR gate answer?

A. “Are the two inputs different?” — output 1 if they differ, 0 if they agree.

Q4. Why is AND irreversible?

A. Output 0 arises from three different inputs (00, 01, 10), so the input cannot be recovered from the output — information is destroyed. Quantum gates will never be allowed to do this.

Q5. What does a parity bit detect, and what does it miss?

A. It detects any *odd* number of flips (in particular any single flip) by making the 1-count come out odd; it misses double flips, and it never says *which* bit flipped.

Q6. A problem needs 2^{50} steps. Why does buying a computer twice as fast barely help?

A. Doubling speed removes a single factor of 2 — one power out of fifty. Exponential problems shrink only when the *exponent* shrinks; that is an algorithm's job, not the hardware's.

Solutions to the Practice Problems

Problem 1 [Basic] $2^4 = 16$ patterns. Counting order: 0000, 0001, 0010, 0011, 0100.

Problem 2 [Basic] Place values from the left: $1 \cdot 8 + 0 \cdot 4 + 1 \cdot 2 + 0 \cdot 1 = 8 + 2 = 10$.

Problem 3 [Basic] **(1)** Largest power of 2 in 19 is 16: digit 1, remainder 3. **(2)** 8 and 4 don't fit: digits 0, 0. **(3)** 2 fits: digit 1, remainder 1. **(4)** 1 fits: digit 1. Result: $19 = 10011_2$. Check: $16 + 2 + 1 = 19$.

Problem 4 [Basic] Rows $(a, b) \rightarrow$ out: $(0, 0) \rightarrow 0$, $(0, 1) \rightarrow 0$, $(1, 0) \rightarrow 0$, $(1, 1) \rightarrow 1$. Rule: output is 1 only when *both* inputs are 1.

Problem 5 [Basic] 1100110 has four 1s (even) $\rightarrow$ parity 0: transmit 11001100. 0000001 has one 1 (odd) $\rightarrow$ parity 1: transmit 00000011. (Both totals are now even.)

Problem 6 [Basic] Count the 1s in 10111010: five — odd. Corrupted: at least one bit flipped. The receiver knows an error occurred but cannot tell which bit (nor whether *three* bits flipped rather than one); it can only request retransmission.

Problem 7 [Standard] $2^7 = 128$ commands. For 200 commands, 7 bits are too few ($128 < 200$) and $2^8 = 256 \geq 200$: 8 bits.

Problem 8 [Standard] $1000000_2 = 2^6 = 64$. Since 111111 is exactly one less than 1000000, its value is $64 - 1 = 63$. Trick: a string of n ones equals $2^n - 1$ — the “odometer” one step before rollover.

Problem 9 [Standard] XOR outputs 0 when the inputs *agree*. Since $a = 1$, agreement forces $b = 1$.

Problem 10 [Standard] Sum $= a \oplus b = 1 \oplus 0 = 1$; carry $= a \wedge b = 0$. Two-bit answer (carry, sum) $= 01_2 = 1$ — correctly, $1 + 0 = 1$.

Problem 11 [Standard] $2^{40} = (2^{10})^4 \approx (10^3)^4 = 10^{12}$ configurations. At 10^9 per second: $10^{12}/10^9 = 10^3$ seconds ≈ 16.7 minutes.

Problem 12 [Standard] Example: flip bits 1 and 2 of 11001100 $\rightarrow$ 00001100, which has two 1s — even: the check passes despite two errors. Any double flip changes the 1-count by -2 , 0 , or $+2$, always preserving evenness, so parity misses *all* 28 double-flip patterns.

Problem 13 [Challenge] Need $2^n \geq 48,000$. $2^{15} = 32,768$ is too small; $2^{16} = 65,536$ suffices: **16 bits**, which could label up to 65,536 seats.

Problem 14 [Challenge] Rows $(a, b) \rightarrow (a \wedge b) \vee (\neg a)$: $(0, 0) \rightarrow 0 \vee 1 = 1$; $(0, 1) \rightarrow 0 \vee 1 = 1$; $(1, 0) \rightarrow 0 \vee 0 = 0$; $(1, 1) \rightarrow 1 \vee 0 = 1$. Output is 0 only for $(a, b) = (1, 0)$. (Students may notice the circuit computes “ a implies b .”)

Problem 15 [Challenge] **(1)** Time now: $2^{80} \approx (10^3)^8 = 10^{24}$ steps at $10^9/s = 10^{15}$ s. **(2)** One day $\approx 8.6 \times 10^4$ s, so we must gain a speed factor $\approx 10^{15}/8.6 \times 10^4 \approx 10^{10} \approx 2^{33}$. **(3)** One doubling per two years: about 33 doublings = **roughly 66 years**. Moral: exponential walls outlast hardware roadmaps; only better *algorithms* (or different physics — this book!) move them meaningfully.

Problem 16 [Challenge] Arrange bits $b_1 b_2 / b_3 b_4$ with row parities $r_1 = b_1 \oplus b_2$, $r_2 = b_3 \oplus b_4$. A single flip breaks exactly one row parity, so the row is located. Adding column parities $c_1 = b_1 \oplus b_3$, $c_2 = b_2 \oplus b_4$: a single flip now breaks one row *and* one column check, and the intersection names the exact bit — location, hence correction, from parities alone, without ever “reading the data.” This is precisely the syndrome logic of Chapter 29.

CHAPTER TWO

Why Quantum Computing Is Different

Teaching notes, self-test, and complete solutions

TEACHING NOTES

This chapter sets the course's intellectual tone: enthusiasm *and* honesty, in the same breath. Two ideas must land. First, interference: amplitudes add *before* squaring, so possibilities can cancel — run the double-slit numbers (new in this edition) slowly, since every algorithm later is that arithmetic scaled up. Second, the honest map: quantum computers are specialists; most computing gains nothing. Students arrive with the “tries everything at once” meme — do not just forbid it, *replace* it (Problem 11 provides the correct sentence; make the class memorize it). The claim-checking checklist deserves a full classroom exercise on a real press release; it returns in the capstones. The history question (Problem 1) works as a two-minute warm-up. Guard rail for discussion: skepticism here is not pessimism — the same chapter that debunks hype also presents Feynman's argument, which is genuinely spectacular.

SELF-TEST — QUESTIONS WITH ANSWERS

Q1. Name the three quantum rules of this chapter.

A. Superposition (amplitudes for several possibilities at once), measurement randomness (one outcome, probabilities from amplitudes), and entanglement (joint states not describable qubit by qubit).

Q2. What is the single mechanism that makes quantum algorithms fast, in one line?

A. Arrange the computation so amplitudes of wrong answers cancel and amplitudes of right answers reinforce — interference, steered on purpose.

Q3. How many amplitudes describe 50 qubits, roughly?

A. $2^{50} \approx 10^{15}$ — a million billion, the reason classical simulation chokes.

Q4. A quantum computer “holds” 2^n amplitudes. Why can't you read them all out?

A. Measurement returns one n -bit pattern, at random; the amplitudes themselves are never printed. (Holevo: n qubits deliver at most n classical bits.)

Q5. Name two problem families with likely quantum advantage and two without.

A. With: simulating molecules/materials, factoring (and certain searches/optimizations). Without: streaming, sort-

ing, spreadsheets, web serving — everyday computing.

Q6. Why does the book insist on checking claims against the *best* classical method?

A. Because “faster than a bad baseline” is not an advantage; classical algorithms keep improving, and honest comparisons are against the strongest opponent (the 2019 supremacy debate is the standing example).

Solutions to the Practice Problems

Problem 1 [Basic] Planck’s quanta — 1900s; Feynman’s proposal — 1980s; Shor’s algorithm — 1990s; cloud quantum computers — 2010s (first public cloud access 2016).

Problem 2 [Basic] (Sample answers.) Superposition: before measurement a qubit can carry weighted possibilities for 0 *and* 1 simultaneously. Measurement: asking collapses to one answer, randomly, with probabilities set by the weights. Entanglement: two qubits can share one joint state whose parts have no individual description — measuring one instantly sharpens predictions about the other.

Problem 3 [Basic] $2^4 = 16$ amplitudes, for patterns 0000, 0001, 0010, 0011, ...

Problem 4 [Basic] Total amplitude $+0.4 + (-0.4) = 0$: the outcome *never happens*. Two open routes produce silence — destructive interference, impossible for ordinary probabilities.

Problem 5 [Basic] Indistinguishable: amplitude $0.6 - 0.6 = 0$, probability 0. Distinguishable: probabilities add instead, $0.36 + 0.36 = 0.72$.

Problem 6 [Basic] The “paradox” proves such a state cannot exist: probabilities over *all* outcomes must total 1, so unitarity (Appendix B’s entrance exam) forbids two paths to one outcome from both carrying such large amplitudes — the same physics that steers amplitude into one detector must drain it from the others. Scaled legally by $\frac{1}{\sqrt{2}}$: constructive $\left|\frac{0.6+0.8}{\sqrt{2}}\right|^2 = 0.98$, versus $0.18 + 0.32 = 0.50$ if the paths are distinguishable. Interference boosted 50% to 98% — redistribution, never creation, of probability.

Problem 7 [Standard] Need $2^n > 10^{21}$. $2^{70} = (2^{10})^7 \approx 10^{21}$: about **70 qubits** — a state too big for every hard drive on Earth, held by a chip smaller than a coin.

Problem 8 [Standard] Storage is not retrieval: Rule 2 says a measurement returns *one* random 300-bit pattern, not the 2^{300} amplitudes. Unread storage is not a hard disk; the amplitudes work only *inside* the computation, through interference.

Problem 9 [Standard] (a) Likely advantage — quantum system simulating a quantum system (Feynman’s argument). (b) None — bandwidth problem, no structure for interference. (c) Likely advantage — Shor’s algorithm. (d) None — sorting is fast classically; no known meaningful quantum speedup in practice.

Problem 10 [Standard] $\sqrt{4 \times 10^6} = 2000$ Grover steps versus $N/2 = 2 \times 10^6$ classical checks on average — a thousandfold reduction, from a square root.

Problem 11 [Standard] Misleading: measurement yields one random answer, so “trying everything” by itself returns garbage. Correct: a quantum computer *interferes* all computational paths so that wrong answers cancel and the right answer’s probability is amplified before the single measurement.

Problem 12 [Standard] $|a + b|^2 = 0.25 + 0.25 + 2(0.5)(0.5) \cos \Delta\varphi = 0.5 + 0.5 \cos \Delta\varphi$. At 120° : $0.5 - 0.25 = 0.25$. At 0° : 1 (bright). At 180° : 0 (dark). The general fringe formula in one line.

Problem 13 [Challenge] (Grading guide — expect five lines resembling:) “The claim gives no error rates, no gate depths, and no algorithm; 2000 physical qubits without error correction support depths far below chemistry’s needs (Chapters 21, 28); no classical baseline is named; ‘next year’ contradicts every published resource estimate; verdict: marketing until benchmarks appear — ask for fidelity numbers and a named molecule.” Credit any answer hitting: missing error rates, missing algorithm/molecule, missing baseline, timeline implausibility.

Problem 14 [Challenge] $1000 \times 100 = 100,000$ physical qubits — about $100\times$ today’s largest machines. Conclusion: useful error-corrected chemistry needs roughly two orders of magnitude of scaling at error rates good enough for correction — a decade-class engineering gap, not a next-year press release.

Problem 15 [Challenge] Ratio: $10,000 \text{ years} / 2.5 \text{ days} = 10^4 \times 365 / 2.5 \approx 1.5 \times 10^6$. A further $100\times$ classical improvement: $2.5 \text{ days} / 100 \approx 36 \text{ minutes}$. Lesson: “quantum advantage” is measured against a moving target; claims must be dated, and margins must be enormous to survive classical progress.

Problem 16 [Challenge] Tagged half (distinguishable): probabilities add: $0.25 + 0.25 = 0.5$. Untagged half (interfering): amplitude $\frac{1}{2} - \frac{1}{2} = 0 \rightarrow$ probability 0. Mixture: $0.5(0.5) + 0.5(0) = 0.25$. Moral: partial which-path information produces partial loss of interference — measurement strength and fringe visibility trade off continuously, not all-or-nothing.

CHAPTER THREE

Qubits and Quantum States

Teaching notes, self-test, and complete solutions

TEACHING NOTES

Notation week: the whole course's grammar (kets, bras, amplitudes, normalization, inner products) is installed here, and fluency matters more than depth. Insist on the conjugate rule for bras from day one — Problems 10 and 14 exist because every class loses someone to a missing conjugate in week five. The two conceptual peaks: normalization as “probabilities must total 1” (not an arbitrary rule), and the new fidelity/orthogonality pair — similarity as $|\langle\phi|\psi\rangle|^2$, perfect distinguishability as orthogonality. Problem 20 previews quantum cryptography and rewards a full class discussion: physics itself forbidding perfect readout is the first genuinely quantum *security* idea students meet. Quick formative check mid-week: dictate three states, ask for legal/illegal verdicts with reasons in under a minute each.

SELF-TEST — QUESTIONS WITH ANSWERS

Q1. What are the two conditions on α, β for $\alpha|0\rangle + \beta|1\rangle$ to be a legal state?

A. They are complex numbers with $|\alpha|^2 + |\beta|^2 = 1$ (normalization). Nothing else — signs and phases are allowed and meaningful.

Q2. Turn $|\psi\rangle = \begin{pmatrix} 0.6i \\ 0.8 \end{pmatrix}$ into its bra.

A. $\langle\psi| = (-0.6i, 0.8)$ — transpose *and* conjugate: the i flips sign.

Q3. What is $\langle\phi|\psi\rangle$ good for, in one line?

A. It measures overlap: its squared modulus is the fidelity (similarity 0 to 1) — and, come Chapter 4, the probability that a $|\phi\rangle$ -question answers “yes” on $|\psi\rangle$.

Q4. When can one measurement tell two states apart with certainty?

A. Exactly when they are orthogonal, $\langle\phi|\psi\rangle = 0$ — antipodal points on the Bloch sphere.

Q5. Why is $0.5|0\rangle + 0.5|1\rangle$ illegal, and what is the legal state with equal weights?

A. $0.25 + 0.25 = 0.5 \neq 1$. The legal equal-weight state is $\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle = |+\rangle$.

Q6. Give the orthogonal partner of $0.6|0\rangle + 0.8|1\rangle$.

A. $0.8|0\rangle - 0.6|1\rangle$ (swap, one sign flip; conjugates if complex). Check: $0.48 - 0.48 = 0$.

Solutions to the Practice Problems

Problem 1 [Basic] $|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$, and $|\psi\rangle = \begin{pmatrix} 0.28 \\ 0.96 \end{pmatrix}$.

Problem 2 [Basic] $0.28^2 + 0.96^2 = 0.0784 + 0.9216 = 1.0000$: normalized. (A Pythagorean pair — students may recognize 7-24-25 scaled.)

Problem 3 [Basic] $\langle\psi| = (0.8, 0.6)$ (real entries: conjugation changes nothing). $\langle\psi|\psi\rangle = 0.64 + 0.36 = 1$.

Problem 4 [Basic] (a) Legal: $0 + 1 = 1$. (b) Illegal: $0.81 + 0.81 = 1.62 \neq 1$. (c) Legal: $\frac{1}{2} + \frac{1}{2} = 1$ — the minus sign is perfectly permitted (it is $|-\rangle$).

Problem 5 [Basic] Partners: $|0\rangle \rightarrow |1\rangle$; $|+\rangle \rightarrow |-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$; $\frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle \rightarrow \frac{1}{2}|0\rangle - \frac{\sqrt{3}}{2}|1\rangle$. Verify the last: $\frac{\sqrt{3}}{2} \cdot \frac{1}{2} - \frac{1}{2} \cdot \frac{\sqrt{3}}{2} = 0$ ✓

Problem 6 [Basic] Antipodes — diametrically opposite points. Checks: $|0\rangle$ (north pole) vs $|1\rangle$ (south pole); $|+\rangle$ ($+x$) vs $|-\rangle$ ($-x$).

Problem 7 [Standard] Length: $\sqrt{9 + 16} = 5$. Normalized: $\frac{3}{5}|0\rangle + \frac{4}{5}|1\rangle = 0.6|0\rangle + 0.8|1\rangle$.

Problem 8 [Standard] $\langle 0|+\rangle = \frac{1}{\sqrt{2}}$ and $\langle 1|+\rangle = \frac{1}{\sqrt{2}}$: equal. Their squares are $\frac{1}{2}$ each — which Chapter 4 reveals as the 50/50 measurement probabilities of $|+\rangle$.

Problem 9 [Standard] Column $\begin{pmatrix} 1/\sqrt{2} \\ i/\sqrt{2} \end{pmatrix}$; bra $\begin{pmatrix} \frac{1}{\sqrt{2}}, -\frac{i}{\sqrt{2}} \end{pmatrix}$ (conjugate!). Normalization: $\frac{1}{2} + \left| \frac{i}{\sqrt{2}} \right|^2 = \frac{1}{2} + \frac{1}{2} = 1$ ✓

Problem 10 [Standard] $\langle +i|-i\rangle = \left(\frac{1}{\sqrt{2}}\right)\left(\frac{1}{\sqrt{2}}\right) + \left(-\frac{i}{\sqrt{2}}\right)^* \dots$ — carefully: the bra of $|+i\rangle$ is $\left(\frac{1}{\sqrt{2}}, -\frac{i}{\sqrt{2}}\right)$, so the product is $\frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} + \left(-\frac{i}{\sqrt{2}}\right) \cdot \left(-\frac{i}{\sqrt{2}}\right) = \frac{1}{2} + \frac{i^2}{2} = \frac{1}{2} - \frac{1}{2} = 0$ ✓ The $\pm i$ pair is orthogonal — the y-axis poles of the Bloch sphere.

Problem 11 [Standard] Error one: probabilities are *squared* amplitudes: $0.7^2 = 0.49$, so it is 49%/49% — which exposes error two: $0.49 + 0.49 = 0.98 \neq 1$, so the state is not even normalized (legal version: $\frac{1}{\sqrt{2}} \approx 0.707$ amplitudes).

Problem 12 [Standard] $\langle +|+i\rangle = \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \cdot \frac{i}{\sqrt{2}} = \frac{1+i}{2}$. Fidelity = $\left| \frac{1+i}{2} \right|^2 = \frac{1^2+1^2}{4} = \frac{1}{2}$: exactly the halfway landmark — neighboring equator points 90° apart.

Problem 13 [Standard] By the conjugate rule, $\langle\phi|\psi\rangle = (\langle\psi|\phi\rangle)^*$, and a complex number and its conjugate share the same modulus: $|z| = |z^*|$. Squaring preserves the equality — fidelity has no direction.

Problem 14 [Standard] Recipe with conjugates: $|\psi^\perp\rangle = \beta^*|0\rangle - \alpha^*|1\rangle = -\frac{i}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle$. Check: $\langle\psi^\perp|\psi\rangle = \left(+\frac{i}{\sqrt{2}}\right)\frac{1}{\sqrt{2}} + \left(-\frac{1}{\sqrt{2}}\right)\frac{i}{\sqrt{2}} = \frac{i}{2} - \frac{i}{2} = 0$ ✓ (the bra conjugated the first entry back to $+i/\sqrt{2}$).

Problem 15 [Challenge] $a^2 + (2a)^2 = 5a^2 = 1 \Rightarrow a = \frac{1}{\sqrt{5}}, b = \frac{2}{\sqrt{5}}$: $|\psi\rangle = \frac{1}{\sqrt{5}}|0\rangle + \frac{2}{\sqrt{5}}|1\rangle$ (probabilities 20%/80%).

Problem 16 [Challenge] $\frac{3}{4} + \frac{|c|^2}{4} = 1 \Rightarrow |c| = 1$. Valid choices: $c = 1, c = i, c = e^{i45^\circ}$ (any unit complex number). They share identical standard-basis statistics (75%/25%) but differ in *relative phase* — physically distinct states that other measurements (Chapter 4's $\pm$ basis, Chapter 5's equator) tell apart.

Problem 17 [Challenge] $\langle\psi'|\psi'\rangle = (e^{i\theta})^* e^{i\theta} \langle\psi|\psi\rangle = e^{-i\theta} e^{i\theta} \langle\psi|\psi\rangle = \langle\psi|\psi\rangle$. For any bra $\langle\phi|$: $\langle\phi|\psi'\rangle = e^{i\theta} \langle\phi|\psi\rangle$, and $|e^{i\theta}z|^2 = |z|^2$ — every squared modulus, hence every probability, is untouched. A global phase is physically invisible.

Problem 18 [Challenge] $\langle+|\psi\rangle = \frac{0.74+0.67}{\sqrt{2}} = \frac{1.41}{1.414} = 0.997$, so fidelity $= 0.997^2 = 0.994$: the device delivers 99.4% and *beats* the 99% spec. (Fine print worth praising if a student spots it: the given amplitudes square-sum to 0.9965 — rounding in the data; renormalizing nudges fidelity to 99.8%, so the verdict stands either way.)

Problem 19 [Challenge] Fidelity to $|0\rangle$ is $|\alpha|^2 = \frac{1}{2}$: all states with $|\alpha| = \frac{1}{\sqrt{2}}$ — the Bloch sphere's *equator*. Gallery members: $|+\rangle, |-\rangle, |+i\rangle, |-i\rangle$ (and every other longitude).

Problem 20 [Challenge] Standard basis: $|0\rangle$ gives “0” always; $|+\rangle$ gives “0” half the time — outcome “0” is ambiguous, so certainty is impossible. Any other basis $\{|\phi\rangle, |\phi^\perp\rangle\}$: certainty would need each input to land on a different basis state with probability 1, i.e. $|0\rangle$ and $|+\rangle$ would be $|\phi\rangle$ and $|\phi^\perp\rangle$ up to phase — but they are not orthogonal ($\langle 0|+\rangle = \frac{1}{\sqrt{2}} \neq 0$), contradiction. Best single-shot success: $\frac{1+\sqrt{1-F}}{2} = \frac{1+\sqrt{1/2}}{2} \approx 85.4\%$. Chapter 15 celebrates this because an eavesdropper faces exactly this limit: non-orthogonal signal states cannot be read reliably — physics itself enforcing secrecy.

CHAPTER FOUR

Superposition and Measurement

Teaching notes, self-test, and complete solutions

TEACHING NOTES

The Born rule is the course's first *physical* law, and the chapter's silent second law matters as much: statistics is how quantum information is read. Push the histogram habit hard — every claim needs shots, every estimate a wobble. The measurement-sandwich lesson (Z-X-Z) reliably produces the semester's best “wait, *what?*” moment; let it breathe, then anchor it with the rule “measurement rewrites the state in its own basis.” The certification lesson (shot budgets, sigma-distances) turns statistics into professional practice; students who meet it here will read hardware datasheets fluently in Chapter 30. Watch for the classic confusions: amplitudes vs probabilities (squares!), “the qubit was secretly decided all along” (defer full rebuttal to Chapter 11, but flag it), and $\sqrt{N}$ wobble applied to frequencies instead of counts. Problems 17 and 20 preview cryptography and certification — both make strong homework-plus-discussion pairs.

SELF-TEST — QUESTIONS WITH ANSWERS

Q1. State the Born rule in one sentence.

A. Measuring $\alpha|0\rangle + \beta|1\rangle$ in the standard basis yields 0 with probability $|\alpha|^2$ and 1 with probability $|\beta|^2$, and the state collapses to the observed outcome.

Q2. After a measurement returns 0, what does an immediate repeat return?

A. 0, with certainty — collapse leaves $|0\rangle$, and repeated same-basis measurements agree.

Q3. What is $P(+)$ for the state $|1\rangle$?

A. $|\langle +|1\rangle|^2 = \frac{1}{2}$: a definite- Z state is a coin flip in the X basis — incompatible questions.

Q4. With 1000 shots, how wobbly is an estimated frequency?

A. About $\pm\sqrt{p(1-p)/1000} \leq \pm 0.016$ (one sigma, worst case at $p = 0.5$).

Q5. Two states give identical standard-basis histograms. Are they the same state?

A. Not necessarily — they may differ in relative phase ($0.6|0\rangle + 0.8|1\rangle$ vs $0.6|0\rangle - 0.8|1\rangle$). A second basis (e.g. $\pm$) is needed to tell them apart.

Q6. Why does the book insist “one shot proves nothing”?

A. Any single outcome is consistent with many states; quantum information lives in probability distributions, which only many shots reveal — with $1/\sqrt{n}$ precision.

Solutions to the Practice Problems

Problem 1 [Basic] $P(0) = 0.8^2 = 0.64$, $P(1) = 0.6^2 = 0.36$; $0.64 + 0.36 = 1$ ✓

Problem 2 [Basic] Both give $P(0) = P(1) = \left| \pm \frac{1}{\sqrt{2}} \right|^2 = \frac{1}{2}$. The minus sign vanishes under squaring — the standard basis is blind to relative phase (the point of the whole $\pm$ -basis lesson).

Problem 3 [Basic] Collapse: the state *becomes* $|1\rangle$. An immediate second measurement gives 1 with probability 1 — the original 25% is history.

Problem 4 [Basic] Expected 1000/1000; wobble $\sqrt{2000} \approx 45$ counts — so anything within roughly ± 45 (one sigma) or ± 90 (two sigma) of 1000 is unremarkable.

Problem 5 [Basic] Budget $n \gtrsim 0.25 (2/\text{margin})^2$: margin 0.02 $\rightarrow n = 0.25 \times 10^4 = 2500$ shots; margin 0.01 $\rightarrow 10,000$ shots. Rule: halving the margin quadruples the shots ($n \propto 1/\text{margin}^2$).

Problem 6 [Basic] $\hat{p} = 385/400 = 0.9625$. Spec wobble: $\sigma = \sqrt{0.98 \times 0.02/400} = 0.007$. Distance: $(0.98 - 0.9625)/0.007 = 2.5$ sigma below spec: reject (or at minimum flag and re-test with more shots) — chance alone does this about once in a hundred runs.

Problem 7 [Standard] Normalization: $\frac{1}{4} + \left| \frac{\sqrt{3}}{2}i \right|^2 = \frac{1}{4} + \frac{3}{4} = 1$ ✓ Then $P(0) = \frac{1}{4}$, $P(1) = \frac{3}{4}$ — the i affects nothing here (its modulus is 1).

Problem 8 [Standard] $|-\rangle$ and $|+i\rangle$ (indeed any equator state) also predict 50/50. Distinguisher: measure in the $\pm$ basis — $|+\rangle$ gives all +, $|-\rangle$ all −, $|+i\rangle$ stays 50/50. (A Y-basis run separates the last from both.)

Problem 9 [Standard] $P(0) = 0.16 \Rightarrow |\beta|^2 = 0.84 \Rightarrow |\beta| \approx 0.917$. A consistent state: $0.4|0\rangle + 0.917|1\rangle$.

Problem 10 [Standard] $\langle +|0\rangle = \frac{1}{\sqrt{2}} \Rightarrow P(+)=\frac{1}{2}$ (and $P(-)=\frac{1}{2}$). A definite answer to the Z-question is a pure coin flip on the X-question: the two questions are incompatible, and certainty in one buys randomness in the other.

Problem 11 [Standard] $\langle +|\psi\rangle = \frac{1.4}{\sqrt{2}} = 0.990 \Rightarrow P(+)=0.98$; $\langle -|\psi\rangle = \frac{-0.2}{\sqrt{2}} = -0.141 \Rightarrow P(-)=0.02$. Sum = 1 ✓ (The state lies close to $|+\rangle$ on the sphere.)

Problem 12 [Standard] Z-measurement of $|1\rangle$: result 1, certain; state $|1\rangle$. X-measurement: $\pm$ each $\frac{1}{2}$, collapsing to $|+\rangle$ or $|-\rangle$. Final Z: either way $P(0) = \frac{1}{2}$. Total $P(\text{final } 0) = \frac{1}{2}$ — the middle question re-randomized a certainty, exactly as when starting from $|0\rangle$.

Problem 13 [Standard] A second Z-measurement re-asks the same question of a collapsed state: it returns the first answer with certainty, changing nothing; the final measurement then also repeats it. $P(\text{final } 1)$ equals the first measurement's $P(1)$ — for the sandwich's $|0\rangle$ start, 0. Rule confirmed: repeated same-basis measurements are consistent; only *incompatible* questions disturb.

Problem 14 [Standard] Budgets scale with $p(1-p)$: $0.98 \times 0.02 = 0.0196$ versus $0.5 \times 0.5 = 0.25$. Ratio $0.25/0.0196 \approx 12.8$: certifying the near-certainty needs ≈ 784 shots where the coin flip needs 10,000 — extreme probabilities are cheap to pin down.

Problem 15 [Challenge] $P(+)$ for $|a\rangle$: $|\frac{0.6+0.8}{\sqrt{2}}|^2 = 0.98$; for $|b\rangle$: $|\frac{0.6-0.8}{\sqrt{2}}|^2 = 0.02$. Procedure: run one histogram in the standard basis (should match 36/64 for both — a sanity check), then one in the $\pm$ basis: $|a\rangle$ shows $\approx 98\%$ +, $|b\rangle \approx 2\%$ — unmistakable with even 100 shots. Two bases together determine what one basis cannot.

Problem 16 [Challenge] Expected 5000 ± 100 (two sigma); observed 5350 sits $350/50 = \dots$ careful: σ of the count is $\sqrt{10000 \times 0.25} = 50$, so 5350 is $350/50 = 7$ sigma out — astronomically incompatible with $|+\rangle$. The data suggest $|\alpha|^2 = 0.535$, i.e. $|\alpha| \approx 0.731$ (a state tilted north of the equator). Verdict: not credible as $|+\rangle$; recalibrate. (Accept $\sqrt{N} = 100$ as the crude wobble: then 3.5 sigma — same verdict.)

Problem 17 [Challenge] Eve's Z-measurement of $|+\rangle$: result 0 or 1, each $\frac{1}{2}$; she forwards $|0\rangle$ or $|1\rangle$ accordingly. Bob measures in the $\pm$ basis: either forwarded state gives $P(-) = \frac{1}{2}$. Total: $P(|-\rangle) = \frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{2}$ — whereas an untouched $|+\rangle$ gives $|-\rangle$ never. Every such “impossible” click is Eve's fingerprint; Chapter 15 builds the full protocol (where random basis choices dilute the fingerprint to the famous 25%).

Problem 18 [Challenge] Per check, escape probability $\approx \sin^2(0.5^\circ) = 7.6 \times 10^{-5}$. Over 90 checks: $P(\text{ever leaves}) \approx 90 \times 7.6 \times 10^{-5} \approx 0.7\%$ (union bound; the checks keep resetting the state to $|0\rangle$). Unwatched, the same 90 steps rotate to the equator: $P(1) = \sin^2 45^\circ = 50\%$. Watching held 99.3% of the population frozen — the Zeno effect, quantified.

Problem 19 [Challenge] Outcome probabilities — sent $|0\rangle$: copy 1 (Z) gives 0 always; copy 2 (X) gives + or — each $\frac{1}{2}$: outcomes $(0, +)$ and $(0, -)$ each $\frac{1}{2}$. Sent $|+\rangle$: copy 1 gives 0/1 each $\frac{1}{2}$; copy 2 gives + always: outcomes $(0, +)$ and $(1, +)$ each $\frac{1}{2}$. Certain identifications: $(1, \cdot)$ — impossible for $|0\rangle$, so it proves $|+\rangle$; $(\cdot, -)$ — impossible for $|+\rangle$, so it proves $|0\rangle$. Each occurs with probability $\frac{1}{2}$ for its state, so half of all runs end in certainty, never wrongly. Ambiguous outcome $(0, +)$: probability $\frac{1}{2}$ either way; guessing there succeeds half the time. Overall success: $\frac{1}{2}(1) + \frac{1}{2}(\frac{1}{2}) = 75\%$ — below the single-copy 85% in raw accuracy, but with a property no single-copy scheme has: when it speaks, it is *certain*. (Cryptographers prize exactly this trade.)

Problem 20 [Challenge] Rotation: apply H, then measure Z (the X-compiler). Spec translates to $p = P(0) = (1 + 0.96)/2 = 0.98$. Budget for margin 0.01 at two sigma: $n \gtrsim 0.98 \times 0.02 \times (2/0.01)^2 \approx 784$ — run 1000. This X-test itself catches the 50/50 $|0\rangle / |1\rangle$ impostor spectacularly: the mixture gives $\langle X \rangle \approx 0$, i.e. $p \approx 0.5$, some 60 sigma from spec. (A Z-only certification would have passed it — both give 50/50 in Z — which is the trap the problem plants.) For completeness against subtler impostors, add a Y-basis run ($S^\dagger$ then H, Chapter 8) with the same budget to pin the last Bloch coordinate near zero.

CHAPTER FIVE

The Bloch Sphere

Teaching notes, self-test, and complete solutions

TEACHING NOTES

Geometry week: the payoff is a mental picture students will consult for the rest of their careers. Three destinations. (1) The dictionary: latitude θ = probabilities, longitude φ = relative phase; drill the conversion formula both directions until routine. (2) Global vs relative phase — the single most examined distinction in the chapter; Problem 17's "store data in a global phase" trap converts confusion into understanding better than any lecture. (3) The two new lessons: mixed states *inside* the ball (arrow length = purity — the honest picture of decoherence) and the flight log (the H-phase-H interferometer), which quietly installs the Ramsey experiment three chapters early. Use physical props: a ball, a skewer, stickers for the six landmarks. Common stumbles: factor-of-two angles ($\theta/2$ in amplitudes!), confusing the equator's longitude with measurement probability, and treating the mixture at the center as "a state we don't know" rather than "a preparation with genuinely less information."

SELF-TEST — QUESTIONS WITH ANSWERS

Q1. What do latitude and longitude encode on the Bloch sphere?

A. Latitude (θ): the 0/1 probability balance, $P(1) = \sin^2(\theta/2)$. Longitude (φ): the relative phase between the amplitudes.

Q2. Where do the six landmark states sit?

A. $|0\rangle / |1\rangle$ at the poles ($\pm z$); $|+\rangle / |-\rangle$ at $\pm x$; $|+i\rangle / |-i\rangle$ at $\pm y$ — three orthogonal pairs, three axes.

Q3. Why is $-|+\rangle$ the same physical state as $|+\rangle$?

A. They differ by a global phase (e^{i180°), which no measurement can see; only *relative* phase between $|0\rangle$ and $|1\rangle$ amplitudes is physical.

Q4. What distinguishes a mixed state from a superposition, operationally?

A. A superposition like $|+\rangle$ answers some basis (X) with certainty; the 50/50 mixture answers *every* basis 50/50 — its Bloch arrow has length 0, and no rotation can restore certainty.

Q5. A state's Bloch arrow has length 0.6. Pure or mixed, and what does the length mean?

A. Mixed. Length 1 = pure; shorter = less purity; the arrow's direction still gives the best-aligned measurement, with visibility scaled by 0.6.

Q6. What does the interferometer H-(phase φ)-H measure, and how?

A. The invisible phase φ : it lands as a visible probability $P(0) = (1 + \cos \varphi)/2$ — phase in, population out.

Solutions to the Practice Problems

Problem 1 [Basic] (a) $|0\rangle$: $\theta = 0^\circ$ (north pole; φ irrelevant). (b) $|1\rangle$: $\theta = 180^\circ$. (c) $|+\rangle$: $(90^\circ, 0^\circ)$. (d) $|+i\rangle$: $(90^\circ, 90^\circ)$.

Problem 2 [Basic] Equator: $P(0) = P(1) = \frac{1}{2}$ regardless of φ . The longitude 137° set only the relative phase — invisible to a standard-basis measurement, decisive for any phase-sensitive one.

Problem 3 [Basic] $\cos(90^\circ) |0\rangle + e^{i0^\circ} \sin(90^\circ) |1\rangle = 0 \cdot |0\rangle + 1 \cdot |1\rangle = |1\rangle$ — the south pole, as it must be.

Problem 4 [Basic] Antipodal: $(|0\rangle, |1\rangle)$ yes; $(|+\rangle, |-\rangle)$ yes; $(|0\rangle, |+\rangle)$ no — they are 90° apart. Antipodal means physically: perfectly distinguishable (orthogonal), the two answers of one measurement.

Problem 5 [Basic] $|0\rangle$ (pole) $\rightarrow$ H $\rightarrow$ equator at $\varphi = 0^\circ \rightarrow$ S $\rightarrow$ equator at $\varphi = 90^\circ \rightarrow$ H $\rightarrow P(0) = (1 + \cos 90^\circ)/2 = \frac{1}{2}$. (The right-angle phase hides completely from this detector — compare Z in the same slot, which drives $P(0)$ to 0.)

Problem 6 [Basic] S and T rotate about the z-axis, and $P(0)$ depends only on the z-coordinate (latitude) — z-turns change longitude only, so no S/T sequence can ever move a histogram.

Problem 7 [Standard] $\cos(60^\circ) |0\rangle + e^{i180^\circ} \sin(60^\circ) |1\rangle = 0.5 |0\rangle - 0.866 |1\rangle$. Probabilities: $P(0) = 0.25$, $P(1) = 0.75$.

Problem 8 [Standard] $\cos(\theta/2) = 0.28 \Rightarrow \theta/2 = 73.7^\circ \Rightarrow \theta \approx 147.5^\circ$ — deep southern hemisphere (mostly $|1\rangle$: indeed $P(1) = 0.92$). Amplitudes real and positive $\Rightarrow \varphi = 0^\circ$.

Problem 9 [Standard] Both have $\cos 35^\circ \approx 0.819$ and $\sin 35^\circ \approx 0.574$: state A $= 0.819 |0\rangle + 0.574 |1\rangle$; state B $= 0.819 |0\rangle + 0.574 i |1\rangle$. A standard-basis histogram cannot distinguish them (same latitude: both 67%/33%); a $\pm$ -basis measurement can: A gives $P(+) = \left| \frac{0.819 + 0.574}{\sqrt{2}} \right|^2 = 0.97$, while B gives $P(+) = \frac{1}{2}$ (its phase points along y).

Problem 10 [Standard] The overall factor $e^{i\pi/3}$ is a *global* phase — physically invisible; strip it. What remains is $\frac{1}{\sqrt{2}}(|0\rangle + i |1\rangle) = |+i\rangle$: the $+y$ pole, $(\theta, \varphi) = (90^\circ, 90^\circ)$.

Problem 11 [Standard] $\sin(\theta/2) = \sqrt{0.9} = 0.949 \Rightarrow \theta/2 = 71.6^\circ \Rightarrow \theta \approx 143^\circ$: well below the equator, close-ish to the south pole — as befits a 90%- $|1\rangle$ state.

Problem 12 [Standard] Both mixtures average two antipodal surface points and land at the *center* of the ball: arrow length 0. With no direction, every measurement axis sees 50/50 — the two recipes produce one identical physical object. (This “different recipes, same state” fact is what density matrices formalize.)

Problem 13 [Standard] $\text{Length} = \sqrt{0.4^2 + 0^2 + 0.4^2} = 0.566 < 1$: mixed. The purity ratio to the surface state in the same direction is 0.566 — the state behaves like that pure state with all interference visibilities scaled by 0.566.

Problem 14 [Standard] $|0\rangle \rightarrow H \rightarrow \text{equator } \varphi = 0^\circ \rightarrow Z \rightarrow \text{equator } \varphi = 180^\circ \rightarrow H \rightarrow P(0) = (1 + \cos 180^\circ)/2 = 0$: the state is $|1\rangle$, certainly. Geographically proven identity: $HZH = X$ (the flight’s net effect flipped the pole).

Problem 15 [Challenge] $\theta = 60^\circ, \varphi = 0^\circ$: $|\psi_1\rangle = \cos 30^\circ |0\rangle + \sin 30^\circ |1\rangle = 0.866 |0\rangle + 0.5 |1\rangle$. Partner point $(120^\circ, 180^\circ)$: $|\psi_2\rangle = \cos 60^\circ |0\rangle + e^{i180^\circ} \sin 60^\circ |1\rangle = 0.5 |0\rangle - 0.866 |1\rangle$. Inner product: $0.866 \times 0.5 - 0.5 \times 0.866 = 0$: orthogonal ✓ General statement: antipodal points on the sphere are exactly the orthogonal state pairs — every measurement is a diameter.

Problem 16 [Challenge] From the angle formula: $|+i\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle)$ and $|-i\rangle = \frac{1}{\sqrt{2}}(|0\rangle - i|1\rangle)$ (longitudes 90° and 270°). $\langle +i|-i\rangle = \frac{1}{2}(1 + (-i)(-i))$ — carefully, bra conjugates: $\frac{1}{2}(1 \cdot 1 + (-i) \cdot (-i)) = \frac{1}{2}(1 + i^2 \dots)$; cleanest: $\frac{1}{2}(1) + \frac{1}{2}(-i)(-i) = \frac{1}{2} - \frac{1}{2} = 0$ ✓ Antipodal $\Rightarrow$ orthogonal, confirmed on the y -axis pair.

Problem 17 [Challenge] $-|+\rangle = e^{i180^\circ} |+\rangle$ differs only by global phase: every probability of every measurement is identical, so no receiver, with any apparatus, can distinguish the two “symbols” — the message is unreadable in principle (Chapter 3, Problem 17 proved the invariance). Minimal fix: put the sign on *one amplitude only* — encode 0 as $|+\rangle$ and 1 as $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$: now the phase is *relative*, the states are antipodal (orthogonal), and one X-basis measurement reads the bit perfectly.

Problem 18 [Challenge] Average the arrows: $0.9(+x) + 0.1(-x) = 0.8$ along $+x$: a mixed state, length 0.8. X-basis: $P(+)= (1 + 0.8)/2 = 0.9$ (equivalently: 90% of shots come from $|+\rangle$). The same-direction pure state (length 1) would give $P(+)= 1$: the missing 0.2 of arrow length is exactly the preparation’s 10% contamination showing up as lost certainty.

Problem 19 [Challenge] (Model answer.) The environment continually “measures” the qubit (Chapter 28), and un-read measurement converts superpositions into mixtures — this chapter’s verdict lesson. A mixture’s arrow is the average of the outcomes’ arrows, which is *shorter* than either: hence decoherence = arrows shrinking toward the center, with length tracking surviving interference visibility. The Ramsey experiment measures precisely that shrinkage over time: its fringe amplitude is the transverse arrow length, decaying as e^{-t/T_2} . One picture — collapsing arrows — thus unifies measurement, mixedness, and the T_2 clock.

Problem 20 [Challenge] $\hat{P}(0) = 3415/4000 = 0.854 \Rightarrow \cos \varphi = 2\hat{P} - 1 = 0.708 \Rightarrow \varphi \approx \pm 45^\circ$ — two candidates, because cosine is even: the interferometer reads only $\cos \varphi$, not the sign. Wobble: $\sigma = \sqrt{0.854 \times 0.146/4000} \approx 0.006$ (so φ is pinned to a few degrees, but its sign is not). Tie-breaker: insert S before the final H (total phase $\varphi + 90^\circ$): candidate $+45^\circ$ predicts $P(0) = (1 + \cos 135^\circ)/2 = 0.146$; candidate

-45° predicts $(1 + \cos 45^\circ)/2 = 0.854$. One 200-shot run decides emphatically. (This two-quadrature read is exactly how laboratories fit both cos and sin of an unknown phase.)

CHAPTER SIX

Single-Qubit Gates I: X, Z, and H

Teaching notes, self-test, and complete solutions

TEACHING NOTES

Gates arrive, and with them the course's first real computations. Two non-negotiables: matrix-vector multiplication must become automatic (short daily drills beat one long session), and the HH interference circuit must be traced by hand by every student — it is the book's minimal quantum computation, and its collapse under a middle measurement (Problem 10) is the cleanest demonstration that measurement destroys interference. The chapter's twin lessons — two reflections make a rotation, and the commutation field guide — are compiler literacy; emphasize the practical rule (diagonal gates slide past each other, H stops everything) since students will debug with it in Chapter 14. Sandwiches ($HZH = X$) deserve choral repetition. Frequent stumble: applying gates right-to-left in writing but left-to-right on diagrams; make the class annotate both orders once, loudly.

SELF-TEST — QUESTIONS WITH ANSWERS

Q1. What condition makes a matrix a legal quantum gate?

A. Unitarity: $U^\dagger U = I$ — lengths (total probability) are preserved, and the gate is reversible.

Q2. What does H do to $|0\rangle$ and to $|+\rangle$?

A. $H|0\rangle = |+\rangle$ and $H|+\rangle = |0\rangle$ — H is its own inverse, swapping the z- and x-axes.

Q3. Why does the circuit H-H on $|0\rangle$ return $|0\rangle$ with certainty?

A. The two paths to $|1\rangle$ carry amplitudes $+\frac{1}{2}$ and $-\frac{1}{2}$ and cancel; the paths to $|0\rangle$ add. Interference, not coincidence.

Q4. Do X and Z commute?

A. No: $XZ = -ZX$ (they anticommute). Order matters up to a sign — and that sign is a relative phase in superpositions.

Q5. What is the sandwich rule?

A. $HZH = X$ and $HXH = Z$: conjugating by H converts between z-type and x-type actions — the basis-change trick used everywhere from measurement compiling to error correction.

Q6. A middle measurement is inserted into an interference circuit. What happens?

A. The superposition collapses; path amplitudes stop adding; the interference pattern is replaced by classical 50/50 statistics. Measurement is an action, not a peek.

Solutions to the Practice Problems

Problem 1 [Basic] $X|1\rangle = |0\rangle$. $X|+\rangle = \frac{1}{\sqrt{2}}(|1\rangle + |0\rangle) = |+\rangle$ (unchanged — $|+\rangle$ is X 's $+1$ eigenvector). $X(0.28|0\rangle + 0.96|1\rangle) = 0.96|0\rangle + 0.28|1\rangle$ (amplitudes swap).

Problem 2 [Basic] $Z|0\rangle = |0\rangle$ and $Z|1\rangle = -|1\rangle$: each output differs from its input by at most a *global* phase — physically unchanged. $Z|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = |+\rangle$: a genuinely new state (the sign was *relative* there).

Problem 3 [Basic] $ZZ = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$: two phase flips cancel.

Problem 4 [Basic] Diagram: $|0\rangle \rightarrow [H] \rightarrow [X] \rightarrow \text{output}$. As matrices, time order reverses: $\text{output} = XH|0\rangle$ — H stands next to $|0\rangle$ because it acts first.

Problem 5 [Basic] No. $XZ = -iY$ while $ZX = +iY$: opposite signs. On $|+\rangle$: $XZ|+\rangle = X|-\rangle = -|-\rangle$, while $ZX|+\rangle = Z|+\rangle = |-\rangle$ — same physical state here but opposite sign, and in superpositions with other terms that relative sign is measurable.

Problem 6 [Basic] Both products equal $\begin{pmatrix} 1 & 0 \\ 0 & e^{i3\pi/4} \end{pmatrix}$ (angles add: $90^\circ + 45^\circ$ in either order): S and T commute (both diagonal). $ST = R_z$ -type gate attaching 135° — in menu terms, $ST = TS =$ the 135° phase gate ($= T^3$).

Problem 7 [Standard] $X|0\rangle = |1\rangle$, then $H|1\rangle = |-\rangle$. Output $|-\rangle$: equator, longitude 180° ($-x$ axis).

Problem 8 [Standard] $H(0.6|0\rangle + 0.8|1\rangle) = \frac{0.6+0.8}{\sqrt{2}}|0\rangle + \frac{0.6-0.8}{\sqrt{2}}|1\rangle = 0.990|0\rangle - 0.141|1\rangle$. Probabilities: $P(0) = 0.98$, $P(1) = 0.02$. Unequal inputs interfere *partially*: strong constructive toward $|0\rangle$, weak destructive toward $|1\rangle$ — nothing cancels exactly unless amplitudes match.

Problem 9 [Standard] $XZ = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, $ZX = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$: unequal (they differ by overall sign). $XZ|+\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} -1 \\ 1 \end{pmatrix} = -|-\rangle$; $ZX|+\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -1 \end{pmatrix} = |-\rangle$. Both are gallery state $|-\rangle$ up to global phase.

Problem 10 [Standard] After first H : $|+\rangle$. Middle measurement: 0 or 1, each $\frac{1}{2}$, collapsing to $|0\rangle$ or $|1\rangle$. Second H sends either to an equal superposition, so the final measurement is 50/50 regardless of branch: total $P(0) = \frac{1}{2}$. The middle measurement destroyed the *coherent minus sign* between paths — the ingredient HH needed to reconstruct certainty.

Problem 11 [Standard] Prediction: $HXH = Z$ (apply H to both sides of $HZH = X$ and use $HH = I$). Verification: $HXH = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \checkmark$

Problem 12 [Standard] $ZX = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = +iY$. The two orders differ by the factor -1 ($XZ = -ZX$). An overall -1 is a global phase: probabilities take moduli squared, so it can never show up alone — only *relative* to another amplitude (which is exactly where anticommutation bites in circuits).

Problem 13 [Standard] $XZ = -iY$; so $XZXZ = (-iY)(-iY) = (-i)^2 Y^2 = (-1)(I) = -I$. The string is the identity times global phase -1 : physically, doing nothing — elegantly.

Problem 14 [Standard] $HZ = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \neq ZH = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$; similarly $HX \neq XH$; but the three-factor product HZH collapses to X (computed in Problem 11's pattern). A sandwich is not a swap because the two outer H's undo each other's basis change around the middle gate — one H alone just moves the disagreement elsewhere.

Problem 15 [Challenge] Group: $HZH Z H Z H = (HZH) Z (HZH) = X Z X$. Then $XZX = (XZ)X = (-iY)X = -i(YX) = -i(iZ) = Z$... carefully: $YX = -iZ$? Compute directly: $XZX = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} = -Z$. So the seven-gate circuit equals $-Z$: on $|0\rangle$ it gives $-|0\rangle$ — physically $|0\rangle$, unchanged. (Conjugating Z by X flips its sign: $XZX = -Z$, a useful identity in its own right.)

Problem 16 [Challenge] Unitary: columns are orthonormal ✓ ($M^\dagger M = I$ by direct multiplication). $M|0\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = |-\rangle$; $MM|0\rangle = M|-\rangle = \frac{1}{2} \begin{pmatrix} 1 & -1 \\ -1 & -1 \end{pmatrix} \dots$ compute: $MM = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, so $MM|0\rangle = -|1\rangle$. H is a *reflection* ($H^2 = I$); moving the minus sign turned M into a genuine 90° rotation about the y-axis ($M = R_y(-90^\circ)$), and quarter-turns are not self-inverse: $M^4 = -I$, not M^2 .

Problem 17 [Challenge] $b = 0$: Bob receives $HH|0\rangle = |0\rangle$, measures 0. $b = 1$: $HZH|0\rangle = X|0\rangle = |1\rangle$, measures 1. One standard measurement reads b perfectly. Mid-flight (between the H's) the bit exists only as a *relative phase* ($|+\rangle$ vs $|-\rangle$ — identical standard-basis statistics); the second H converts phase back to population. An eavesdropper measuring mid-flight in the standard basis collapses either carrier state to $|0\rangle / |1\rangle$ (learning nothing — both give 50/50) and destroys the phase, so Bob's readout becomes a coin flip: tampering scrambles the channel. (BB84's engine, in one qubit.)

Problem 18 [Challenge] Prediction: reflections about axes 45° apart compose to a rotation by 90° about the mutual perpendicular — here the **y-axis**. Test: $HZ|0\rangle = H|0\rangle = |+\rangle$; a $+90^\circ$ y-turn indeed carries the north pole to the $+x$ point $|+\rangle$ ✓ (So $HZ = R_y(90^\circ)$ up to global phase — students may verify against the R_y matrix of the next chapter.)

Problem 19 [Challenge] $XH = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$ and $HZ = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$: equal ✓ Then $XH X H = (XH)(XH) = (HZ)(HZ) = H(ZH)Z \dots$ simplest: $(HZ)^2$; from Problem 18, HZ is a 90° y-rotation, so its square is the 180° y-rotation: $XH X H = -iY$ — a single Pauli times phase. (Matrix check: $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ ✓)

Problem 20 [Challenge] $S' = HSH$ (then $S'H = HSH \cdot H = HS$ automatically). By the sandwich rule, conjugating the z-quarter-turn S by H yields the x-axis quarter turn — the “ $\sqrt{X}$ ” gate, $\frac{1}{2} \begin{pmatrix} 1+i & 1-i \\ 1-i & 1+i \end{pmatrix}$. Verify on $|0\rangle$: $HS|0\rangle = H|0\rangle = |+\rangle$; and $S'H|0\rangle = HSH H|0\rangle \dots = HS|0\rangle$ trivially — more instructively, $S'|+\rangle$: an x-rotation leaves the x-axis state $|+\rangle$ fixed, so $S'H|0\rangle = |+\rangle$ ✓ Compilers call this “pushing a phase gate through a Hadamard”: z-turns become x-turns.

CHAPTER SEVEN

Single-Qubit Gates II: Y, S, T, and Rotations

Teaching notes, self-test, and complete solutions

TEACHING NOTES

The gate menu completes, and two engineering ideas debut that outlast the course: *compiling* (any gate from H and T; Euler's three turns) and *cost* (the T-count — students should leave knowing why fault-tolerant machines bill by the T). Phase arithmetic ($S^2 = Z$, angles add) makes excellent mental math drills. The Euler lesson rewards a geometric telling before the algebra: three dials, longitude–latitude–longitude, cover the sphere. The universality discussion ($\{H, S\}$ traps you in six states; T's 45° escapes) is a lovely class investigation — let students discover the finite orbit themselves (Problem 17) before naming it. Recurrent stumbles: the $\theta/2$ in rotation matrices (again!), confusing $T^2 = S$ with $2T$ -angles, and treating global phase as physical (Problems 14 and 20 settle it constructively).

SELF-TEST — QUESTIONS WITH ANSWERS

Q1. What do S and T do, in phase language?

A. They attach a phase to $|1\rangle$ only: S a quarter turn (90°), T an eighth (45°). $T^2 = S$, $S^2 = Z$.

Q2. Which single family of gates changes latitude on the Bloch sphere?

A. Rotations with an x- or y-component ($R_x, R_y, H\dots$). Pure z-turns (Z, S, T, R_z) move longitude only — histograms cannot see them.

Q3. State the universality result for one qubit.

A. H and T alone approximate any single-qubit gate to any desired accuracy (with efficiently short sequences — Solovay–Kitaev); adding CNOT extends this to all multi-qubit circuits.

Q4. Why is the T gate “expensive” later in the book?

A. Error-corrected Clifford gates (H, S, CNOT) are cheap in the surface code; T needs distilled magic states, costing orders of magnitude more — so compiled T-count prices fault-tolerant algorithms.

Q5. Give the ZYZ Euler statement.

A. Any single-qubit $U = e^{i\gamma} R_z(\alpha) R_y(\beta) R_z(\delta)$: three turns about two fixed axes, plus an unphysical global phase.

Q6. What is $S^\dagger$, and when is it used?

A. The -90° z-turn (phase $-i$ on $|1\rangle$), undoing S ; it appears in the Y-measurement compiler ($S^\dagger$ then H) and wherever a phase must be unwound.

Solutions to the Practice Problems

Problem 1 [Basic] $Y|1\rangle = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -i \\ 0 \end{pmatrix} = -i|0\rangle$: physically the state $|0\rangle$, with global phase $-i$ factored out.

Problem 2 [Basic] $S|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle) = |+\rangle$ — the quarter-turn slides the equator point from $+x$ to $+y$.

Problem 3 [Basic] $S^2: |1\rangle$ picks up $90^\circ + 90^\circ = 180^\circ$, i.e. the factor -1 : that is $Z \checkmark T^4: 4 \times 45^\circ = 180^\circ$: Z again $\checkmark$

Problem 4 [Basic] $R_y(90^\circ)$: the y-axis turn moves the north pole along the xz -meridian to the equator. Check: $R_y(90^\circ) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$, and applied to $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ gives $\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = |+\rangle \checkmark$

Problem 5 [Basic] $Z: (\alpha, \beta, \delta) = (180^\circ, 0^\circ, 0^\circ)$ (or $(0^\circ, 0^\circ, 180^\circ)$ — pure z-turns are flexible). $S: (90^\circ, 0^\circ, 0^\circ)$. $R_y(30^\circ): (0^\circ, 30^\circ, 0^\circ)$.

Problem 6 [Basic] The Born rule takes squared moduli, and $|e^{i\gamma}z|^2 = |z|^2$ for every amplitude at once — so a global phase changes no probability of any measurement.

Problem 7 [Standard] T adds 45° to the relative phase: $T|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle + e^{i(180^\circ+45^\circ)}|1\rangle) = \frac{1}{\sqrt{2}}(|0\rangle + e^{i225^\circ}|1\rangle)$: $\varphi = 225^\circ$.

Problem 8 [Standard] $R_y(90^\circ)|0\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = |+\rangle$.

Problem 9 [Standard] $\sin^2(\theta/2) = 0.36 \Rightarrow \sin(\theta/2) = 0.6 \Rightarrow \theta/2 = 36.9^\circ \Rightarrow \theta \approx 73.7^\circ$. Prepared state: $\cos(36.9^\circ)|0\rangle + \sin(36.9^\circ)|1\rangle = 0.8|0\rangle + 0.6|1\rangle$.

Problem 10 [Standard] $S^\dagger = \begin{pmatrix} 1 & 0 \\ 0 & -i \end{pmatrix}$ (conjugate the i); $SS^\dagger = \begin{pmatrix} 1 & 0 \\ 0 & i(-i) \end{pmatrix} = I \checkmark$ On the phase circle $S^\dagger$ attaches -90° (equivalently 270°) to $|1\rangle$: the quarter-turn in reverse.

Problem 11 [Standard] After $H: |+\rangle$; after $T: \frac{1}{\sqrt{2}}(|0\rangle + e^{i45^\circ}|1\rangle)$; after H : amplitudes $\frac{1 \pm e^{i45^\circ}}{2}$. $P(0) = \frac{|1+e^{i45^\circ}|^2}{4} = \frac{2+\sqrt{2}}{4} = 0.854$; $P(1) = \frac{2-\sqrt{2}}{4} = 0.146$. Compare: $H-Z-H$ gives 0/1 (full flip), $H-H$ gives 1/0 (full return): the 45° phase steers the interference to a *partial* split — the phase dial made visible.

Problem 12 [Standard] $225^\circ = 180^\circ + 45^\circ$: minimum T-count 1 with ZT (or SST — same T-count, more gates). Worse expansion: T^5 (five T's, T-count 5) — same physics, five times the fault-tolerant price.

Problem 13 [Standard] $-45^\circ \equiv 315^\circ \pmod{360^\circ}$: the same z-rotation. As a single menu gate both are $T^\dagger$. (Equivalently ZST reaches 315° with T-count 1 — accept either, prefer $T^\dagger$.)

Problem 14 [Standard] $R_z(180^\circ) = \begin{pmatrix} -i & 0 \\ 0 & i \end{pmatrix} = -iZ$; multiplying: $R_y(90^\circ)R_z(180^\circ) = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}(-i)\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = -i\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = -iH$ ✓ And $(-iH)(-iH) = (-i)^2 H^2 = -I$: as a circuit, the double application acts as the identity on all measurement statistics (global -1), exactly as $HH = I$ demands physically.

Problem 15 [Challenge] $XZ = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = -iY$ ✓ So Y is, physically, “phase-flip then bit-flip” (Z then X, reading right to left) — one gate doing both Pauli jobs. The $-i$ is global: harmless alone, bookkeeping-relevant only inside larger products (where it recombines into relative phases).

Problem 16 [Challenge] $R_x(45^\circ) = HTH$ (sandwich: H converts the z-turn T into the same-angle x-turn; T’s global-phase difference from $R_z(45^\circ)$ stays global). Checkpoint: $|+\rangle$ lies on the x-axis, and any x-rotation fixes its own axis: $HTH|+\rangle = HT|0\rangle = H|0\rangle = |+\rangle$ ✓

Problem 17 [Challenge] From $|+\rangle$, S-steps move longitude in 90° jumps: $\{0^\circ, 90^\circ, 180^\circ, 270^\circ\}$ — four equator points; H swaps the x- and z-axes, reaching only the poles from these. The reachable set is the six *landmark states* (the octahedron), a finite orbit: no sequence of 90° machinery ever lands between them. T’s 45° breaks the grid — and (the deep part) repeated H/T sequences generate *irrational* rotations whose multiples fill the sphere densely: fine-grained angles from coarse parts. That is universality; the finite $\{H, S\}$ world is the Clifford group.

Problem 18 [Challenge] $R_x(90^\circ) = HSH$ (sandwich again), T-count 0 — a Clifford rotation, cheap even fault-tolerantly. On $|0\rangle$: $SH|0\rangle \dots$ carefully in circuit order: first H: $|+\rangle$; then S: $|+i\rangle$; then H: $\frac{1+i}{2}|0\rangle + \frac{1-i}{2}|1\rangle$, which is e^{i45° times $\frac{1}{\sqrt{2}}(|0\rangle - i|1\rangle) = |-i\rangle$: a y-meridian (equator, $\varphi = -90^\circ$) point ✓ — exactly where a 90° x-turn drops the north pole.

Problem 19 [Challenge] Best single menu gate for 60° : T at 45° (or ST at 135° — farther). Equator states with phase gap δ have fidelity $\cos^2(\delta/2)$: for $\delta = 15^\circ$, $F = \cos^2 7.5^\circ = 0.983 < 0.999$: fails. Requirement: $\cos^2(\delta/2) \geq 0.999 \Rightarrow \delta \leq 3.6^\circ$. An H/T compile must therefore land within 3.6° of the target angle — the accuracy budget that Solovay-Kitaev-style sequences (a dozen or so gates here) are built to meet.

Problem 20 [Challenge] $R_x(180^\circ) = \cos 90^\circ I - i \sin 90^\circ X = -iX$, so $X = iR_x(180^\circ)$: equal up to global phase i ✓ Both send $|0\rangle \rightarrow |1\rangle$ (one exactly, one as $-i|1\rangle$). In any circuit ending in measurement, the global i is invisible — hardware may freely implement X as the plain 180° pulse. But *controlling* the gate promotes the phase: controlled- $R_x(180^\circ)$ applies the $-i$ only on the control’s $|1\rangle$ branch, creating a measurable *relative* phase compared with a true controlled-X — the difference is an unwanted controlled- $S^\dagger$ -squared... precisely the kickback arithmetic of Chapter 10. Moral: global phases die alone but resurrect under control.

CHAPTER EIGHT

Observables and Expectation Values

Teaching notes, self-test, and complete solutions

TEACHING NOTES

This chapter converts histograms into *numbers with algebra* — the language of every variational and ML chapter later. Three anchors: $\langle Z \rangle = P(0) - P(1)$ (a histogram folded to one number), the Bloch connection (averages *are* coordinates — tomography demystified), and the rotate-then-read compiler (X: prefix H; Y: prefix $S^\dagger H$), which students should rehearse until reflexive since Chapters 20–27 use it constantly. The uncertainty lesson upgrades the course slogan into a theorem; the equality case (equator states) is worth a full worked demonstration at the board. Shot budgeting (Problems 13, 16) is the professional skill — assign at least one budgeting problem as graded homework. Watch for: sign errors in $\langle Z \rangle$ (counts to ± 1), forgetting the conjugate in $\langle Y \rangle$ hand-computations, and the perennial “uncertainty = bad apparatus” misconception.

SELF-TEST — QUESTIONS WITH ANSWERS

Q1. Convert: 700 zeros and 300 ones in 1000 shots. $\langle Z \rangle$?

A. $\langle Z \rangle = 0.7 - 0.3 = +0.4$ (wobble about $\pm 1/\sqrt{1000} \approx 0.03$).

Q2. What are the three Bloch coordinates operationally?

A. Three expectation values — $\langle X \rangle, \langle Y \rangle, \langle Z \rangle$ — each estimated from its own compiled histogram; together they locate the state (tomography).

Q3. How do you measure X on hardware that only measures Z?

A. Apply H, then measure Z: $HXH = Z$ maps X's eigenstates onto the standard basis. Convert counts as usual.

Q4. What is ΔP for a Pauli, in terms of $\langle P \rangle$?

A. $\Delta P = \sqrt{1 - \langle P \rangle^2}$: outcomes are ± 1 , so certainty ($\langle P \rangle = \pm 1$) means zero spread and vice versa.

Q5. State the qubit uncertainty relation.

A. $\Delta Z \Delta X \geq |\langle Y \rangle|$ — a floor under simultaneous sharpness, a property of the state, not the instrument.

Q6. Why does VQE care about this chapter?

A. A molecular energy is $\sum_i c_i \langle P_i \rangle$: a weighted list of exactly these compiled histogram averages, shot noise and

all.

Solutions to the Practice Problems

Problem 1 [Basic] (a) $+1$; (b) -1 ; (c) $|-\rangle$: 50/50 in Z, so 0; (d) $0.64 - 0.36 = +0.28$.

Problem 2 [Basic] $\hat{P}(0) = 0.67$, so $\langle Z \rangle \approx 0.67 - 0.33 = +0.34 \pm 0.045$ (bar $1/\sqrt{500}$).

Problem 3 [Basic] Circuit: — [H] — meter. Counts convert exactly as for Z: $\langle X \rangle = P(0_{\text{after H}}) - P(1_{\text{after H}})$.

Problem 4 [Basic] The arrow points along $-y$: the state is $|-i\rangle = \frac{1}{\sqrt{2}}(|0\rangle - i|1\rangle)$.

Problem 5 [Basic] $\langle Z \rangle = 0.9 - 0.1 = 0.8$; $\Delta Z = \sqrt{1 - 0.64} = 0.6$. Such states live on the circle of latitude with z -coordinate 0.8 — high on the northern hemisphere, any longitude.

Problem 6 [Basic] All three spreads zero would need $\langle X \rangle = \langle Y \rangle = \langle Z \rangle = \pm 1$ simultaneously — a Bloch vector of length $\sqrt{3} > 1$. Forbidden: pure states have length exactly 1 (and mixed states less). One certainty is the most a qubit ever grants.

Problem 7 [Basic] $\langle X \rangle$: prefix H, measure, $\hat{P}(0) - \hat{P}(1)$. $\langle Y \rangle$: prefix $S^\dagger$ then H, measure, same conversion. (Mnemonic: undo the phase, then undo the superposition.)

Problem 8 [Basic] $H|0\rangle = |+\rangle$: histogram 50/50. Conversion: $\langle X \rangle = 0.5 - 0.5 = 0$ ✓ — a pole state has no x -lean, and the compiled histogram says so through maximal randomness.

Problem 9 [Standard] $\langle Z \rangle = |\frac{1}{\sqrt{2}}|^2 - |\frac{i}{\sqrt{2}}|^2 = 0$. $\langle X \rangle = 2 \operatorname{Re}(\alpha^* \beta) = 2 \operatorname{Re}(\frac{1}{\sqrt{2}} \cdot \frac{i}{\sqrt{2}}) = 0$. With $(\langle X \rangle, \langle Z \rangle) = (0, 0)$ and purity $|r| = 1$: $\langle Y \rangle = \pm 1$; the $+i$ phase gives $+1$ — the state is $|+i\rangle$, the $+y$ pole.

Problem 10 [Standard] $\langle Z \rangle = \cos^2 \frac{\theta}{2} - \sin^2 \frac{\theta}{2} = \cos \theta$ (the double-angle identity). This is why VQE's one-qubit example writes its energy directly as $E(\theta) = c \cos \theta + \dots$

Problem 11 [Standard] Length: $\sqrt{0.71^2 + 0.70^2} = 0.997 \approx 1$: pure (within shot noise). Location: the xz -meridian, northern hemisphere, leaning toward $-x$: $\theta \approx 45^\circ$ tilted *backwards* — reachable as $R_y(-45^\circ)|0\rangle$ (a -45° y -turn). (Accept $R_y(315^\circ)$; the point is the sign.)

Problem 12 [Standard] (a) $|\Phi^+\rangle$: outcomes 00/11 only — always agree: $\langle ZZ \rangle = +1$. (b) $|\Psi^+\rangle$: 01/10 only: -1 . (c) $|+\rangle|+\rangle$: four outcomes equally: agreements and disagreements balance: 0.

Problem 13 [Standard] Per term: $n \approx (1/0.02)^2 = 2500$ shots; per energy evaluation: $40 \times 2500 = 10^5$; per 200-step run: 2×10^7 shots. At 10^4 shots/s: $2000 \text{ s} \approx 33 \text{ minutes} (\approx 0.6 \text{ h})$ — *before* gradient evaluations multiply it further (Chapter 25's pricing); grouping (Chapter 20) divides it back down.

Problem 14 [Standard] $\langle Z \rangle = 0.28$; $\langle X \rangle = 2(0.8)(0.6) = 0.96$; $\langle Y \rangle = 0$ (real amplitudes). Spreads: $\Delta Z = \sqrt{1 - 0.0784} = 0.960$; $\Delta X = \sqrt{1 - 0.9216} = 0.28$; $\Delta Y = 1$. Check: $\Delta Z \Delta X = 0.269 \geq 0 = |\langle Y \rangle|$ ✓ (comfortably above a zero floor).

Problem 15 [Standard] $\sum_P \Delta P^2 = \sum_P (1 - \langle P \rangle^2) = 3 - (r_x^2 + r_y^2 + r_z^2) = 3 - 1 = 2$ for pure states. Total spread is conserved: sharpening one observable necessarily blunts the others — the uncertainty seesaw as an exact identity.

Problem 16 [Standard] Wobble $1/\sqrt{n} \leq 0.03 \Rightarrow n \approx 1100$ per basis (one sigma); three bases: ≈ 3300 shots. (Two-sigma confidence: quadruple, $\approx 13,000$.)

Problem 17 [Standard] Prediction: $P(0) = \frac{1+\langle X \rangle}{2} = \frac{1.96}{2} = 0.98$. Direct: $\langle +|\psi \rangle = \frac{0.6+0.8}{\sqrt{2}} = 0.990$, squared = 0.98 ✓

Problem 18 [Challenge] $(\langle \psi | A | \psi \rangle)^* = \langle \psi | A^\dagger | \psi \rangle = \langle \psi | A | \psi \rangle$ (first step: conjugating reverses the sandwich and daggers A ; second: $A^\dagger = A$). A number equal to its own conjugate is real. Physically it must be: the average of real outcomes (± 1 , or any eigenvalues) weighted by real probabilities cannot be complex.

Problem 19 [Challenge] $(\langle Z \rangle, \langle X \rangle)$: at 0° : $(1, 0)$; at 45° : $(0.707, 0.707)$; at 90° : $(0, 1)$. Each pair satisfies $\langle X \rangle^2 + \langle Z \rangle^2 = 1$ ✓ The identity says the two averages share one unit of “sharpness”: both can be moderately known, neither pinned while the other is free — complementarity (Bohr), the uncertainty principle’s older name, peeking through as a circle.

Problem 20 [Challenge] Angle differences: $A_1 B_1$: 45° ; $A_2 B_1$: 45° ; $A_2 B_2$: 45° ; $A_1 B_2$: 135° . Correlations: 0.707, 0.707, 0.707, -0.707 . Score: $S = 0.707 + 0.707 + 0.707 - (-0.707) = 4 \times 0.707 = 2\sqrt{2} \approx 2.83$ ✓ The minus sign lands on the single anti-correlated pair, so all four terms pull upward — the settings are engineered for exactly that. $2.83 > 2$: no local hidden-answer model survives; the 2022 Nobel number, reproduced on one line.

Problem 21 [Challenge] Equator: $\langle Z \rangle = 0 \Rightarrow \Delta Z = 1$; $\langle X \rangle = \cos \varphi \Rightarrow \Delta X = |\sin \varphi|$. Product: $|\sin \varphi|$, maximized at $\varphi = 90^\circ$ or 270° (the states $|\pm i\rangle$), where it equals 1. But the floor there is $|\langle Y \rangle| = |\sin \varphi| = 1$ too: *equality* — indeed equality holds at every equator point. The product-floor gap is zero on the whole equator; it opens only off-equator (e.g. the xz -meridian at $\theta = 45^\circ$: product $\frac{1}{2}$, floor 0). Teaching point: the equator is the minimum-uncertainty family for the (Z, X) pair.

Problem 22 [Challenge] The observable’s axis lies in the xz -plane, 45° from z ; the prefix $R_y(-45^\circ)$ rotates that axis onto z , after which a standard measurement reads it. Prediction for $|0\rangle$: $P(0) = \frac{1+\langle \hat{n} \rangle}{2} = \frac{1+1/\sqrt{2}}{2} = 0.854$. Check: the observable’s $+1$ eigenstate is $(\cos 22.5^\circ, \sin 22.5^\circ)$, and $|\langle \lambda_+ | 0 \rangle|^2 = \cos^2 22.5^\circ = 0.854$ ✓ — the same 85.4% that keeps appearing wherever axes meet at 45° .

CHAPTER NINE

Multi-Qubit Systems and Tensor Products

Teaching notes, self-test, and complete solutions

TEACHING NOTES

The chapter where quantum computing's power source — exponential state space — becomes arithmetic. Drill the tensor product as bracket expansion ("amplitudes multiply, labels join") until it is algebra, not ceremony. The three load-bearing skills: probability tables with marginals, the partial-measurement rule (filter and renormalize — students who master it here find teleportation easy), and the Feynman-wall estimates (2^n bytes arithmetic — assign Problem 19 and watch conviction form). The W-state lesson quietly teaches that "entangled" has *kinds*; do not skip its robustness comparison, which networks and error correction both echo. Persistent stumbles: writing $|0\rangle|1\rangle$ amplitudes into the wrong slots (fix with the 00, 01, 10, 11 column order chanted aloud), and adding amplitudes across *different* patterns (they add only within a pattern; across patterns, probabilities add).

SELF-TEST — QUESTIONS WITH ANSWERS

Q1. How many amplitudes for 8 qubits? For 30?

A. $2^8 = 256$; $2^{30} \approx 10^9$ — a gigabyte-scale list from thirty qubits.

Q2. Expand $|+\rangle \otimes |1\rangle$.

A. $\frac{1}{\sqrt{2}}|01\rangle + \frac{1}{\sqrt{2}}|11\rangle$: A undecided, B certainly 1.

Q3. What is the marginal rule?

A. $P(A=1) = \text{sum of } |a_{xy}|^2 \text{ over patterns whose A-digit is 1}$ — add probabilities of consistent patterns.

Q4. State the partial-measurement rule in one line.

A. Keep only amplitudes consistent with the observed digit, divide by the square root of their total probability.

Q5. How do you spot a product state from its four amplitudes?

A. It factors: $a_{00}a_{11} = a_{01}a_{10}$ (equivalently, rows of the 2×2 amplitude grid are proportional). Any failure of that identity means entanglement.

Q6. Why can't a 60-qubit state be stored classically?

A. $2^{60} \approx 10^{18}$ amplitudes $\times$ 16 bytes = 1.6×10^{19} bytes — beyond exabyte data centers; each added qubit doubles the bill.

Solutions to the Practice Problems

Problem 1 [Basic] $|00\rangle = (1, 0, 0, 0)^T$, $|01\rangle = (0, 1, 0, 0)^T$, $|10\rangle = (0, 0, 1, 0)^T$, $|11\rangle = (0, 0, 0, 1)^T$.

Problem 2 [Basic] $|1\rangle \otimes |+\rangle = \frac{1}{\sqrt{2}}|10\rangle + \frac{1}{\sqrt{2}}|11\rangle$: A certainly 1, B a fair coin.

Problem 3 [Basic] $P(01) = |\frac{1}{2}|^2 = \frac{1}{4}$. $P(A=0) = P(00) + P(01) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$.

Problem 4 [Basic] $2^5 = 32$ amplitudes. First three patterns: 00000, 00001, 00010; last: 11111.

Problem 5 [Basic] Amplitudes sit on 00 and 10: $P(A=0) = P(A=1) = \frac{1}{2}$; after either outcome B collapses to $|0\rangle$. B's state never depended on A's result — the state was the product $|+\rangle \otimes |0\rangle$: perfectly uncorrelated.

Problem 6 [Basic] $P(B=1) = |0.6|^2 = 0.36$ (only pattern 11 has $B = 1$). Afterward the state is $|11\rangle$ — A snapped to 1 as well: that is correlation.

Problem 7 [Basic] Norm: $3 \times \frac{1}{3} = 1$ ✓ $P(\text{exactly one } 1) = P(001) + P(010) + P(100) = 1$; $P(000) = 0$. The W state *always* shows exactly one excitation.

Problem 8 [Basic] C reads 1 only via pattern 001: $P(C=1) = \frac{1}{3}$, collapsing A,B to $|00\rangle$. $P(C=0) = \frac{2}{3}$, keeping $\frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$ — a Bell state — after renormalizing $\frac{1}{\sqrt{3}}(|01\rangle + |10\rangle)$ by $\sqrt{2/3}$.

Problem 9 [Standard] Multiply out: $0.48|00\rangle + 0.36|01\rangle + 0.64|10\rangle + 0.48|11\rangle$. Probabilities $0.2304 + 0.1296 + 0.4096 + 0.2304 = 1.0000$ ✓

Problem 10 [Standard] Table: $P(00) = \frac{1}{2}$, $P(10) = \frac{1}{4}$, $P(11) = \frac{1}{4}$ (pattern 01 absent). $P(A=1) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$; $P(B=0) = \frac{1}{2} + \frac{1}{4} = \frac{3}{4}$.

Problem 11 [Standard] $X \otimes I$: flip the first digit: $\frac{1}{\sqrt{2}}|10\rangle + \frac{1}{\sqrt{2}}|11\rangle = |1\rangle \otimes |+\rangle$. $I \otimes X$: flip the second: $\frac{1}{\sqrt{2}}|01\rangle + \frac{1}{\sqrt{2}}|00\rangle = |0\rangle \otimes |+\rangle$ — the same B-superposition, opposite definite A: slot operators act strictly in their slot.

Problem 12 [Standard] $(H \otimes I)|10\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |10\rangle)$... careful: $H|1\rangle = |-\rangle$, so the result is $|-\rangle \otimes |0\rangle = \frac{1}{\sqrt{2}}|00\rangle - \frac{1}{\sqrt{2}}|10\rangle$. Applying $H \otimes I$ again: $H|-\rangle = |1\rangle$, restoring $|10\rangle$ ✓

Problem 13 [Standard] $P(000) = \frac{1}{2}$. Middle qubit is 1 only in pattern 111: $P = \frac{1}{2}$. All three agree in both surviving patterns: $P(\text{agree}) = 1$ — the GHZ signature: individually random, jointly unanimous.

Problem 14 [Standard] Measuring $A = 0$: keep $\frac{1}{2}(|00\rangle + |01\rangle)$, renormalize by $\sqrt{1/2}$: $|0\rangle \otimes \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = |0\rangle |+\rangle$; B is $|+\rangle$, exactly as before; likewise for $A = 1$. Moral: in a product state, measuring one factor never touches the other — correlation, not measurement, is what transmits influence.

Problem 15 [Standard] Norm: $\frac{3+1+4}{8} = 1$ ✓ $P(A=0) = \frac{3+1}{8} = \frac{1}{2}$. Collapsed state: $\frac{1}{2}(\sqrt{3}|00\rangle + |01\rangle)$, i.e. $|0\rangle \otimes (\frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle)$. $P(B=0 | A=0) = \frac{3}{4}$.

Problem 16 [Standard] A-first: $P(A=0) = 0.64$ then $B = 0$ certain; $P(A=1) = 0.36$ then $B = 1$ certain: histories 00: 0.64, 11: 0.36. B-first: identically $P(B=0) = 0.64 \rightarrow A = 0$; $P(B=1) = 0.36 \rightarrow A = 1$. Same joint statistics either way ✓ — measurement order never matters for outcome distributions.

Problem 17 [Standard] $|W_4\rangle = \frac{1}{2}(|0001\rangle + |0010\rangle + |0100\rangle + |1000\rangle)$. Each marginal: $P(q=1) = \frac{1}{4}$. Measuring any one qubit as 0 (probability $\frac{3}{4}$) leaves the other three sharing one excitation equally: exactly $|W_3\rangle$.

Problem 18 [Challenge] Matching: $ad = 0$ forces $a = 0$ or $d = 0$. If $a = 0$: $ac = \frac{1}{\sqrt{2}}$ fails. If $d = 0$: $bd = \frac{1}{\sqrt{2}}$ fails. No assignment exists: the state is not a product — entangled, proven by four lines of algebra.

Problem 19 [Challenge] Need $16 \times 2^n \leq 10^{18}$, i.e. $2^n \leq 6.3 \times 10^{16} \approx 2^{56}$: the exabyte center holds about **55–56 qubits** of state vector. “We simulate any quantum computer” is thus a statement with an expiry near 50-odd qubits (clever compression buys a few more on special circuits) — which is precisely why the supremacy experiments camped at 53.

Problem 20 [Challenge] (a) Even numbers end in 0: four of eight patterns: $P = \frac{1}{2}$. (b) Each shot yields *one* pattern — three bits of information, not eight patterns. The eight patterns are the *space* of possible answers; per-shot information is $\log_2 8 = 3$ bits (Holevo’s bound, in miniature).

Problem 21 [Challenge] Before: $P(B=1) = \frac{1}{2}$. After Alice measures, Bob’s outcome is 1 in the branch where Alice saw 1 (probability $\frac{1}{2}$) and 0 otherwise: averaged, $P(B=1) = \frac{1}{2} \cdot 1 + \frac{1}{2} \cdot 0 = \frac{1}{2}$ — unchanged. Bob’s *local* statistics are the marginal, and marginals are untouched by remote measurement; only when Alice’s result arrives classically can Bob’s *conditional* knowledge sharpen. Hence: correlations yes, messages no — relativity is safe.

Problem 22 [Challenge] Suppose $|W\rangle = (a|0\rangle + b|1\rangle) \otimes |\chi\rangle_{BC}$ with $|\chi\rangle = \sum_{yz} c_{yz} |yz\rangle$. The amplitude of $|1yz\rangle$ is $b c_{yz}$: since $|100\rangle$ has amplitude $\frac{1}{\sqrt{3}} \neq 0$, both $b \neq 0$ and $c_{00} \neq 0$. The amplitude of $|0yz\rangle$ is $a c_{yz}$: patterns $|001\rangle, |010\rangle$ nonzero force $a \neq 0, c_{01}, c_{10} \neq 0$. But then $|101\rangle$ would have amplitude $b c_{01} \neq 0$ — while W gives it zero. Contradiction; no A-versus-BC split exists, and by symmetry no split at all: genuine three-party entanglement.

CHAPTER TEN

Controlled Gates

Teaching notes, self-test, and complete solutions

TEACHING NOTES

The entangler arrives. Insist on the branch method (split the control's superposition, act branchwise, recombine) — it converts every controlled-gate computation into two easy ones and sets up kickback perfectly. Three summits: CNOT creating entanglement from superposition (the two-gate Bell circuit — have every student trace it), oracle-building (algorithms stop being magic the day students build their own oracle from logic; the OR sign-algebra worked example is the model), and phase kickback (the “control that gets painted” — run the $|+\rangle|-\rangle$ computation live, slowly, twice). The realitycheck about “the oracle knows the answer” addresses the single most common misunderstanding of Grover's algorithm — worth ten minutes of discussion now, repaid in Chapter 18. Stumbles: applying the CNOT to amplitudes instead of *labels* (it permutes basis states!), and forgetting that CZ is symmetric.

SELF-TEST — QUESTIONS WITH ANSWERS

Q1. What does CNOT do to basis states, in one formula?

A. $|c\rangle|t\rangle \rightarrow |c\rangle|t \oplus c\rangle$: target XORs in the control; control untouched (on basis states).

Q2. Which gate creates entanglement in the Bell circuit, H or CNOT?

A. CNOT — H only makes single-qubit superposition; the CNOT correlates the two qubits into a non-product state.

Q3. Why is CZ “symmetric”?

A. It multiplies only the $|11\rangle$ amplitude by -1 — a joint condition; control and target labels are interchangeable.

Q4. Bit oracle vs phase oracle in one line each.

A. Bit: writes $f(x)$ into a helper, $|x\rangle|a\rangle \rightarrow |x\rangle|a \oplus f(x)\rangle$. Phase: stamps the sign, $|x\rangle \rightarrow (-1)^{f(x)}|x\rangle$ — no helper, no cleanup.

Q5. State the kickback phenomenon.

A. If the target is an eigenstate of the controlled operation, the target never changes; the eigenvalue's phase attaches to the control's $|1\rangle$ branch instead.

Q6. What does Toffoli compute, and why does it matter classically?

A. Target $\oplus$ (AND of the two controls): a reversible AND — with it (plus X), any classical logic runs on a quantum computer.

Solutions to the Practice Problems

Problem 1 [Basic] $|00\rangle \rightarrow |00\rangle; |01\rangle \rightarrow |01\rangle; |10\rangle \rightarrow |11\rangle; |11\rangle \rightarrow |10\rangle$ — target flips exactly when the control is 1.

Problem 2 [Basic] CNOT: dot on the control wire joined to $\oplus$ on the target. CZ: two dots joined (no arrow — symmetric, no “target”). SWAP: two $\times$ symbols joined. (Grade the labeling: CNOT is the only one of the three where “control/target” is meaningful.)

Problem 3 [Basic] $\frac{1}{\sqrt{2}}|10\rangle + \frac{1}{\sqrt{2}}|11\rangle \rightarrow \frac{1}{\sqrt{2}}|11\rangle + \frac{1}{\sqrt{2}}|10\rangle$ — the same state ($|1\rangle \otimes |+\rangle$): with control definitely 1, the target's $|+\rangle$ is X-flipped, and $X|+\rangle = |+\rangle$.

Problem 4 [Basic] $(1, 1, 1) \rightarrow (1, 1, 0)$: both controls are 1, so the target flips. A second application flips it back: $(1, 1, 1)$ — Toffoli is self-inverse.

Problem 5 [Basic] Mark $|10\rangle$: X on the *second* qubit, CZ, X on the second qubit again. Trace: $|00\rangle \rightarrow |01\rangle \rightarrow +|01\rangle \rightarrow |00\rangle; |01\rangle \rightarrow |00\rangle \rightarrow + \rightarrow |01\rangle; |10\rangle \rightarrow |11\rangle \rightarrow -|11\rangle \rightarrow -|10\rangle$ ✓; $|11\rangle \rightarrow |10\rangle \rightarrow + \rightarrow |11\rangle$. Only $|10\rangle$ acquires the minus.

Problem 6 [Basic] $|11\rangle|0\rangle$: first CNOT (control x_1) flips helper $\rightarrow |11\rangle|1\rangle$; second (control x_2) flips again $\rightarrow |11\rangle|0\rangle$: helper $= 1 \oplus 1 = 0 = f(11)$ ✓ $|10\rangle|0\rangle$: first flips $\rightarrow |10\rangle|1\rangle$; second does nothing: helper $= 1 = f(10)$ ✓

Problem 7 [Basic] Branches: $\frac{1}{\sqrt{2}}(|0\rangle|+\rangle + |1\rangle X|+\rangle) = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)|+\rangle = |+\rangle|+\rangle$: unchanged. $|+\rangle$ carries X-eigenvalue +1, and a kicked-back phase of +1 is no phase at all.

Problem 8 [Basic] $\frac{1}{\sqrt{2}}(|0\rangle|-\rangle - |1\rangle X|-\rangle) \dots$ carefully: $|-\rangle|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)|-\rangle$; the $|1\rangle$ branch applies $X|-\rangle = -|-\rangle$, turning $-|1\rangle|-\rangle$ into $+|1\rangle|-\rangle$: result $|+\rangle|-\rangle$. The control flipped $|-\rangle \rightarrow |+\rangle$ — the eigenvalue -1 kicked back onto an already-negative branch.

Problem 9 [Standard] Output: $0.6|00\rangle + 0.8|11\rangle$. Factoring test: $a_{00}a_{11} = 0.48 \neq 0 = a_{01}a_{10}$: not a product — **entangled**. One CNOT on a superposed control manufactured correlation.

Problem 10 [Standard] CNOT sends $|11\rangle \rightarrow |10\rangle$: result $\frac{1}{\sqrt{2}}(|00\rangle + |10\rangle) = |+\rangle \otimes |0\rangle$ — a product. A second CNOT *disentangles* what the first entangled: entanglement is created and destroyed reversibly by the same gate.

Problem 11 [Standard] Control B: flips A when B = 1: $|00\rangle \rightarrow |00\rangle, |01\rangle \rightarrow |11\rangle, |10\rangle \rightarrow |10\rangle, |11\rangle \rightarrow |01\rangle$.

Matrix (columns = images of 00, 01, 10, 11): rows/cols in standard order give $\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$.

Problem 12 [Standard] $|10\rangle$: CNOT(A $\rightarrow$ B): $|11\rangle$; CNOT(B $\rightarrow$ A): $|01\rangle$; CNOT(A $\rightarrow$ B): $|01\rangle$ ✓ $|01\rangle \rightarrow |11\rangle \rightarrow |10\rangle$ ✓ $|00\rangle$ fixed throughout; $|11\rangle \rightarrow |10\rangle \rightarrow |10\rangle \rightarrow |11\rangle$ ✓ Three CNOTs = SWAP.

Problem 13 [Standard] Expand: $|+\rangle|-\rangle = \frac{1}{2}(|00\rangle - |01\rangle + |10\rangle - |11\rangle)$; CZ flips the last sign: $\frac{1}{2}(|00\rangle - |01\rangle + |10\rangle + |11\rangle)$. Factor test: $a_{00}a_{11} = \frac{1}{4} \neq -\frac{1}{4} = a_{01}a_{10}$: **entangled** — CZ, fed superpositions on both wires, entangles (equivalently: it is CNOT in H-disguise, Problem 18).

Problem 14 [Standard] Gate list: X on x_1 , X on x_2 ; Toffoli into helper; X on x_1 , X on x_2 ; X on helper. Trace $|00\rangle|0\rangle$: X's $\rightarrow |11\rangle|0\rangle$; Toffoli $\rightarrow |11\rangle|1\rangle$; X's restore $\rightarrow |00\rangle|1\rangle$; final X on helper $\rightarrow |00\rangle|0\rangle$: helper = 0 = OR(0, 0) ✓ (The construction computes NOT-AND-of-NOTs = OR.)

Problem 15 [Standard] The H's convert the target's X-action into a Z-action: sandwiching the Toffoli's controlled-controlled-X between H's on the target gives controlled-controlled- $HXH = \text{CCZ}$. Since Z on $|1\rangle$ is the only sign change, CCZ flips exactly $|111\rangle$'s sign — the desired mark.

Problem 16 [Standard] Bit style: when the *value* $f(x)$ must be used or copied — Deutsch's circuit queries (Chapter 17), arithmetic in Shor's algorithm. Phase style: when only interference matters — Grover's marking (Chapter 18). Phase oracles alter no register and write no helper, so there is nothing to uncompute — the sign lives in the amplitudes themselves.

Problem 17 [Standard] Target $|1\rangle$: branches $\frac{1}{\sqrt{2}}(|0\rangle|1\rangle + |1\rangle(-1)|1\rangle) = |-\rangle|1\rangle$: control flipped to $|-\rangle$. Target $|0\rangle$: Z acts trivially: $|+\rangle|0\rangle$ unchanged. Both restated jointly: CZ negates exactly the $|11\rangle$ amplitude — present in the first case (hence the sign enters the control's superposition), absent in the second.

Problem 18 [Challenge] Trace (control, target): $|00\rangle$: H $\rightarrow|0\rangle|+\rangle$; CZ acts trivially on $|0\rangle$ -control branch: $|0\rangle|+\rangle$; H $\rightarrow|00\rangle$ ✓ $|01\rangle$: $\rightarrow|0\rangle|-\rangle \rightarrow|0\rangle|-\rangle \rightarrow|01\rangle$ ✓ $|10\rangle$: $\rightarrow|1\rangle|+\rangle$; CZ flips the sign of the $|11\rangle$ component: $|1\rangle|-\rangle$; H $\rightarrow|11\rangle$ ✓ $|11\rangle$: $\rightarrow|1\rangle|-\rangle \rightarrow|1\rangle|+\rangle \rightarrow|10\rangle$ ✓ Exactly CNOT's table: $\text{CNOT} = (I \otimes H) \text{CZ} (I \otimes H)$.

Problem 19 [Challenge] Wire $c = 1$: the target outputs $1 \oplus (a \wedge b) = \text{NAND}(a, b)$, with a, b passing through unchanged. NAND is classically universal, so any Boolean circuit compiles into Toffolis (plus X's for constants) — hence a quantum computer contains classical computing as the special case of basis-state inputs: nothing classical is lost by going quantum.

Problem 20 [Challenge] Base: the input $|0 \cdots 0\rangle$ is a product. Step: a single-qubit gate U on qubit k maps $|\phi_1\rangle \otimes \cdots \otimes |\phi_k\rangle \otimes \cdots \rightarrow |\phi_1\rangle \otimes \cdots \otimes (U|\phi_k\rangle) \otimes \cdots$ — still a product. By induction the state stays a product forever. Tracking cost: n qubits $\times 2$ complex amplitudes $\approx 4n$ real numbers — versus $2 \cdot 2^n$ for general states. Conclusion: without entangling gates there is no exponential space, hence no quantum advantage — CNOT is not optional.

Problem 21 [Challenge] $Z \otimes I = \text{diag}(1, 1, -1, -1)$; $I \otimes Z = \text{diag}(1, -1, 1, -1)$; CZ = $\text{diag}(1, 1, 1, -1)$. Product: $\text{diag}(1 \cdot 1 \cdot 1, 1 \cdot (-1) \cdot 1, (-1) \cdot 1 \cdot 1, (-1)(-1)(-1)) = \text{diag}(1, -1, -1, -1)$ ✓ — the OR mark.

Multiplying by CZ once more: $\text{diag}(1, -1, -1, +1)$: signs on 01 and 10 only — the **XOR** oracle (mark where the bits differ).

Problem 22 [Challenge] Prepare control $|+\rangle$, target $|1\rangle$ (T's eigenstate with phase 45°). Controlled-T: $\frac{1}{\sqrt{2}}(|0\rangle|1\rangle + e^{i45^\circ}|1\rangle|1\rangle)$ — the phase kicked onto the control. Final H on the control: $P(0) = \frac{1+\cos 45^\circ}{2} = 0.854$. Measuring the control's histogram reads T's eigenphase — phase estimation's engine in three gates (Chapter 19 adds precision digits by repeating with $U^2, U^4, \dots$).

CHAPTER ELEVEN

Entanglement and Bell States

Teaching notes, self-test, and complete solutions

TEACHING NOTES

The chapter students remember for life — teach it with the drama it deserves, and with this book’s discipline: every “spooky” claim immediately cashed into arithmetic. The four Bell states and the factory/decoder circuits are mechanical skills; drill them briskly. Spend the saved time on the two crown jewels: the CHSH score (run the worked computation to $2\sqrt{2}$, then the eavesdropper’s collapse to $\sqrt{2}$ — the classroom CHSH game from the activity makes the classical ceiling *felt*) and no-signalling (Problem 19’s dam alarm exposes the exact broken step; insist on it). The monogamy lesson pays forward to cryptography — the GHZ-pair computation ($S_{\text{pair}} = \sqrt{2}$) is the quantitative heart, and Problem 22’s “unlocking” resolves the apparent paradox beautifully in class discussion. Stumbles: confusing correlation with communication (relentlessly separate them), and sign errors in Ψ^- traces.

SELF-TEST — QUESTIONS WITH ANSWERS

Q1. Write the four Bell states.

A. $\Phi^\pm = \frac{1}{\sqrt{2}}(|00\rangle \pm |11\rangle)$, $\Psi^\pm = \frac{1}{\sqrt{2}}(|01\rangle \pm |10\rangle)$ — agree/disagree, plus/minus phase.

Q2. What two gates manufacture a Bell state from $|00\rangle$?

A. H on the first qubit, then CNOT — superpose, then correlate.

Q3. What does $|S| \leq 2$ mean, and who violates it?

A. It bounds every local pre-written-answer explanation of correlations; entangled qubits reach $2\sqrt{2} \approx 2.83$, so no such explanation exists.

Q4. Alice measures her half of Φ^+ . What can Bob detect?

A. Nothing: his marginal stays 50/50. Correlations appear only when results are *compared* classically — no faster-than-light messages.

Q5. Why can’t three qubits be pairwise maximally entangled?

A. Monogamy: maximal entanglement between A and B leaves zero for anyone else; inside GHZ, every pair even tests classical ($S_{\text{pair}} = \sqrt{2}$).

Q6. What drags a measured CHSH score below $2\sqrt{2}$ in practice?

A. Noise and detector imperfections (visibility V : $S = 2\sqrt{2}V$), and any eavesdropping — full intercept-resend collapses it to $\sqrt{2}$.

Solutions to the Practice Problems

Problem 1 [Basic] H on q_0 : $|1\rangle \rightarrow |-\rangle$: $\frac{1}{\sqrt{2}}(|00\rangle - |10\rangle)$; CNOT: $\frac{1}{\sqrt{2}}(|00\rangle - |11\rangle) = |\Phi^-\rangle$.

Problem 2 [Basic] $P(01) = P(10) = \frac{1}{2}$, $P(00) = P(11) = 0$. $P(\text{agree}) = 0$: perfect anti-correlation.

Problem 3 [Basic] Z on A: term by term, $|00\rangle$ untouched, $|11\rangle \rightarrow -|11\rangle$ (A's digit is 1): $|\Phi^+\rangle \rightarrow |\Phi^-\rangle$ ✓ (Z on B works equally — one local phase flip toggles the $\pm$.)

Problem 4 [Basic] Not credible: $|\Phi^+\rangle$ assigns pattern 01 probability zero — the word is “forbidden” (or “zero”). 260 counts there is not noise; it is a different (or badly decohered) state.

Problem 5 [Basic] $S = 0.71 + 0.69 + 0.70 - (-0.72) = 2.82$. Inconsistent with any local hidden-answer model (> 2); comfortably consistent with quantum mechanics ($\leq 2\sqrt{2} = 2.83$) — in fact a superb experiment.

Problem 6 [Basic] All correlators +1: $S = 1 + 1 + 1 - 1 = 2$ — exactly the ceiling, not beyond. Perfect classical correlation saturates but never breaks the bound; breaking it needs the impossible-to-pre-write correlations of entanglement.

Problem 7 [Basic] GHZ pair: $E = \cos 45^\circ \cos 45^\circ = 0.5$. Bell pair: $E = \cos(45^\circ - 45^\circ) = 1$. In words: inside GHZ the pair agrees only as strongly as classical coins; a true Bell pair at equal settings agrees *always*.

Problem 8 [Basic] Monogamy of entanglement. Tell-tale: run pairwise CHSH tests — genuine pairwise maximal entanglement would need every pair to score $2\sqrt{2}$, which monogamy forbids (any one pair at $2\sqrt{2}$ forces the third player's correlators to pure noise).

Problem 9 [Standard] $|\Psi^-\rangle$ anti-correlates: Alice 0 $\Rightarrow$ Bob's state $|1\rangle$, he measures 1 (certainly). Alice 1 $\Rightarrow$ Bob holds $|0\rangle$, measures 0.

Problem 10 [Standard] Marginals: $P(A=0) = P(B=0) = \frac{1}{2}$ (patterns 00 and 11 each $\frac{1}{2}$). The minus sign is a *relative phase between patterns*; standard-basis probabilities square it away, jointly and marginally. It becomes visible to the X-basis correlation measurement (or the decoder circuit): Φ^+ and Φ^- give opposite $\langle XX \rangle$.

Problem 11 [Standard] $|01\rangle$: H gives $\frac{1}{\sqrt{2}}(|01\rangle + |11\rangle)$; CNOT: $\frac{1}{\sqrt{2}}(|01\rangle + |10\rangle) = |\Psi^+\rangle$. $|11\rangle$: $\frac{1}{\sqrt{2}}(|01\rangle - |10\rangle) = |\Psi^-\rangle$. Table: 00 $\rightarrow \Phi^+$, 01 $\rightarrow \Psi^+$, 10 $\rightarrow \Phi^-$, 11 $\rightarrow \Psi^-$ — two classical bits select among four Bell states (superdense coding's seed).

Problem 12 [Standard] $\Phi^+ = \frac{1}{\sqrt{2}}(1, 0, 0, 1)$, $\Psi^+ = \frac{1}{\sqrt{2}}(0, 1, 1, 0)$: inner product = 0. Orthogonality (of all four, pairwise) is what makes “which Bell state?” a legal measurement question with reliably distinguishable answers — the Bell measurement of teleportation.

Problem 13 [Standard] Ψ^+ : CNOT $\rightarrow \frac{1}{\sqrt{2}}(|01\rangle + |11\rangle) = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)|1\rangle$; H on A $\rightarrow |0\rangle|1\rangle$: readout 01 ✓
 Ψ^- : CNOT $\rightarrow \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)|1\rangle$; H $\rightarrow |1\rangle|1\rangle$: readout 11 ✓

Problem 14 [Standard] $2\sqrt{2}V > 2 \Rightarrow V > 1/\sqrt{2} = 0.707$. Below 70.7% visibility the test cannot certify anything — a hard experimental floor.

Problem 15 [Standard] Differences: $a-b = -45^\circ$, $a'-b = 45^\circ$, $a'-b' = 135^\circ$, $a-b' = 45^\circ$: correlators 0.707, 0.707, -0.707, 0.707. $S = 0.707 + 0.707 + (-0.707) - 0.707 = 0$. Moral: the violation is not automatic from entanglement — the four settings must be *aimed* so the minus lands on the lone anti-correlated pair; measurement design is half the experiment.

Problem 16 [Standard] $2\sqrt{2} - f(2\sqrt{2} - \sqrt{2}) > 2 \Rightarrow f < \frac{2\sqrt{2}-2}{\sqrt{2}} = 2 - \sqrt{2} \approx 0.586$. A spy could touch up to $\sim 58\%$ of pairs before the CHSH average alone dips under 2 — but intercepted pairs also produce *errors in the key rounds* (25%-style fingerprints), which the error-rate check catches at far smaller f ; the two tests together leave no comfortable window.

Problem 17 [Standard] The classical model: one shared coin decides both “true answers” (++ or --, half/half); each reader independently reports its true answer with probability set by its angle so its *average* report is $\cos \theta$ times the truth. Independence factorizes the expectation: $E = \cos \theta_A \cos \theta_B$ — exactly the GHZ pair’s correlations. Since a classical model reproduces them, the CHSH bound applies: $S_{\text{pair}} \leq 2$ for every choice of settings, no computation needed.

Problem 18 [Challenge] Circuit: H on q_0 ; CNOT($q_0 \rightarrow q_1$); CNOT($q_0 \rightarrow q_2$) (or $q_1 \rightarrow q_2$). Trace: $|000\rangle \rightarrow \frac{1}{\sqrt{2}}(|000\rangle + |100\rangle) \rightarrow \frac{1}{\sqrt{2}}(|000\rangle + |110\rangle) \rightarrow \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle)$ ✓ — superpose once, then chain the correlation outward.

Problem 19 [Challenge] Bob’s local statistics are his marginal, and the no-signalling computation shows the marginal is *identical* whether or not (and in whatever basis) Alice has measured: 50/50 before, 50/50 after. Watching any number of his qubits’ statistics, Bob sees one unchanging distribution — there is no “change” to detect. The broken step is the phrase “watches for the change”: measurement by Alice changes the *conditional* state given her result (which Bob lacks), never the unconditional statistics Bob actually observes.

Problem 20 [Challenge] If A–B are in $|\Phi^+\rangle$, the pair’s state is pure and specific: A’s role is fully spoken for as “the partner of B”; writing a second pure pairing A–C would require the total state to factor as (A entangled with B) *and* (A entangled with C) simultaneously — but a state of A jointly determined by two different partners cannot factor either way; the algebra of Problem 18-style matching fails on every split. For cryptography: when Alice–Bob verify (near-)maximal entanglement via CHSH, monogamy guarantees Eve’s systems carry (near-)zero correlation with their outcomes — the eavesdropper is the one who “fails to get a copy.”

Problem 21 [Challenge] $B_b, B_{b'} \in \{\pm 1\}$: either $B_b = B_{b'}$ (second bracket ± 2 , first 0) or $B_b = -B_{b'}$ (first

bracket ± 2 , second 0). Either way $S = A_{(\cdot)} \times (\pm 2) = \pm 2$: every deterministic answer sheet scores exactly ± 2 , so $|S| \leq 2$. A probabilistic strategy is a weighted average of sheets, and averages of numbers in $[-2, 2]$ stay in $[-2, 2]$. QED — and the quantum 2.83 therefore certifies that no sheets, however clever, were behind the data.

Problem 22 [Challenge] Rewrite C's basis: $|0\rangle_C = \frac{1}{\sqrt{2}}(|+\rangle + |-\rangle)$, $|1\rangle_C = \frac{1}{\sqrt{2}}(|+\rangle - |-\rangle)$. GHZ becomes $\frac{1}{2}[(|00\rangle + |11\rangle)_{AB} |+\rangle_C + (|00\rangle - |11\rangle)_{AB} |-\rangle_C]$. Measuring C in X: outcome $+$ (probability $\frac{1}{2}$) leaves A,B in $|\Phi^+\rangle$; outcome $-$ leaves $|\Phi^-\rangle$ — a perfect Bell pair either way (given C's announced result). No monogamy violation: the entanglement was always *collective*; converting it into A-B pair entanglement required *consuming* C by measurement — afterward there are only two entangled parties, exactly as monogamy demands. (This “entanglement unlocking” is a real protocol step in multiparty quantum networks.)

CHAPTER TWELVE

Quantum Circuits and Circuit Design

Teaching notes, self-test, and complete solutions

TEACHING NOTES

The engineering chapter of Part Four: reading diagrams, computing them reliably, shrinking them, and costing them. The column method (write the whole state after every column) prevents ninety percent of student errors — enforce it in graded work. Three ideas need emphasis beyond mechanics: no-cloning (run the linearity proof, then Problem 13’s basis-state loophole to sharpen it), the uncomputation sandwich (the superposition worked example — dirty scratch killing interference — is the conceptual heart; Problem 21 makes it a lab), and the cost sheet (width/size/depth), which every later budget quotes. The identity toolbox rewards drill: five minutes of “simplify this string” at the start of three consecutive classes builds real fluency. Stumbles: matrix order vs diagram order (the mistake box exists for a reason — test it), and forgetting that only *same-column* gates tensor together.

SELF-TEST — QUESTIONS WITH ANSWERS

Q1. In a circuit diagram, what are columns, and what do double lines carry?

A. Columns are simultaneous time-steps (their count is the depth); double lines carry classical bits produced by measurements.

Q2. State the no-cloning theorem.

A. No quantum machine can copy an *unknown* state: $|\psi\rangle|0\rangle \rightarrow |\psi\rangle|\psi\rangle$ for all $|\psi\rangle$ contradicts linearity. (Known basis states: CNOT copies fine.)

Q3. Why uncompute scratch qubits?

A. Dirty scratch stays entangled with the data and destroys its interference; compute-copy-uncompute returns scratch to $|0\rangle$, provably disentangled.

Q4. Width, size, depth — one phrase each.

A. Qubits used; total gates; parallel time-steps (the one that races the coherence clock).

Q5. Give two toolbox identities and one use.

A. E.g. $HH = I$ (cancel duplicates), $T^2 = S$ (merge phase ladders), H-sandwich reverses a CNOT (re-route on

limited wiring).

Q6. What does one SWAP cost in CNOTs, and why does it matter?

A. Three. Chips with sparse wiring insert SWAPs to move qubits together, so poor connectivity silently multiplies the two-qubit gate bill.

Solutions to the Practice Problems

Problem 1 [Basic] Columns: 5 (H; CNOT; T; H on q_1 ; the measurement column) — depth 5 as drawn (accept 4 if measurements are excluded; the convention should be stated). Two-qubit gates: one, the CNOT. Double lines carry the two classical measurement results.

Problem 2 [Basic] X on q_1 : $|00\rangle \rightarrow |01\rangle$. CNOT ($q_0 = 0$): does nothing. Output $|01\rangle$.

Problem 3 [Basic] $HH = I$; $TTTT = T^4 = Z$. The whole sequence is the single gate **Z**.

Problem 4 [Basic] No machine can duplicate an unknown quantum state ($|\psi\rangle \rightarrow |\psi\rangle |\psi\rangle$ is impossible for all $|\psi\rangle$). Forbidden everyday operation: copy-paste (equally: backup, forward-and-keep, screenshot) of quantum data.

Problem 5 [Basic] $|10, 0, 0\rangle$: Toffoli (controls 1,0): scratch stays 0; CNOT (scratch 0): y stays 0; Toffoli: nothing. Output $|10, 0, 0\rangle$: $y = 0 \oplus f(10) = 0$ ✓ $|10, 0, 1\rangle$: identical path, y stays $1 = 1 \oplus 0$ ✓ Scratch clean both times.

Problem 6 [Basic] Measuring the scratch collapses *the data too*: by the partial-measurement rule, only data branches consistent with the observed scratch value survive. If scratch held $f(x)$, the data register collapses onto $\{x : f(x) = \text{observed}\}$ — superposition destroyed, algorithm ruined. Reset-by-measurement is exactly the “dirty blackboard” failure, made permanent.

Problem 7 [Basic] Width 3, size 4. Dependencies force the order: columns $[H(q_1)]$, $[CNOT\ q_1 \rightarrow q_2]$, $[CNOT\ q_0 \rightarrow q_1]$, $[H(q_0)]$ — depth 4, duration $4 \times 60 = 240$ ns. (No compression: each gate waits on the previous one’s qubit.)

Problem 8 [Basic] Times: $6\ \mu\text{s}$ vs $0.72\ \mu\text{s}$. Decay factors: $e^{-6/80} = 0.928$ vs $e^{-0.72/80} = 0.991$: the shallow circuit keeps about 6–7 percentage points more fidelity from time-decay alone — before counting the identical gate-error toll. Depth, not size, buys survival.

Problem 9 [Standard] Columns on $|00\rangle$: H’s $\rightarrow \frac{1}{2} \sum_{xy} |xy\rangle$; CZ flips the $|11\rangle$ sign; H’s again: collecting terms gives $|00\rangle$ — the same as CNOT $|00\rangle$. Running all four basis states reproduces CNOT’s full table: you have re-derived Chapter 10’s challenge, $CNOT = (I \otimes H) CZ (I \otimes H)$ (equivalently the H-sandwich toolbox row).

Problem 10 [Standard] Before: 5 columns ($[H,Z], [T,X], [CNOT], [T,X], [H,Z]$). Slides: T (diagonal) commutes past the CNOT’s *control*; X commutes past its *target*: bring the two T’s and two X’s adjacent: $T^2 = S$ fires on

q_0 ; $XX = I$ fires on q_1 . After: $[H,Z],[S],[CNOT],[H,Z]$ — depth 4 (and two gates fewer). Identities: control-diagonal slide, target-X slide, $T^2 = S$, $XX = I$.

Problem 11 [Standard] From $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$: T on q_0 phases the $|1\cdot\rangle$ term: $\frac{1}{\sqrt{2}}(|00\rangle + e^{i45^\circ}|11\rangle)$; H on q_1 : $|0\rangle \rightarrow |+\rangle$, $|1\rangle \rightarrow |-\rangle$: $\frac{1}{2}(|00\rangle + |01\rangle + e^{i45^\circ}|10\rangle - e^{i45^\circ}|11\rangle)$. All four probabilities 25%; the phases await a phase-sensitive measurement.

Problem 12 [Standard] Design 1: H on each qubit (depth 1, no two-qubit gates): uniform 25% histogram. Design 2: $H(q_0)$, CNOT, then X on q_1 : $\Phi^+ \rightarrow \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle) = \Psi^+$: histogram 50/50 on 01/10. The second histogram's perfect anti-correlation cannot come from any product state — hence no two-qubit-gate-free design exists for it.

Problem 13 [Standard] The proof applies the hypothetical cloner to a *superposition* and finds linearity forcing cross-terms that the target $|\psi\rangle|\psi\rangle$ lacks — the failure lives entirely in the superposition step. For the two known basis states alone, no superposition is involved: CNOT (blank target) is a perfect basis-state copier, $|x\rangle|0\rangle \rightarrow |x\rangle|x\rangle$ — exactly why classical data survives on quantum hardware while unknown quantum states cannot be backed up.

Problem 14 [Standard] Sequence: $X(x_1), X(x_2)$; Toffoli $\rightarrow s$; $X(x_1), X(x_2); X(s)$; CNOT($s \rightarrow y$); then uncompute in mirror: $X(s); X(x_1), X(x_2)$; Toffoli; $X(x_1), X(x_2)$. Trace $|00, 0, 0\rangle$: X's $\rightarrow 11$; Toffoli: $s = 1$; X's restore inputs; $X(s)$: $s = 0$; CNOT: $y \oplus = 0$; uncompute returns everything: output $|00, 0, 0\rangle$, $y = \text{OR}(0, 0) = 0$ ✓

Problem 15 [Standard] Compute: Toffoli($x_1x_2 \rightarrow s_1$); Toffoli($s_1x_3 \rightarrow s_2$); copy: CNOT($s_2 \rightarrow y$); uncompute: Toffoli($s_1x_3 \rightarrow s_2$), Toffoli($x_1x_2 \rightarrow s_1$) — reverse order because s_2 's uncomputation needs s_1 still intact: undoing must peel the dependency stack from the top, exactly like nested function calls returning.

Problem 16 [Standard] $G + 1 + G = 81$ gates; cleanup fraction $40/81 \approx 49\%$ — nearly half the circuit exists to erase its own footprints. (The price of clean interference.)

Problem 17 [Standard] Columns: $[H(q_0)]$; $[CNOT\ 0 \rightarrow 1]$; $[CNOT\ 0 \rightarrow 2, 1 \rightarrow 3]$; $[CNOT\ 0 \rightarrow 4, 1 \rightarrow 5, 2 \rightarrow 6, 3 \rightarrow 7]$. Size = $1 + 7 = 8$ gates; depth = 4 versus the chain's 8 — doubling spreads entanglement in $\log_2$ time, the standard trick for wide states on parallel hardware.

Problem 18 [Challenge] Circuit: $H(q_0)$; CNOT($q_0 \rightarrow q_1$) — q_2 untouched: $\frac{1}{\sqrt{2}}(|000\rangle + |110\rangle)$ ✓ One more gate for the minus: Z on q_0 (or on q_1): the $|11\cdot\rangle$ term (that qubit = 1) acquires -1 : $\frac{1}{\sqrt{2}}(|000\rangle - |110\rangle)$ ✓

Problem 19 [Challenge] (Model letter.) “Your service requires transforming an unknown customer state $|\psi\rangle$ into $|\psi\rangle|\psi\rangle$; the no-cloning theorem (Wootters–Zurek 1982, a two-line linearity argument) proves no physical device does this, so the advertisement is not an engineering challenge but an impossibility claim. If your customers' data is classical — definite basis states — a CNOT-based copier is legitimate, and we would be happy to certify that limited service instead.”

Problem 20 [Challenge] Depth A: 1; depth B: 8. Same output, but B holds every qubit exposed eight steps: its time-decay factor is e^{-8t_g/T_1} versus e^{-t_g/T_1} — run A. The quantum die was designed depth-1 for exactly

this reason: maximum randomness, minimum exposure — good circuits are wide and shallow when physics permits.

Problem 21 [Challenge] (a) CNOT copies the basis components: $|+\rangle|0\rangle \rightarrow \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ — data and scratch entangled. H on the data: $\frac{1}{2}[(|0\rangle + |1\rangle)|0\rangle + (|0\rangle - |1\rangle)|1\rangle]$: $P(\text{data} = 0) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$ — the interference that once forced $P(0) = 1$ is gone; the scratch “knows” the path. (b) Second CNOT first: returns $|+\rangle|0\rangle$ (scratch disentangled); H gives $|0\rangle|0\rangle$: $P(0) = 1$ restored ✓ Decoherence and its cure, in four gates.

Problem 22 [Challenge] Route: $\text{SWAP}(q_0, q_1)$, $\text{SWAP}(q_1, q_2)$ — the payload sits at q_2 ; $\text{CNOT}(q_2 \rightarrow q_3)$; then SWAP back twice to restore the layout: $3 + 3 + 1 + 3 + 3 = 13$ CNOTs (7 if the layout needn’t be restored). Error toll: $(1 - 0.006)^{13} = 0.925$ versus 0.994 for the single gate the algorithm “wanted” — one logical CNOT cost thirteen physical ones and $12\times$ the error. Moral: match circuits to the chip’s wiring map (or choose hardware whose connectivity matches the algorithm); connectivity is a first-class budget line, not a detail.

CHAPTER THIRTEEN

Quantum Teleportation and Superdense Coding

Teaching notes, self-test, and complete solutions

TEACHING NOTES

The chapter where everything before it works one shift together — superposition, entanglement, partial measurement, no-cloning, no-signalling. Insist students trace the full four-outcome algebra once by hand (painful, transformative); afterwards the protocol table does the work. Keep hammering the resource ledger: one Bell pair + two classical bits per qubit teleported; the classical bits' light-speed limit is what saves relativity (Problem 20 shows even correction *timing* is flexible — information timing is not). The swapping and fidelity lessons are network engineering: the $(2f + 1)/3$ formula and the classical ceiling $2/3$ turn headlines into numbers — assign Problem 22's distillation economics as the synthesis. Stumbles: believing the qubit “travels” (nothing physical moves but two bits), forgetting Alice's original is destroyed (no-cloning demands it), and sign slips in the Ψ^- branch.

SELF-TEST — QUESTIONS WITH ANSWERS

Q1. Teleportation's resource ledger, per qubit?

A. Consumed: one shared Bell pair and two classical bits (and Alice's original state, destroyed by measurement). Delivered: the exact state $|\psi\rangle$ at Bob's side.

Q2. Why doesn't teleportation violate no-cloning?

A. Alice's copy is destroyed by her Bell measurement at the very moment Bob's becomes recoverable — the state is moved, never duplicated.

Q3. Why can't teleportation send signals faster than light?

A. Until Alice's two classical bits arrive, Bob's qubit is (to him) a featureless 50/50 mixture — all four repair branches averaged. The protocol completes only at classical speed.

Q4. Superdense coding in one line?

A. Alice's one local gate (I, X, Z, XZ) steers the shared pair among the four Bell states; sending her qubit lets Bob's Bell measurement read two bits.

Q5. What does entanglement swapping achieve?

A. A Bell measurement on the two middle qubits entangles the two outer qubits that never met — the elementary step of quantum repeaters.

Q6. What fidelity certifies genuine quantum teleportation?

A. Average fidelity above the classical ceiling $2/3$ (measure-and-rebuild's best). Real experiments report ≈ 0.8 – 0.9 ; $F_{\text{out}} = (2f + 1)/3$ ties it to pair quality.

Solutions to the Practice Problems

Problem 1 [Basic] Start: the mystery qubit $|\psi\rangle$, one shared Bell pair, a classical channel. Consumed: the Bell pair (measured away) and $|\psi\rangle$'s presence at Alice's side; the classical channel carries two bits but survives for reuse.

Problem 2 [Basic] Outcome 01: Bob holds $\alpha|1\rangle + \beta|0\rangle$ — amplitudes swapped: the repair is X.

Problem 3 [Basic] Message 10 (in this book's table): apply Z. The pair becomes $|\Phi^-\rangle$, which Bob's decoder reads as 10.

Problem 4 [Basic] If Alice kept $|\psi\rangle$ while Bob gained it, the protocol would have produced two copies of an unknown state — forbidden by the no-cloning theorem. Her Bell measurement's collapse is the mechanism that enforces the law.

Problem 5 [Basic] A–D share $|\Psi^+\rangle$; the standardizing correction is X on D.

Problem 6 [Basic] Each swap broadcasts 2 bits: 4 bits total must reach the ends (or wherever corrections are applied). The two Bell measurements act on disjoint qubit pairs — commuting operations — so their time order is irrelevant; only the classical bookkeeping must eventually meet.

Problem 7 [Basic] $F_{\text{out}} = 1$ (perfect protocol); 0.95 (the premium-service threshold); 0.90 (a good real link); $\frac{2}{3}$ (the classical ceiling — an unentangled “pair” adds nothing).

Problem 8 [Basic] With no entanglement, Alice just measures $|\psi\rangle$ once (best single-shot estimate, Chapter 4) and phones Bob a description; Bob rebuilds his best guess. Optimizing that strategy over all input states yields average fidelity exactly $\frac{2}{3}$ — the number any true teleportation must beat.

Problem 9 [Standard] Keeping the $|10\rangle$ terms of the pre-measurement expansion leaves Bob's ket $\frac{1}{2}(\alpha|0\rangle - \beta|1\rangle)$; renormalizing (probability $\frac{1}{4}$) gives $\alpha|0\rangle - \beta|1\rangle$. Applying Z: $\alpha|0\rangle + \beta|1\rangle = |\psi\rangle$ ✓

Problem 10 [Standard] With $|\psi\rangle = |0\rangle$ the four outcomes remain equally likely ($\frac{1}{4}$ each — the Bell measurement's statistics never depend on $|\psi\rangle$; Problem 11 turns this into a security feature). Bob's branches hold $|0\rangle, |0\rangle, |0\rangle$ -up-to-sign etc.; each table repair returns exactly $|0\rangle$ — corrections work state-independently, which is the protocol's whole design.

Problem 11 [Standard] The outcome probabilities are $\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}$ whatever α, β are — change $|\psi\rangle$ and nothing in the table’s statistics moves. Eve’s two bits are therefore statistically independent of the state: pure randomness without Bob’s qubit. Physically, the state’s information lives in the *correlation* between the bits and the entangled partner, not in the bits alone — entanglement is the encryption.

Problem 12 [Standard] ZX versus XZ differ by a global -1 (anticommutation): the produced state is $-\lvert\Psi^-\rangle \simeq \lvert\Psi^-\rangle$ — physically identical, so the decoder still prints 11. The order made only an unobservable global-phase difference.

Problem 13 [Standard] Two hops: 2 Bell pairs and 4 classical bits (and two Bell measurements). Chain of k hops: k pairs, $2k$ bits — linear cost per hop, which is precisely the repeater’s economy: entanglement is spent hop by hop, never transmitted end to end.

Problem 14 [Standard] Each swap succeeds with $\frac{1}{2}$: four independent swaps succeed together with $(\frac{1}{2})^4 = \frac{1}{16} \approx 6\%$ of attempts — why deterministic (matter-based) Bell measurements and memories are worth so much to network designers.

Problem 15 [Standard] (a) $T_{100} = 10^{-2} = 1\%$. (b) $T_{50}^2 = (10^{-1})^2 = 1\%$: identical — splitting alone buys nothing when both halves must be lucky *simultaneously*. Memory changes the logic from AND-in-one-attempt to eventually-both: each half needs only ~ 10 attempts independently, so pairs arrive in tens of attempts instead of hundreds — the exponential becomes additive (Chapter 16 builds the full budget).

Problem 16 [Standard] Before the announcement, each end’s qubit shows the maximally mixed 50/50 statistics in every basis — averaged over the four unannounced outcomes, A-D correlations are exactly zero. Only the middle station’s 2 classical bits (light-speed) convert the raw correlation into a usable, known Bell state.

Problem 17 [Standard] $f = \frac{1}{4} + \frac{3}{4}(0.9)(0.9) = \frac{1}{4} + 0.6075 = 0.8575$; $F_{\text{out}} = (2 \times 0.8575 + 1)/3 = 0.905$. (Working with the excess keeps the “useless floor” $f = \frac{1}{4}$ fixed, as it must be.)

Problem 18 [Challenge] Bob’s half of pair one is a qubit like any other; teleportation moves *any* unknown qubit faithfully, including one that happens to be entangled with Alice’s. Whatever was true of Bob’s qubit — including every correlation with Alice — is now true of Charlie’s qubit, so Alice-Charlie share exactly $\lvert\Phi^+\rangle$. This is the key to networks because entanglement can thus be *extended* through intermediate stations without any quantum particle surviving the whole distance — only short hops plus classical bits.

Problem 19 [Challenge] (Model note.) “The two classical bits carry only the *correction index*, not the state: without the pre-shared entangled qubit they are two random bits (Problem 11). The claim ignores the Bell pair — a quantum resource distributed in advance at full quantum cost. Holevo’s theorem caps a lone qubit’s classical capacity at one bit, so no compression of classical data occurs in either direction. Correct ledger: teleportation = 1 ebit + 2 cbits $\rightarrow$ 1 qubit moved; superdense = 1 ebit + 1 qubit sent $\rightarrow$ 2 cbits — resource conversions, not compression.”

Problem 20 [Challenge] Bob’s premature “00 guess” applies I — nothing; when the true bits arrive he applies the true correction: net effect identical, trivially. The instructive cases: had he guessed wrong (say applied

X, truth Z), the final composite $Z X X$ -type products collapse by $XX = ZZ = I$ to the needed correction times at most a global ± 1 (from sliding Z past X) — harmless. So corrections may be applied, undone, and reapplied in any order at any time; what cannot be advanced is *knowing which* correction — the information, not the operation, obeys the speed limit.

Problem 21 [Challenge] Expand $|\Phi^+\rangle_{AB} |\Phi^+\rangle_{CD} = \frac{1}{2}(|0000\rangle + |0011\rangle + |1100\rangle + |1111\rangle)$ (order A,B,C,D). Project B,C onto $\frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$: middle digits read 00, 01, 10, 11 across the four terms; the projector's components pick $+\frac{1}{\sqrt{2}}$ times the BC= 01 term and $-\frac{1}{\sqrt{2}}$ times the BC= 10 term: from $|0011\rangle$ (BC = 01): $+\frac{1}{2\sqrt{2}}|01\rangle_{AD}$; from $|1100\rangle$ (BC = 10): $-\frac{1}{2\sqrt{2}}|10\rangle_{AD}$. Unnormalized $\frac{1}{2\sqrt{2}}(|01\rangle - |10\rangle)_{AD}$: squared length $\frac{1}{8} + \frac{1}{8} = \frac{1}{4}$ ✓; renormalized: $|\Psi^-\rangle_{AD}$ ✓

Problem 22 [Challenge] $f' = \frac{0.64}{0.64+0.04} = 0.941$. Fidelities: before $F = (2(0.8)+1)/3 = 0.867$; after $F = (2(0.941)+1)/3 = 0.961$. Cost: 2 pairs per attempt at $\approx \frac{1}{2}$ success $\Rightarrow$ about 4 raw pairs per distilled pair. Verdict: yes — $0.961 \geq 0.95$ meets the target, at a $4\times$ rate penalty; without distillation the service fails (0.867). (Students proposing a second round: $f'' = 0.996$, ≈ 16 raw pairs — overkill for this spec.)

CHAPTER FOURTEEN

Programming Quantum Computers

Teaching notes, self-test, and complete solutions

TEACHING NOTES

The hands-on chapter — run it in a computer lab if at all possible (Appendix E gets Python novices aboard in one session). Grading advice: require submitted code to *print its own checks* (state vectors at checkpoints, counts with expected values alongside) — the debugging lesson’s assertion discipline, enforced from day one. The three debugging techniques (simulator print, deterministic inputs, basis-change checks) and the readout-correction arithmetic are professional skills; the histogram-signature worked example (reversed CNOT) is the archetype — reproduce it live, then let students hunt seeded bugs in pairs. Qiskit’s reversed bit-order (Problem 11) costs every cohort an hour eventually — spend five minutes now instead. Emphasize throughout: simulate first, verify, then hardware; never debug on the queue.

SELF-TEST — QUESTIONS WITH ANSWERS

Q1. Why can simulators print the state while hardware cannot?

A. Simulators are classical programs holding all 2^n amplitudes in memory; hardware states collapse when read (and no-cloning forbids copying for inspection).

Q2. First test input for any new circuit, and why?

A. Basis states: on them many circuits act classically (truth tables), so outputs are deterministic — any randomness is already a bug report.

Q3. A phase bug is invisible to standard-basis counts. The fix?

A. Measure in a rotated basis (append H, or $S^\dagger H$): every phase relationship becomes a plain probability in some basis.

Q4. What two calibration numbers drive readout correction?

A. The flip rates $\epsilon_{0 \rightarrow 1}$ and $\epsilon_{1 \rightarrow 0}$, measured by preparing $|0\rangle$ and $|1\rangle$ and counting the lies; correction inverts the known mixing.

Q5. How many shots to see a 5-point probability difference at 2σ ?

A. ≈ 400 : $\text{margin} = 2 \times 0.5/\sqrt{n} \leq 0.05 \Rightarrow n \geq 400$.

Q6. Why test the simulator-verified circuit again on hardware?

A. Hardware adds what simulators omit unless told: gate errors, readout errors, decoherence, crosstalk — the histogram’s forbidden-bin fraction measures them.

Solutions to the Practice Problems

Problem 1 [Basic] `minus = (ket0 - ket1)/np.sqrt(2)` and `minus2 = H @ ket1`. Check: `np.allclose(minus, minus2)` — prints `True`.

Problem 2 [Basic] $|1\rangle \otimes |0\rangle = |10\rangle$: the vector $(0, 0, 1, 0)$.

Problem 3 [Basic] `qc = QuantumCircuit(2); qc.h(0); qc.cx(0, 1); qc.measure_all()`.

Problem 4 [Basic] $|\beta|^2 \approx 0.755 \Rightarrow |\beta| \approx 0.87$.

Problem 5 [Basic] The H was omitted (the state stayed $|0\rangle$). Fix: add `qc.h(0)` before the measurement.

Problem 6 [Basic] With q_1 starting in $|1\rangle$: H on q_0 gives $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)|1\rangle$; CNOT: $\frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$ — measured: 50% 01, 50% 10, exactly the observed histogram. (An X left over from a previous experiment, or a mis-reset, does this.)

Problem 7 [Basic] $m_0 = 0.50$: $t_0 = (0.50 - 0.03)/0.94 = 0.5$ — unchanged (symmetric flips cancel at the midpoint). $m_0 = 0.97$: $t_0 = 0.94/0.94 = 1.00$ — a three-point correction. Extremes correct most: flips have nowhere to cancel.

Problem 8 [Basic] Circuit A: prepare $|0\rangle$ (do nothing), measure many shots: the fraction reading 1 is $\epsilon_{0 \rightarrow 1}$. Circuit B: X then measure: the fraction reading 0 is $\epsilon_{1 \rightarrow 0}$. Re-run at every recalibration because readout errors drift with the machine’s tuning (Chapter 28’s weather-report rule).

Problem 9 [Standard] `import numpy as np; S = np.array([[1, 0], [0, 1]]); Z = np.array([[1, 0], [0, -1]]); print(np.allclose(S @ S, Z))` — prints `True`.

Problem 10 [Standard] Two independent $|+\rangle$ ’s: uniform over four patterns: ≈ 200 each of 00, 01, 10, 11 ($\pm$ shot noise $\approx \pm 14$). This reuses Chapter 9’s “combining two superpositions” worked example.

Problem 11 [Standard] Qiskit prints $q_{n-1} \cdots q_1 q_0$ — reversed from this book: our $|01\rangle$ ($q_0 = 0, q_1 = 1$) prints as “10”. The reversed-CNOT Bell circuit still yields perfectly *correlated* qubits, and the strings 00/11 are palindromes — identical under reversal — so the ordering trap produced no visible symptom. (Teach: always test with an *asymmetric* state like $|01\rangle$ to expose conventions.)

Problem 12 [Standard] Prediction: $\{000: \approx 800, 111: \approx 800\}$, nothing else. Construction: `qc = QuantumCircuit(3); qc.h(0); qc.cx(0, 1); qc.cx(1, 2); qc.measure_all()`.

Problem 13 [Standard] Forbidden fraction: $240/1000 = 24\%$ — five times the Bell circuit's 4.7%. One more qubit and one more CNOT mean more gates, more readout taxes, and longer exposure: errors compound multiplicatively with size and depth ($(1 - p)^d$ falling), so wider/deeper circuits corrupt disproportionately.

Problem 14 [Standard] With $|\psi\rangle = |0\rangle$ (order: ψ , A, B): after entangling A-B: $|0\rangle \otimes \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$, i.e. $\frac{1}{\sqrt{2}}(|000\rangle + |011\rangle)$. After Alice's CNOT ($\psi \rightarrow A$): unchanged (control is 0): same state. After Alice's H (on ψ): $\frac{1}{2}(|000\rangle + |011\rangle + |100\rangle + |111\rangle)$ — the pre-measurement checkpoint. (Assert each with `np.allclose`.)

Problem 15 [Standard] After H: true $|+\rangle$ gives 100% 0; the 50/50 mixture is basis-blind and still gives 50/50. Even 20 shots decide overwhelmingly: twenty straight zeros from a mixture has probability $2^{-20} \approx 10^{-6}$ — any 1s reveal the impostor immediately.

Problem 16 [Standard] Need the gap $0.05 > 2\sigma = 2 \times 0.5/\sqrt{n} = 1/\sqrt{n}$: $\sqrt{n} > 20 \Rightarrow n > 400$ shots.

Problem 17 [Standard] $\langle Z \rangle = 2P(0) - 1$ is linear in $P(0)$, and the mixing shrinks $P(0)$'s distance from $\frac{1}{2}$ by $(1 - 2\epsilon)$; the offset cancels in the $2P - 1$ conversion, leaving pure scaling: $\langle Z \rangle_{\text{true}} = \langle Z \rangle_{\text{meas}} / (1 - 2\epsilon) = 0.62/0.96 = 0.646$. (Same inflation factor un-lies CHSH scores and VQE energies — always with its noise-amplification tax.)

Problem 18 [Challenge] (Grading checklist.) Code defines `ket0`, `H`, `Z`; prints the state after each column: $(1, 0) \rightarrow (0.707, 0.707) \rightarrow (0.707, -0.707) \rightarrow (0, 1)$ — matching $H-Z-H = X$. Sampling loop: `probs = np.abs(state)**2; rng.choice(['0', '1'], size=1000, p=probs); counts ≈ 1000 × 1` — and Qiskit's version returns `{'1': 1000}`. Full credit requires the printed checkpoints *and* the agreement sentence.

Problem 19 [Challenge] Steps: build 3-qubit circuit; prepare $|\psi\rangle$ on q_0 (e.g. `qc.ry(angle, 0)`; `qc.h(1)`; `qc.cx(1, 2)` (share the pair); `qc.cx(0, 1)`; `qc.h(0)`; `qc.measure(0, c0)`; `qc.measure(1, c1)`; conditionals: with `qc.if_test((c1, 1))`: `qc.x(2)` and with `qc.if_test((c0, 1))`: `qc.z(2)`; finally measure q_2 in chosen bases. Convincing evidence: *tomography-style statistics* on q_2 — e.g. prepare $|\psi\rangle = R_y(66^\circ)|0\rangle$, then show q_2 's Z-basis and X-basis histograms match $|\psi\rangle$'s predicted P 's (within shot noise) across all four measurement branches — amplitudes are never read, but their statistical fingerprints are.

Problem 20 [Challenge] (Model answer — accept any defensible trio.) Circuit 1: single-qubit H + measure on every qubit (readout + 1q calibration; grade on distance from 50/50). Circuit 2: Bell pair (adds one CNOT; grade on forbidden-bin fraction). Circuit 3: depth-20 CNOT ladder returning to $|0 \cdots 0\rangle$ ideally (grades compounding under depth; the survival fraction estimates effective p). Health grade: e.g. $G = \text{survival}_3$ weighted by circuit-2 forbidden fraction — one number summarizing readout, entangling quality, and depth tolerance. (Chapter 30's "quantum volume" is the professionals' refined version.)

Problem 21 [Challenge] (i) Standard counts: 00/11 only, $\approx 50/50$ — catches wiring/initialization bugs (wrong correlations). (ii) X-basis parity: H on both, then measure: Φ^+ stays Φ^+ , so counts remain 00/11 — an even-parity pattern; catches phase bugs like a stray Z (which would flip the parity signature) that check (i) cannot see. (iii) Marginals 50/50 per qubit — catches biased/broken single-qubit behavior or readout asymmetry even when joint patterns look plausible. Three views, three distinct failure classes.

Problem 22 [Challenge] **Mixing:** $\begin{pmatrix} m_0 \\ m_1 \end{pmatrix} = \begin{pmatrix} 0.99 & 0.04 \\ 0.01 & 0.96 \end{pmatrix} \begin{pmatrix} t_0 \\ t_1 \end{pmatrix}$. Determinant 0.95; inverse $\frac{1}{0.95} \begin{pmatrix} 0.96 & -0.04 \\ -0.01 & 0.99 \end{pmatrix}$. Correcting (0.90, 0.10): $t_0 = (0.96 \times 0.90 - 0.04 \times 0.10)/0.95 = 0.905$; $t_1 = 0.095$. Symmetric formula at $\epsilon = 2.5\%$: $t_0 = (0.90 - 0.025)/0.95 = 0.921$ — overestimating t_0 by ≈ 0.016 , a bias several times the typical shot noise. Asymmetry is worth two calibration numbers.

CHAPTER FIFTEEN

Quantum Cryptography and Communication

Teaching notes, self-test, and complete solutions

TEACHING NOTES

The chapter with the field's most deployable technology — teach it as engineering resting on two theorems (no-cloning; incompatible bases) rather than as magic. The BB84 card-game activity is the semester's most reliable hit: play it *before* the formal protocol, and the sifting/fingerprint logic installs itself. Keep three honesty flags flying: QKD needs authenticated classical channels (Problem 13 — unauthenticated quantum crypto falls to a classical impersonator); intrinsic noise vs eavesdropping is a statistical judgment (Problem 11); and the deployment race is PQC-first (Mosca's clock, Problems 7/17/22 — assign at least one, since it is the argument students will actually use professionally). The E91 lesson connects to Chapter 11's CHSH machinery; its round-sorter arithmetic makes good quick quizzes. Stumbles: forgetting that only *checked* bits reveal Eve (the $(3/4)^n$ uses check bits, not all bits), and conflating QBER sources.

SELF-TEST — QUESTIONS WITH ANSWERS

Q1. Why does Eve's measurement leave marks in BB84?

A. She must guess bases; wrong guesses (half) collapse the state in the wrong basis, randomizing Bob's matched-basis results — 25% errors where there should be none.

Q2. What survives sifting, and why half?

A. Rounds where Bob's random basis matched Alice's — about half. Mismatched rounds are discarded as meaningless.

Q3. Detection probability after checking n bits against an intercept-resend Eve?

A. $1 - (3/4)^n$: each checked bit independently exposes her with probability $1/4$.

Q4. What does E91 add over BB84?

A. The source needn't be trusted: a CHSH score near $2\sqrt{2}$ certifies the pairs untouched (monogamy), moving security from device trust toward measured statistics.

Q5. State Mosca's inequality and its use.

A. If $L + M > T$ (secrecy lifetime + migration time vs. time-to-quantum-computer), migration is already late —

harvest-now-decrypt-later makes recorded traffic retroactively vulnerable.

Q6. QKD or PQC — which is the default defense, and why?

A. PQC: it deploys by software on existing networks and protects everything now; QKD is complementary hardware for special links needing physics-level assurance.

Solutions to the Practice Problems

Problem 1 [Basic] About half: Bob's random basis matches Alice's random basis half the time; only matched rounds carry reliable bits.

Problem 2 [Basic] Wrong-basis round: outcomes 0 or 1, each $\frac{1}{2}$ (a $|+\rangle$ read in Z is a coin flip). The round is discarded at sifting — bases differed.

Problem 3 [Basic] $1 - (3/4)^4 = 68.4\%$; $1 - (3/4)^8 = 90.0\%$. (Twelve bits, as in the chapter's table: 96.8%.)

Problem 4 [Basic] No-cloning (Chapter 12) — Eve cannot copy the qubit and study it later; and measurement disturbance of non-orthogonal states / incompatible bases (Chapters 4 and 8) — reading in the wrong basis changes the state detectably.

Problem 5 [Basic] Nine equally likely angle pairs, 500 rounds each: key = $2/9 \rightarrow 1000$ bits; CHSH tests = $4/9 \rightarrow 2000$ rounds; discards = $3/9 \rightarrow 1500$.

Problem 6 [Basic] Abort — $S = 1.90 < 2$ certifies nothing. Indistinguishable causes: an innocent noisy channel, or a full intercept-resend attack (which gives $S \approx 1.41$ – 2 depending on fraction). The protocol needn't distinguish: in both cases the pairs cannot certify privacy, and the only safe response is identical — discard and retry (no key material was exposed by aborting).

Problem 7 [Basic] (a) $L + M = 15$: late for any $T < 15$ — i.e. if a code-breaking machine could arrive within 15 years, the bank must start now. (b) $L + M = 3$: late only if $T < 3$ — the news site can safely wait and migrate with routine upgrades.

Problem 8 [Basic] Adversaries can record today's encrypted traffic cheaply and decrypt it the day a quantum computer exists — so data is exposed *now* if its secrecy must outlive that day. This justifies counting L , the secrecy lifetime, inside the deadline: protection must begin while the data still has L years of sensitivity ahead.

Problem 9 [Standard] Sifted ≈ 1000 ; minus 200 checked $\rightarrow 800$; error-correction and privacy-amplification overhead 15%: final key ≈ 680 bits.

Problem 10 [Standard] On sifted rounds Eve guessed Alice's basis correctly half the time (knows the bit exactly) and wrongly half the time (her record is uncorrelated: right with probability $\frac{1}{2}$): total $\frac{1}{2} + \frac{1}{4} = 75\%$ of sifted bits known to Eve. A passed spot check bounds Eve's *interference*, not her partial knowledge from luck —

privacy amplification exists to crush that 75%-correlated shadow key into provably-near-zero knowledge of the final key.

Problem 11 [Standard] Day one: 3.2% against 3% intrinsic is statistically unremarkable — proceed (error-correct, amplify, log). Day two: 24% is the intercept-resend fingerprint ($\approx 25\%$, or an equally fatal fault) — abort, alarm, investigate; no key is issued either way until QBER returns to baseline.

Problem 12 [Standard] An amplifier is a copier: strengthening light means producing many photons in the state of one, which for unknown quantum states is exactly what no-cloning forbids — amplified QKD signals are either measurably disturbed or classical. Deployed workarounds: trusted relays (decrypt-re-encrypt at guarded stations — pragmatic, physics-lite) and, prospectively, quantum repeaters (swap + distill, Chapter 16), which extend entanglement without ever copying.

Problem 13 [Standard] Unauthenticated, Eve can sit in the middle and run two honest QKD sessions: one with Alice (posing as Bob), one with Bob (posing as Alice); each session passes every quantum test, and Eve simply re-encrypts traffic between her two keys, reading everything. No quantum principle objects — each link genuinely is secure against third parties; the failure is identity, not physics. The fix is classical: a short pre-shared secret (or classical PKI) to *authenticate* the public discussion — QKD then *grows* that small secret into unlimited fresh key, which is its honest description.

Problem 14 [Standard] $\text{QBER} \leq 11\% \Rightarrow V \geq 0.78 \Rightarrow S \geq 0.78 \times 2\sqrt{2} = 2.21$.

Problem 15 [Standard] Definite (unentangled) pairs are a local hidden-answer model made physical: each qubit carries a pre-set disposition for every setting, so Chapter 11's theorem caps their CHSH score at 2 — for every possible choice of definite states and mixtures thereof. The test demands S near 2.83; the cheap source tops out at 2 and is caught, always. (This is why the E91 test cannot be spoofed classically, even by the source's builder.)

Problem 16 [Standard] BB84: trust Alice's and Bob's devices (state preparation and measurement must be characterized); the source *is* Alice; statistic: QBER on checked bits. E91: devices still Alice's and Bob's, but the source may be anyone's — hostile included; statistic: the CHSH score S (plus QBER). E91 removes the trusted-source assumption entirely — certification by measured nonlocality instead of by provenance.

Problem 17 [Standard] After waiting five years: M still 5-ish, L unchanged for *new* data, so the clock reads the same $18 > T-5$ -adjusted comparison — but five years of traffic was harvested in the meantime, carrying obligations to $T' = 25 - 5 = 20$ against $L + M = 18$: margin a razor-thin 2 years, purchased for zero benefit (migration costs the same later). Waiting bought nothing, cost five years of exposed archives, and bet the institution on a forecast's error bars.

Problem 18 [Challenge] Half the qubits intercepted $\Rightarrow$ error rate $e = \frac{1}{2} \times 25\% = 12.5\%$ on sifted bits. Catch probability $1 - (1 - 0.125)^n \geq 0.99$: $n \geq \ln 0.01 / \ln 0.875 = 34.5 \rightarrow 35$ check bits. Moral: partial attacks dilute the fingerprint linearly but never erase it — they only raise the (still modest) check-bit budget; combined with Problem 10's logic, Eve's knowledge shrinks proportionally too.

Problem 19 [Challenge] (Model.) “PQC protects all your traffic — including internet-facing links QKD cannot reach — by software rollout, against both harvest-now and future attacks; deploy it everywhere, immediately. A 40 km metro fiber is comfortably within commercial QKD range (~ 0.2 dB/km: 8 dB), so a link to the DR site is feasible if physics-level key assurance for bulk replication justifies the hardware. Harvest-now applies to your longest-lived data either way — PQC migration cannot wait on the QKD decision. Note that QKD still requires authenticated classical channels and guarded endpoints; it moves trust, not the need for it. Verdict: PQC now across the estate; pilot the QKD link only after PQC is done, and only on condition of a measured, monitored QBER baseline and an endpoint-security audit.”

Problem 20 [Challenge] (Grading guide — the paragraph should contain:) a passed CHSH test at S near $2\sqrt{2}$ implies the two measured qubits are (near-)maximally entangled *with each other*; monogamy then forbids any third system — including anything retained inside the untrusted source — from carrying correlation with the outcomes; therefore the key bits (matched-angle rounds) are statistically private *regardless of who built or controls the source*, whose only power is to lower S and abort the session. Credit mentions of: statistics-based certification, monogamy as the quantitative step, and abort-on-low- S as the safe failure mode.

Problem 21 [Challenge] $2\sqrt{2} - f(2\sqrt{2} - \sqrt{2}) \geq 2.70 \Rightarrow f \leq (2.828 - 2.70)/1.414 = 0.091$: about 9% of pairs. Key bits known: $\leq 0.091 \times 2000 \approx 180$ (fewer in expectation, by Problem 10’s factor). Privacy amplification hashes the 2000 bits down to a shorter final key in a way that provably reduces Eve’s 180-bit shadow to exponentially small knowledge of the result.

Problem 22 [Challenge] (Grading guide — expect eight lines shaped like:) chosen L (justified by data type) and M (justified by system count); three Mosca verdicts, e.g. $L+M = 22$: late for $T = 15$, late for 25, safe only under 40; allocation: PQC universally now (FIPS 203–205), QKD only for the named crown-jewel link if any; endpoint sentence acknowledging that stolen endpoints defeat both. Full credit requires every claim numbered or chapter-cited, per the problem’s own rule.

CHAPTER SIXTEEN

The Quantum Internet

Teaching notes, self-test, and complete solutions

TEACHING NOTES

A synthesis chapter: loss arithmetic (Appendix C's exponentials), swapping (Chapter 13), fidelity and distillation (Chapter 13's lessons), and CHSH-certified privacy (Chapters 11, 15) assemble into network engineering. Teach it as budget-building: the 600-km worked example is the template, and the design activity (plan a national link) makes an outstanding graded group project — regional city pairs give it local reality. Keep two honest distinctions sharp: trusted relays versus true repeaters (who can read the key?), and fiber's exponential versus the satellite's geometric loss (why space wins long-haul). The memory-as-hero moral is the chapter's deepest point — students expect exotic physics and find that the bottleneck is a good quantum RAM. Stumbles: multiplying segment probabilities when memories make them independent-in-time, and treating distillation as free (always bill the $4\times$).

SELF-TEST — QUESTIONS WITH ANSWERS

Q1. Why can't we amplify quantum signals in fiber?

A. Amplification is copying, and no-cloning forbids copying unknown states — loss must be outsmarted (repeaters), not out-shouted.

Q2. The repeater chain's three-step recipe?

A. Divide the distance into segments; store segment successes in quantum memories; swap neighboring pairs into one long pair.

Q3. What does entanglement distillation trade?

A. Quantity for quality: about four raw pairs (two consumed, half the attempts fail) yield one higher-fidelity pair, using only local gates plus classical messages.

Q4. Trusted relay vs. quantum repeater, in one line?

A. A relay decrypts and re-encrypts — it knows the key and must be guarded; a repeater only swaps entanglement — monogamy keeps the key unknowable to it.

Q5. Why do satellites beat fiber at 1000+ km?

A. Fiber loss is exponential in distance (10^{-20} at 1000 km); a satellite link pays mostly beam-spreading, tens of dB — geometry beats exponentials.

Q6. What will the quantum internet actually carry?

A. Entanglement and keys: physics-certified security, teleported states, networked processors and sensors — not faster movie downloads (Holevo caps qubit bandwidth).

Solutions to the Practice Problems

Problem 1 [Basic] $T(25) = 10^{-0.5} \approx 32\%$; $T(150) = 10^{-3} = 0.1\%$; $T(300) = 10^{-6}$. Doubling squares: $T(2L) = T(L)^2$ — the defining signature of exponential loss.

Problem 2 [Basic] An amplifier works by producing many photons in the state of the one that arrived — a cloning machine, forbidden for unknown quantum states by the no-cloning theorem (Chapter 12). Classical repeaters escape because classical bits are *known*, *orthogonal* states (0 or 1), which CNOT-style copying handles legally.

Problem 3 [Basic] The quantum *memory time*: each station must hold its successful half faithfully for the hundreds of microseconds the neighboring segments need — and the T_1/T_2 protocols of Chapter 28 are exactly what measure it.

Problem 4 [Standard] Eight segments at $T = 1\%$: still ≈ 100 attempts = 100 μs each; chain waits for the slowest of eight — same few-hundred-microsecond scale (slightly worse than six, logarithmically); swaps: 7. Grew linearly: stations, swaps, classical bits. Stayed constant: per-segment physics — which is the whole point of dividing.

Problem 5 [Standard] Round two: $f'' = \frac{0.941^2}{0.941^2 + 0.059^2} = \frac{0.8855}{0.8890} = 0.996$. Cost: $\approx 4^2 = 16$ raw pairs per twice-distilled pair. Teleportation: $F_{\text{out}} = (2 \times 0.996 + 1)/3 = 0.997$.

Problem 6 [Standard] $f'(0.5) = \frac{0.25}{0.25 + 0.25} = 0.5$: a fixed point — no improvement, ever. $f'(0.45) = \frac{0.2025}{0.2025 + 0.3025} = 0.40$: below half, distillation makes pairs worse. $f = \frac{1}{2}$ is the separable boundary of Chapter 13 ($F_{\text{out}} = \frac{2}{3}$, the classical ceiling): pairs with no quantum advantage contain nothing for distillation to concentrate.

Problem 7 [Standard] A trusted relay physically holds the *decrypted key bits* between its two QKD links — a guard-worthy room. A repeater station holds only halves of entangled pairs and performs Bell measurements whose two-bit outcomes are correction indices, not key material; a passed end-to-end CHSH test plus monogamy (Chapter 11) guarantees the station's systems carry no correlation with the final key.

Problem 8 [Standard] $2000/100 = 20$ stations (19 intermediate, plus endpoints). Each needs: a quantum memory (hold one half while the neighbor retries) and a Bell-measurement (swapping) capability — plus the unglamorous classical control channel.

Problem 9 [Standard] (a) 80 km fiber: 16 dB $\rightarrow T \approx 2.5\%$ — healthy attempt rates deliver keys continuously: use fiber, always-on. (b) 1200 km fiber: 240 dB $\rightarrow 10^{-24}$: never; satellite passes at tens of dB deliver key in nightly bursts: use satellite (with buffering), fiber only as a relay backbone. Hybrid rule: fiber below a few hundred km, satellite (or future repeater chains) beyond.

Problem 10 [Challenge] Each segment's completion time is geometric with mean $1/(qR) = 100\ \mu\text{s}$; the maximum of k such clocks runs $\approx (\ln k + 1)$ times the mean (the standard slowest-of- k growth): for $k = 6$: $(1.79 + 1) \times 100 \approx 280\ \mu\text{s}$ versus $100\ \mu\text{s}$ — memories must last $\sim 3\times$ a single segment's expectation (plus swap and classical-confirmation time). Design consequence: memory time budgets are set by the *slowest* link, and grow only logarithmically with chain length — good news, carefully earned.

Problem 11 [Challenge] One distillation round: $200/4 = 50$ good pairs per second at $f' = 0.941$; each teleportation consumes one: **50 teleportations/s**, at $F_{\text{out}} = (2 \times 0.941 + 1)/3 = 0.961$ — margin $+0.011$ over the 0.95 target. (A student noting shot-to-shot fluctuation and proposing a small safety factor on the rate deserves praise.)

Problem 12 [Challenge] (Model audit.) (1) *Endpoints*: what protects keys after delivery — HSMs, audited premises? Evidence: an endpoint security assessment; physics certifies transit only. (2) *Relays*: how many trusted relays does the national path traverse, and who guards each? Evidence: the route map with relay custody — every relay reads the key. (3) *Availability*: what key rate through weather, daylight, and pass schedules? Evidence: a year of measured key-delivery statistics, not best-night numbers. (4) *Alternative*: why does this beat NIST post-quantum cryptography deployed by software at near-zero marginal cost? Evidence: a threat model naming what QKD stops that PQC does not — for this customer. “Unbreakable” survives only if all four return solid answers; in 2026, question 4 is usually the fatal one.

CHAPTER SEVENTEEN

First Algorithms: Deutsch–Jozsa and Bernstein–Vazirani

Teaching notes, self-test, and complete solutions

TEACHING NOTES

Small algorithms, permanent lessons. Insist every student traces Deutsch's four-function table by hand (the activity) — it is the one place the full kickback-plus-interference mechanism fits on a page. Repeat the mistake-box mantra until it sticks: one query yields *one bit*, but a global bit. The honesty lesson (randomized classical algorithms tame the DJ promise problem) belongs in class, not a footnote — students who learn it here read algorithm claims correctly forever. Bernstein–Vazirani doubles as the course's cleanest hardware benchmark: if you have cloud access, run it live and let the histogram's error bars teach Chapter 28 early. The partner-oracle activity (build an oracle from CNOTs, extract the secret in one shot) is reliably the most-remembered class of the algorithms unit. Stumbles: forgetting the helper's $|-\rangle$ preparation, and reading the DJ verdict backwards (all-zeros = constant).

SELF-TEST — QUESTIONS WITH ANSWERS

Q1. The Deutsch problem and both costs?

A. Is a one-bit function constant or balanced? Classical: 2 queries, provably; quantum: 1.

Q2. What does the $|-\rangle$ helper accomplish?

A. Kickback: querying with helper $|-\rangle$ converts the oracle into sign stamps $(-1)^{f(x)}$ on the register — values become phases.

Q3. The DJ verdict rule?

A. Measure all-zeros $\Rightarrow$ constant; anything else $\Rightarrow$ balanced. The all-zeros amplitude averages every stamp.

Q4. What does Bernstein–Vazirani output, and after how many queries?

A. The hidden string s itself, with certainty, after one query — versus n queries classically.

Q5. Did the quantum computer learn $f(0)$ and $f(1)$ simultaneously?

A. No — it learned one global property (agree/disagree) and nothing else; Holevo's bound is never cheated.

Q6. Which separation survives error-tolerant classical algorithms?

A. BV's factor of n (unconditional); DJ's exponential collapses if tiny error is tolerated. The template, not the toy scores, is what led to Shor.

Solutions to the Practice Problems

Problem 1 [Basic] Constant: $f \equiv 0$, $f \equiv 1$. Balanced: identity, NOT. One query reveals one value — say $f(0) = 0$ — which is consistent with both a constant ($f \equiv 0$) and a balanced (identity) function: the category stays open.

Problem 2 [Basic] The helper must be in X 's -1 eigenstate $|-\rangle$ for kickback to stamp $(-1)^{f(x)}$. With $|+\rangle$ (eigenvalue $+1$) the “stamps” are all $+1$: the register state never changes, the final H returns $|0\rangle$ regardless of f , and the algorithm learns nothing.

Problem 3 [Basic] Stamps for $f \equiv 0$: both $+$: register stays $|+\rangle$; final H : $H|+\rangle = |0\rangle$. Readout 0 — constant ✓

Problem 4 [Standard] Stamps: $(-1)^{f(0)} = -1$ on $|0\rangle$, $(-1)^{f(1)} = +1$ on $|1\rangle$: state $\frac{1}{\sqrt{2}}(-|0\rangle + |1\rangle) = -\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = -|-\rangle$. The overall -1 is a *global* phase (Chapter 5's distinction) — physically the state is $|-\rangle$, and $H|-\rangle = |1\rangle$: balanced ✓

Problem 5 [Standard] Stamps by x_2 (the second bit): $(+, -, +, -)$ on $(00, 01, 10, 11)$. This is “the x_2 wave”: the final H 's concentrate all amplitude on $|01\rangle$ — readout 01, certainly (not all-zeros: balanced, and the pattern names the dependent bit).

Problem 6 [Standard] $s = 110$ selects bits 1 and 2: signs = $+$ for $x \in \{000, 001, 110, 111\}$ and $-$ for $\{010, 011, 100, 101\}$ (parity of the first two bits). Certain readout: $|110\rangle$.

Problem 7 [Standard] Exact DJ classical worst case: $2^{19} + 1 = 524,289$ queries; BV classical: 20; quantum: 1 and 1. Against a 10^{-9} -tolerant skeptic, DJ's separation collapses (a few dozen random samples decide with overwhelming confidence); BV's clean 20:1 survives every tolerance — the honest scoreboard.

Problem 8 [Standard] Input $x = 110$: CNOT from register bit 1 (value 1) flips the helper $\rightarrow 1$; CNOT from bit 3 (value 0) does nothing: helper = $1 = s \cdot x = 1 \cdot 1 \oplus 0 \cdot 1 \oplus 1 \cdot 0$ ✓ The phase-oracle version (helper in $|-\rangle$) leaves the helper in $|-\rangle$ throughout — kickback moves the information into signs, so there is nothing written that needs uncomputing.

Problem 9 [Standard] Success fidelity $\approx 903/1000 = 90.3\%$. The near-misses $(001, 111, 100)$ differ from $s = 101$ by single bit flips — consistent with per-qubit *readout errors* (~ 1 -3% each, Chapter 28) and single-qubit gate errors/relaxation during the circuit; their roughly equal sizes across qubits is itself diagnostic.

Problem 10 [Challenge] Let the queried set be Q , $|Q| = 2^{n-1}$, all answers 0. Candidate one: $f \equiv 0$ (constant) — consistent. Candidate two: the function that is 0 exactly on Q and 1 on the complement — balanced

(each half has 2^{n-1} elements) and consistent with every observation. Two consistent functions in different categories $\Rightarrow$ no certain verdict with 2^{n-1} queries; hence $2^{n-1}+1$ is required.

Problem 11 [Challenge] $H^{\otimes n} |x\rangle$ has amplitude $2^{-n/2}(-1)^{x \cdot y}$ on $|y\rangle$; at $y = 0 \cdots 0$ every sign is $+$. Acting on the stamped state $2^{-n/2} \sum_x (-1)^{f(x)} |x\rangle$, the $|0 \cdots 0\rangle$ amplitude is $2^{-n} \sum_x (-1)^{f(x)}$. Constant: all stamps equal: amplitude ± 1 — all-zeros certain. Balanced: half $+$, half $-$: amplitude 0 — all-zeros impossible ✓

Problem 12 [Challenge] Final amplitude on $|y\rangle$: $2^{-n} \sum_x (-1)^{s \cdot x} (-1)^{y \cdot x} = 2^{-n} \sum_x (-1)^{(s \oplus y) \cdot x}$. If $s \oplus y \neq 0$, pick a bit position where it is 1 and pair each x with x -flipped-there: the pair's terms cancel, so the sum is 0. If $y = s$, every term is $+1$: amplitude 1. One bar, at s , with certainty — interference doing exact string extraction.

CHAPTER EIGHTEEN

Grover's Search Algorithm

Teaching notes, self-test, and complete solutions

TEACHING NOTES

The most *visualizable* algorithm — exploit that. The human-amplitude-amplification activity (eight students, paper amplitudes) teaches oracle-reflect-repeat better than any slide; run it before the algebra, then let the $N = 4$ perfect case confirm it. Insist on three honesty stations: overshoot (more is not safer — Problems 10, 16, 19), the unstructured-only clause (binary search demolishes Grover on sorted data — Problem 13), and the sequential-depth reality (Problem 12 collides the gate count with Chapter 28's budgets). The multi-target and amplitude-amplification lessons generalize the machinery; their $\sin^2((2k+1)\theta)$ bookkeeping is one formula students should own cold. The security implication (halved key bits, AES-256 — Problem 20) connects to Chapter 15's policy story and makes a strong discussion close. Stumbles: computing the mean over marked-only amplitudes (it is over *all* N), and quoting $\sqrt{N}$ as “exponential speedup.”

SELF-TEST — QUESTIONS WITH ANSWERS

Q1. Grover's speedup, and its fine print?

A. $O(\sqrt{N})$ oracle calls versus $O(N)$ — quadratic, not exponential; only for *unstructured* search; calls run sequentially.

Q2. The two moves of one Grover round?

A. Oracle: flip the marked amplitude's sign. Diffusion: reflect every amplitude about the mean ($a \rightarrow 2\bar{a} - a$).

Q3. Optimal round count for N , one target?

A. $k \approx \frac{\pi}{4}\sqrt{N}$, then STOP — success $P(k) = \sin^2((2k+1)\theta)$, $\sin \theta = 1/\sqrt{N}$, and it *falls* past the peak.

Q4. What changes with M marked items?

A. $\sin \theta = \sqrt{M/N}$: more needles, bigger steps, fewer rounds ($k \approx \frac{\pi}{4}\sqrt{N/M}$); measurement returns one marked item at random.

Q5. Why doesn't Grover break sorted databases or AES outright?

A. Sorted data has structure (binary search: $\log N$); for ciphers it only halves effective key bits — AES-256 stays comfortable, which is why standards moved there.

Q6. What is amplitude amplification?

A. Grover's machinery applied to any coherent success-flagging subroutine: success probability p needs $O(1/\sqrt{p})$ rounds instead of $O(1/p)$ retries.

Solutions to the Practice Problems

Problem 1 [Basic] Classical average: $N/2 = 50$ checks. Grover: $\frac{\pi}{4}\sqrt{100} \approx 8$ rounds.

Problem 2 [Basic] Amplitudes $(0.5, 0.5, -0.5, 0.5)$ (sign flipped on 10). Probabilities: still 0.25 each — the phase is invisible until interference converts it (that is the diffusion step's job).

Problem 3 [Basic] Mean $\bar{a} = (0.5 + 0.5 - 0.5 + 0.5)/4 = 0.25$. Reflection: unmarked $\rightarrow 2(0.25) - 0.5 = 0$; marked $\rightarrow 0.5 + 0.5 = 1$. Amplitudes $(0, 0, 1, 0)$: certainty in one round.

Problem 4 [Basic] $k = \frac{\pi}{4}\sqrt{16} = 3.14 \rightarrow 3$ rounds.

Problem 5 [Basic] $\sin \theta = \sqrt{2/4} = \frac{1}{\sqrt{2}}$; $\theta = 45^\circ$; $P(1) = \sin^2 135^\circ = \frac{1}{2}$. Random guessing also succeeds with $M/N = \frac{1}{2}$. Conclusion: at the ratio $M/N = \frac{1}{2}$, Grover gains *nothing* — amplification needs a minority to amplify.

Problem 6 [Basic] $M = 1$: $k = \frac{\pi}{4} \times 4 = 3.14 \rightarrow 3$. $M = 4$: $\sin \theta = \frac{1}{2}$, $\theta = 30^\circ$, $k_{\text{opt}} \approx 1.57 \rightarrow$ and indeed $P(1) = \sin^2 90^\circ = 1$: exact certainty at one round (the lucky ratio $M/N = \frac{1}{4}$).

Problem 7 [Basic] $\theta = \arcsin \sqrt{0.04} = \arcsin 0.2 = 11.5^\circ$; $k = \frac{\pi}{4 \times 0.2} = 3.9 \rightarrow 4$; $P(4) = \sin^2(9 \times 11.5^\circ) = \sin^2 103.8^\circ = 0.94$.

Problem 8 [Basic] (1) Coherence: the subroutine must run as a measurement-free unitary — violated by any internal mid-circuit measurement (or unclean scratch, Chapter 14). (2) A success flag implementable as a phase oracle — violated when “success” cannot be checked reversibly (e.g. requires consulting an external database mid-run).

Problem 9 [Standard] Oracle: $(0.5, -0.5, 0.5, 0.5)$. Mean = 0.25; reflect: $(0, 1, 0, 0)$: certainty on 01 ✓ The arithmetic never referenced *which* entry was marked — only that exactly one was.

Problem 10 [Standard] Oracle on $(0, 0, 0, 1)$: $(0, 0, 0, -1)$. Mean = -0.25 . Reflect: unmarked $\rightarrow -0.5$; marked $\rightarrow -0.5 + 1 \rightarrow 0.5$: amplitudes $(-0.5, -0.5, -0.5, 0.5)$, so $P(11) = 0.25$ — down from 1. The success-vs- k oscillation figure predicted exactly this crash past the peak.

Problem 11 [Standard] $P(0) = \sin^2 20.7^\circ = 0.125$; $P(1) = \sin^2 62.1^\circ = 0.78$; $P(2) = \sin^2 103.5^\circ = 0.945$ — matching the worked example's 12.5% $\rightarrow$ 78% $\rightarrow$ 94.5% staircase ✓

Problem 12 [Standard] $k = \frac{\pi}{4}\sqrt{2^{20}} = \frac{\pi}{4} \times 1024 \approx 804$ rounds $\times$ $(50 + 30)$ gates $\approx$ **64,000 sequential gates**

(plus initialization). The collision: Chapter 28's survival budget — $(1 - p)^{64000}$ demands $p \sim 10^{-6}$, three orders below today's hardware: Grover at useful sizes awaits error correction.

Problem 13 [Standard] Linear: $\approx 5 \times 10^5$ average. Binary search: $\log_2 10^6 = 20$ probes. Grover: $\frac{\pi}{4} \times 10^3 \approx 785$ oracle calls. Ranking: binary (20) $\ll$ Grover (785) $\ll$ linear (5×10^5). Moral: *structure beats quantum* — Grover's domain is problems with no exploitable order, and a sorted phone book is not one.

Problem 14 [Standard] $\sin \theta = \sqrt{2/64} = 0.177$; $\theta = 10.2^\circ$; $k_{\text{opt}} = \frac{\pi}{4} \sqrt{32} = 4.4 \rightarrow 4$; $P(4) = \sin^2(9 \times 10.2^\circ) = \sin^2 91.6^\circ = 0.999$. Classical: $(1 - \frac{2}{64})^n \leq \frac{1}{2} \Rightarrow n \approx 22$ draws.

Problem 15 [Standard] $k \propto \sqrt{N/M}$: replacing $M \rightarrow 4M$ divides k by $\sqrt{4} = 2$ ✓ At $M = N/4$: $k \approx \frac{\pi}{4} \times 2 = 1.57$ — one round, and (from $\theta = 30^\circ$) that single round is *exact*: the lucky-ratio family.

Problem 16 [Standard] $\theta = 30^\circ$: $P(3) = \sin^2(7 \times 30^\circ) = \sin^2 210^\circ = 0.25$ — from certainty at $k = 1$ down to 25%. Correction note: "Grover rotates rather than accumulates: past the optimum every extra round *removes* success probability — stop at k_{opt} (here, one)."

Problem 17 [Standard] $p = 10^{-2}$: classical 100 vs. $2k = 16$: ratio 6.3. $p = 10^{-4}$: 10^4 vs. 158: ratio 63. $p = 10^{-6}$: 10^6 vs. 1572: ratio 636. Each hundredfold rarer multiplies the advantage tenfold — the $\sqrt{1/p}/(\pi/2)$ scaling, confirmed.

Problem 18 [Challenge] Oracle flips 011 and 101: amplitudes six at $+0.354$, two at -0.354 . Mean $= (6 - 2) \times 0.354/8 = 0.177$. Reflect: unmarked $\rightarrow 2(0.177) - 0.354 = 0$; marked $\rightarrow 0.354 + 0.354 = 0.707$. Total success: $2 \times 0.707^2 = 1.00$ — certainty, split evenly between the two targets. (And indeed $\sin \theta = \sqrt{2/8} = \frac{1}{2}$: the lucky ratio again.)

Problem 19 [Challenge] In the rotation picture the state vector turns by 2θ per round; at $k = 4$ (for $\theta = 20.7^\circ$) it has swung 186° — nearly opposite the marked axis, projecting less on it than the initial 12.5%: $P = \sin^2 186^\circ \approx 0.01$. Fixes: (1) stop at $k_{\text{best}} = 2$ ($P = 0.945$); (2) measure-and-repeat: attempts succeed with 0.945, so expected attempts $= 1/0.945 = 1.06$, expected oracle calls $\approx 2 \times 1.06 \approx 2.1$ — robust even when N (hence k_{best}) is only approximately known.

Problem 20 [Challenge] $b = 128$: $\approx 2^{64} = 1.8 \times 10^{19}$ rounds; at $10^9/\text{s}$: $1.8 \times 10^{10} \text{ s} \approx 580$ years — and each round is a full sequential AES evaluation on error-corrected hardware. Policy conclusion: Grover halves security bits, so AES-128's quantum margin (64 bits) is uncomfortable on decade timescales while AES-256's (128 bits) is untouchable — standards therefore recommend AES-256 for new systems without declaring AES-128 broken today.

Problem 21 [Challenge] Define $|m\rangle$ = normalized superposition of the M marked items, $|u\rangle$ = of the $N - M$ unmarked. The uniform start is $\sin \theta |m\rangle + \cos \theta |u\rangle$ with $\sin \theta = \sqrt{M/N}$ (its marked mass). The oracle reflects about $|u\rangle$ (flips $|m\rangle$'s sign); diffusion reflects about the start; two reflections whose mirrors meet at angle θ compose to a rotation by 2θ in the $\{|m\rangle, |u\rangle\}$ plane (Chapter 6's two-reflections rule). After k rounds the angle from $|u\rangle$ is $(2k+1)\theta$, and the marked-axis projection gives $P(k) = \sin^2((2k+1)\theta)$ ✓

Problem 22 [Challenge] $\theta = 30^\circ$: $P(1) = \sin^2 90^\circ = 1$ — certainty at one round; $P(2) = \sin^2 150^\circ = 0.25$. The formula $k_{\text{opt}} \approx \pi/(4 \times 0.5) = 1.57$ “rounds up” to 2 — straight past the exact peak at $k = 1$. Rule: compute P at both neighboring integers and take the better — rounding conventions are no substitute for evaluating the sine.

CHAPTER NINETEEN

The QFT, Phase Estimation, and Shor's Algorithm

Teaching notes, self-test, and complete solutions

TEACHING NOTES

The summit chapter — pace it as three ascents. (1) QFT as “rhythm to frequency”: the hand-computable staircase (the $|101\rangle$ worked example) removes the mystique; make every student climb one staircase themselves (Problem 5). (2) QPE as kickback-plus-inverse-QFT — link explicitly back to Chapter 10’s phase meter; the binary-fraction readout (Problem 4) is a two-minute drill. (3) Shor as period-finding plus classical book-keeping: run the full 15-story with $a = 7$ and let Problems 10–11 cover the cooperative and unlucky draws. Close with resource honesty — the falling estimates (20M $\rightarrow$ 1M qubits) and Mosca-style urgency; Problems 19–20 make superb take-home essays, and Problem 20 (the “compiled demo” audit) inoculates against the field’s most common demo trick. Stumbles: mixing the QFT’s phase step x/N with QPE’s readout $\varphi = \text{register}/2^m$, and forgetting the final swap in QFT circuits.

SELF-TEST — QUESTIONS WITH ANSWERS

Q1. What does the QFT do to a hidden period?

A. Converts “repeats every r steps” into measurable spikes at multiples of N/r — rhythm into frequency, readable by one measurement.

Q2. The QFT’s gate cost for n qubits?

A. $O(n^2)$: n Hadamards plus $\sim n^2/2$ controlled phase turns (plus swaps) — exponentially smaller than the $2^n \times 2^n$ matrix it implements.

Q3. What does QPE estimate, and with what precision?

A. An eigenphase φ of a unitary, read as an m -bit binary fraction from m counting qubits — each extra qubit adds one bit of precision.

Q4. Shor’s pipeline in one line?

A. Random $a \rightarrow$ quantum period-finding of $a^x \bmod N$ (QPE on modular multiplication) $\rightarrow$ classical $\gcd(a^{r/2} \pm 1, N)$ delivers the factors.

Q5. Why doesn’t Shor break AES or hash functions?

A. Its exponential speedup is specific to period/algebraic structure (factoring, discrete logs); symmetric ciphers lack it — they face only Grover's square root.

Q6. Where do the resource estimates stand?

A. RSA-2048: $\sim 20\text{M}$ physical qubits (2021 estimate) falling under 1M (2025), for hours of below-threshold operation — versus today's 10^3 noisy qubits: real gap, shrinking from both sides.

Solutions to the Practice Problems

Problem 1 [Basic] Period 5 in length 20 $\Rightarrow$ spikes at multiples of $20/5 = 4$: frequencies 0, 4, 8, 12, 16.

Problem 2 [Basic] $2, 4, 8, 16 \equiv 1$: the cycle is $2, 4, 8, 1, \dots$ — period $r = 4$.

Problem 3 [Basic] $x = 0$: all phases $e^0 = 1$: $\frac{1}{2}(1, 1, 1, 1)$ — the uniform superposition $|+\rangle|+\rangle$ (frequency zero = flat rhythm).

Problem 4 [Basic] $\varphi = 0.101_2 = \frac{1}{2} + \frac{1}{8} = 0.625$.

Problem 5 [Basic] Step = $3/8$ turn = 135° per unit k : phases $0^\circ, 135^\circ, 270^\circ, 45^\circ, 180^\circ, 315^\circ, 90^\circ, 225^\circ$ (reducing mod 360°).

Problem 6 [Basic] All amplitudes $\frac{1}{\sqrt{8}}e^0 = \frac{1}{\sqrt{8}}$: the uniform superposition $H^{\otimes 3}|000\rangle$. Frequency zero is the constant note — every pattern equally, no phase variation.

Problem 7 [Basic] $10^6/10^3 = 1000$ logical qubits — roughly the right *order* for the algorithm proper, but the 2025 million-qubit estimate already *includes* its error correction and factories: such a machine is approximately the requirement itself (i.e. fraction ≈ 1), which is exactly why the estimate made news.

Problem 8 [Basic] Algorithm (logical qubits and gate counts — this chapter); error correction (physical-per-logical overhead — Chapter 29's ledger); magic-state factories (the T-gate supply — Chapter 7's T-count economics). Multiply the three for the machine size.

Problem 9 [Standard] $x = 1$, step = $\frac{1}{4}$ turn = 90° : amplitudes $\frac{1}{2}(1, i, -1, -i)$ — precisely the two-qubit kickback state of the QPE worked example (phases $0^\circ, 90^\circ, 180^\circ, 270^\circ$).

Problem 10 [Standard] $a = 2, r = 4$: $2^2 = 4$: $\gcd(4 - 1, 15) = 3$; $\gcd(4 + 1, 15) = 5$. Factors $3 \times 5 = 15$ ✓ — a second cooperative witness.

Problem 11 [Standard] $14 \equiv -1 \pmod{15}$: powers $14, 1, 14, 1, \dots$: $r = 2$. Then $14^{r/2} = 14 \equiv -1$: the “ $a^{r/2} \equiv -1$ ” unlucky case — $\gcd(14 + 1, 15) = 15$ and $\gcd(14 - 1, 15) = 1$, both trivial. The screening test rejects $a = 14$ and redraws.

Problem 12 [Standard] With $m = 2$ the readouts are the fractions 0, 0.25, 0.5, 0.75; the nearest to 0.3 are

0.25 (mostly) and 0.5 (sometimes) — the distribution peaks at the closest fraction with leakage to neighbors. More counting qubits refine the grid (2^{-m} spacing), so the peak sharpens onto a better approximation of 0.3.

Problem 13 [Standard] $n = 10$: ≈ 50 controlled phases. $n = 2048$: $\approx 2048^2/2 \approx 2.1$ million — large but *polynomial*. Grover on a 2048-bit space: $\sqrt{2^{2048}} = 2^{1024}$ calls — exponential and impossible forever. The comparison is the whole story of structured versus unstructured speedups.

Problem 14 [Standard] Step 180° : amplitudes $\frac{1}{2}(1, -1, 1, -1)$. Factor: $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = |+\rangle |-\rangle$ ✓ (first factor from the unchanged sign across the top bit, second from the alternating sign).

Problem 15 [Standard] Hadamards: 5. Controlled phases: $4 + 3 + 2 + 1 = 10$. Swaps: 2. Smallest turn: $360^\circ/2^5 = 11.25^\circ$.

Problem 16 [Standard] Omitting the controlled-T loses the 45° contribution that the bottom digit owes the top — output phases drift from the true staircase by up to 45° on half the components. Because deeper qubits contribute geometrically smaller turns ($22.5^\circ, 11.25^\circ, \dots$), truncating the *smallest* ones costs little — the approximate QFT keeps only the big turns and sacrifices almost nothing at realistic precision.

Problem 17 [Standard] Requirements falling $\sim 20\times/4$ yr; hardware growing $\sim 10\times/4$ yr: the $\sim 10^3$ -fold gap closes at a combined $\sim 200\times$ per 4 years — naively one more 4–6 year cycle to overlap. Distrust both ways: requirement-cuts may saturate (algorithms cannot improve forever), and hardware scaling may stall at error-rate or engineering walls — extrapolation gives a *watch schedule*, not a forecast; re-evaluate annually (Mosca's clock already told you what to do meanwhile).

Problem 18 [Challenge] $7^4 - 1 \equiv 0 \pmod{15}$ and $7^4 - 1 = (7^2 - 1)(7^2 + 1) = 48 \times 50$. So 15 divides 48×50 while dividing neither factor alone ($48 = 3 \times 16$, $50 = 5 \times 10$): 15's prime 3 must sit inside 48 and its prime 5 inside 50 — hence $\gcd(48, 15) = 3$ and $\gcd(50, 15) = 5$ extract them. The screened unlucky cases are exactly those where the split fails: r odd (no $a^{r/2}$) or $a^{r/2} \equiv -1$ (all of 15 divides one factor).

Problem 19 [Challenge] (Model.) Records must be secret to ~ 2076 ; harvest-now means today's leaked ciphertext is readable the day RSA falls. The gap (10^6 good vs 10^3 – 10^4 noisy qubits) is 2–3 orders, and optimistic doubling every 2–3 years crosses it within 15–25 years — comfortably inside the 50-year secrecy window, with estimate-side improvements accelerating the date. Therefore classical RSA does *not* protect these records; probability-weighted, exposure is near-certain within their lifetime. Policy line: migrate hospital systems and re-encrypt archives to NIST post-quantum standards now (FIPS 203–205), treating RSA as legacy transport only.

Problem 20 [Challenge] (Model review.) “An honest small-scale Shor runs the full pipeline — superposition, modular exponentiation as circuits, inverse QFT, classical post-processing — with no step informed by the answer; a compiled demo hard-codes the period into the gates, reducing the ‘algorithm’ to printing a known result. This demo's period-dependent circuit fails that test. The checklist question that exposes it: *what grows when N grows?* — an honest implementation's modular-arithmetic circuits scale polynomially and dominate cost, while a compiled one must be redesigned per instance using the answer. Ask the vendor for their circuit

generator applied to a fresh, unfactored N . Verdict: demonstration of gate control, not of Shor's algorithm."

Problem 21 [Challenge] With wires (top, bottom) and $\text{QFT}_4 = \frac{1}{2} [i^{jk}]_{j,k=0}^3$: multiply $H(\text{top}) = H \otimes I$, then controlled-S (control bottom, target top — either labeling consistent with the chapter's circuit), then $I \otimes H$, then SWAP. Patient 4×4 bookkeeping yields $\frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -1 & 1 & -1 \\ 1 & -i & -1 & i \end{pmatrix}$ ✓ — five elementary gates equal a Fourier transform. (Grade the intermediate matrices; sign slips in the controlled-S column are the common error.)

Problem 22 [Challenge] (Model note.) "Factoring a 48-bit number on 10 qubits is plausible only for special-form integers or compiled circuits — general Shor on 48 bits needs $\sim 100+$ logical qubits for modular arithmetic. The claim does not extrapolate because the resource that explodes is error-corrected circuit *depth times width*: RSA-2048 requires thousands of logical qubits sustained through billions of T-gates, i.e. the $\sim$ million-physical-qubit machines of the published estimates. Nothing in a 48-bit demo tests that regime. The single measurement that would change my assessment: their machine's *logical* error rate per operation, at scale — the ledger number (Chapter 29) from which everything else follows."

CHAPTER TWENTY

Variational Algorithms, Optimization, and Quantum Machine Learning

Teaching notes, self-test, and complete solutions

TEACHING NOTES

The near-term algorithms chapter — and the course’s best laboratory for *scientific temperament*: everything here is measurable, budgetable, and honestly uncertain. Run the one-dial VQE loop live (the $E(\theta) = \cos \theta + 0.35 \sin \theta$ example): students turning a single knob and watching the energy floor appear learn more than a lecture on ansätze. Three load-bearing lessons deserve full time: the variational bound (energies never lie low — Problems 3, 7, 8), the measurement economics (grouping, shot allocation, budgets — Problems 14–17, the professional craft), and the trap catalogue (ansatz misses, barren plateaus, restarts — Problems 18, 19, 23). QAOA’s MaxCut is a lovely classroom game (score splits by hand) — and pair it with the “classical heuristics are ahead at scale” honesty. Problems 20–21 build the audit muscles the ML part will exercise. Stumbles: treating measured E as exact (always attach shot noise), and confusing the optimizer’s floor with the ansatz’s floor.

SELF-TEST — QUESTIONS WITH ANSWERS

Q1. The hybrid loop’s division of labor?

A. Quantum: prepare the parameterized state and measure the cost (energy) — the part classical machines cannot do. Classical: digest results, choose new dials, decide convergence.

Q2. State the variational principle and its practical gift.

A. $E_{\text{trial}} \geq E_0$ for every trial state: measured energies are upper bounds, so lower is provably better — progress is monotone truth.

Q3. How is a many-term energy actually measured?

A. Term by term: $\langle H \rangle = \sum c_i \langle P_i \rangle$, batching qubit-wise-commuting terms into shared measurement settings; shots per term set the error bars.

Q4. What is QAOA’s target problem shape?

A. Combinatorial optimization encoded as an Ising-style cost (MaxCut the flagship): superpose assignments, steer amplitude toward high-scoring patterns.

Q5. Name the three failure modes to check before trusting a variational result.

A. Ansatz can't reach the answer (bound floor above truth); barren plateau/shot noise (no resolvable gradient); optimizer stuck locally (restart spread).

Q6. Why do restarts make sense here more than in classical fitting?

A. The bound makes “keep the best” rigorous: the lowest converged energy is the closest-to-truth claim, never an overfit fluke.

Solutions to the Practice Problems

Problem 1 [Basic] Quantum partner: runs the ansatz at the current dials and returns measured expectation values (the energy estimate). Classical partner: the optimizer — compares, proposes new dials, declares convergence.

Problem 2 [Basic] $E(0^\circ) = 1$; $E(90^\circ) = 0.35$; $E(180^\circ) = -1$.

Problem 3 [Basic] The variational principle: a trial state's average energy is a weighted average of eigen-energies, all $\geq E_0$ — so no measurement of any ansatz state can average below the true ground energy.

Problem 4 [Basic] Links leaving node 0: 0-1 and 0-3: cut = 2 (the other four links stay inside the big camp).

Problem 5 [Basic] $\{ZI, IZ, ZZ\}$: yes — all diagonal, one standard-basis run serves all. $\{ZZ, XX\}$: no — Z and X clash on both qubits. $\{XI, IX, XX\}$: yes — one X-setting (H on both) serves all three.

Problem 6 [Basic] $\langle ZZ \rangle = (420 + 410 - 80 - 90)/1000 = 0.66$; $\langle ZI \rangle = (420 + 80 - 90 - 410)/1000 = 0.00$.

Problem 7 [Basic] Estimate: the best, $E = -0.55$ (the bound guarantees the true ground is at or below it). The tight repeats at -0.55 and -0.41 suggest two distinct local valleys, each found repeatedly — a rugged landscape; restart more around the -0.55 basin.

Problem 8 [Basic] By the variational principle a lower energy is a state provably closer (in energy) to the truth — there is no “fitting the noise” below the floor, because the floor is physical, not statistical.

Problem 9 [Standard] $E(\theta) = 0.5 \cos \theta - \sin \theta$. $E(90^\circ) = -1$; $E(120^\circ) = -0.25 - 0.866 = -1.116$; $E(150^\circ) = -0.433 - 0.5 = -0.933$. The minimum sits between 90° and 150° , near 120° (exact: $-\sqrt{1.25} = -1.118$ at $\theta \approx 117^\circ$).

Problem 10 [Standard] $\langle Z \rangle = 0.62 - 0.38 = 0.24$, wobble $\approx 1/\sqrt{1000} = 0.03$. Energy differences of 0.01 are below this noise — unresolvable at 1000 shots; distinguishing them reliably needs $\sim 10\times$ more shots (or smarter allocation).

Problem 11 [Standard] A link crosses iff its endpoints are in different camps — a property unchanged by swapping the camp labels: $\text{cut}(\text{split}) = \text{cut}(\text{swap})$. Consequence: a successful QAOA histogram shows two mirror peaks (e.g. 01010 and 10101), equal by symmetry — seeing both is a health check, not a bug.

Problem 12 [Standard] The gains have collapsed ($-1.058 \rightarrow -1.059$: one part in a thousand) while the known exact floor of the toy is -1.060 : within shot noise of done — stop, report $E = -1.059$ (bound: ground ≤ -1.059), and spend remaining budget on error bars, not iterations.

Problem 13 [Standard] (1) Variational classifiers/kernels on classical data: runs on today's machines; no proven advantage — the field's workhorse and its hype source. (2) Quantum-linear-algebra speedups (recommendation-style): partially *dequantized* by Tang — the cautionary tale. (3) Learning from *quantum* data: proven exponential advantages (2022 experiments) — most experts' safest bet.

Problem 14 [Standard] Two settings: $\{ZZ, ZI\}$ — plain standard-basis readout; estimate both from the same counts. $\{IX\}$ — H on qubit 2 (the X-compiler on that qubit alone), then standard readout of qubit 2's $\pm$ statistics.

Problem 15 [Standard] $\langle H \rangle_{|01\rangle} = 0.5(-1) + 0.3(0) + 0.2(+1) = -0.3$. Ranking toward the minimum: exact ground $-0.861 < \text{singlet } -0.8 < |01\rangle -0.3 < |\Phi^+\rangle +0.8$: third of four.

Problem 16 [Standard] Allocate 700 shots to the $\{ZZ, ZI\}$ setting and 300 to $\{XX\}$. Equal splitting over-polishes the light-weight term: error contributions scale as $|c| \times \sigma$, so shots should chase weight — precision where the Hamiltonian actually pays.

Problem 17 [Standard] One run: $60 \times 2000 \times 150 = 1.8 \times 10^7$ shots. Twelve restarts: 2.16×10^8 . At a 10^8 cap: $\lfloor 10^8 / 1.8 \times 10^7 \rfloor = 5$ restarts fit.

Problem 18 [Challenge] The real-amplitude ansatz has $\langle Y \rangle \equiv 0$: VQE dutifully reports minimum 0 — while the truth is -1 at $|-i\rangle$, unreachable. One added gate fixes it: an R_z (or S) after the R_y , giving the ansatz complex phases and access to the y -axis. Moral: VQE finds the lowest energy *within the ansatz's reach* — the bound is honest, but the reach is your design responsibility.

Problem 19 [Challenge] Resolving a gradient of 10^{-4} against shot noise $\approx 1/\sqrt{n}$ needs $1/\sqrt{n} \lesssim 10^{-4}$: $n \gtrsim 10^8$ shots *per evaluation* — days of machine time per optimizer step. Conclusion: “bigger ansatz” is self-defeating past the plateau threshold; the professional response is structure — shallow, problem-shaped circuits and smart initialization (Chapter 25's lesson) — not shot-budget heroics.

Problem 20 [Challenge] (Model protocol.) Instances: a public, pre-registered benchmark set (not instances the quantum side selected). Tuning: equal hyperparameter-search budget for QAOA *and* the classical heuristic, tuned by separate teams. Timing: end-to-end wall-clock including compilation, queue, shots, and post-processing — not per-iteration cherry-picks. Metric: best cut found within a fixed time budget, plus approximation ratio, over ≥ 30 instances with error bars. Repetitions: multiple seeds and hardware sessions, all reported. The omission that most flatters quantum: *an untuned classical baseline* — classical MaxCut heuristics are ferociously good when someone bothers to tune them.

Problem 21 [Challenge] (1) Baseline: what accuracy does a tuned classical CNN reach on the same split? (2) Encoding: how were X-rays loaded into qubits, and what did that stage cost (Chapter 22's invoice)? (3) End-to-end timing: shots, queue, and repetitions included — what is the wall-clock per image? (4) Simulability: does a laptop simulation of the same circuit reproduce the result (if yes, nothing quantum happened)? The baseline answer changes assessments most: 98% means nothing until the classical number beside it exists.

Problem 22 [Challenge] $(H \otimes H) |\Phi^+\rangle = \frac{1}{\sqrt{2}}(|++\rangle + |--\rangle)$; expanding $|\pm\pm\rangle$ into the standard basis, cross terms cancel and $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ returns: the state is unchanged ✓ Since $HZH = X$, measuring Z-agreement after the H's is measuring X-agreement before them: standard-basis correlation of the rotated state $= \langle XX \rangle$ of the original — the X-setting recipe, proven.

Problem 23 [Challenge] (Model paragraph.) Converged-but-spread: diagnosis local minima; test: cluster the restart energies (distinct tight clusters); spend: more restarts seeded near the best cluster. Non-converging noisy traces: diagnosis shot noise swamping the landscape; test: repeat one evaluation at fixed dials — if repeats scatter as $1/\sqrt{n}$ predicts, noise is the story; spend: more shots per evaluation (or grouping). Flat gradients from step one: diagnosis barren plateau / over-deep ansatz; test: measure the gradient norm at several random dial settings (Chapter 25's pre-flight); spend: neither restarts nor shots — *shrink and restructure* the ansatz, then retry. The rule: diagnose before purchasing, because each remedy is waste for the other two diseases.

Quantum Simulation: The Original Killer App

Teaching notes, self-test, and complete solutions

TEACHING NOTES

The founding application, taught as countable engineering. The single-qubit Trotter worked example is the chapter's spine — students can *check* it (Appendix E makes it a 15-line script; the app activity plots the $1/n$ error shrinkage). Insist on the two-sided honesty: simulation is the best-founded quantum application (Lloyd's theorem, FeMoco's humanitarian stakes) *and* its bills only clear on error-corrected machines — the magnet-bill worked example makes both halves numerical. The commutator-as-price idea ($\|[A, B]\|$ in the bound) rewards a blackboard moment: commuting terms cost nothing, and that is why grouping mattered in Chapter 20 too. The analog-simulator lesson prevents a common blind spot (special-purpose machines are real science now). Stumbles: dropping the t^2 in the error bound (time is quadratic!), and forgetting that parallel columns cut depth, not gate count.

SELF-TEST — QUESTIONS WITH ANSWERS

Q1. Feynman's argument in one line?

A. Nature runs quantum dynamics that classical bookkeeping cannot (2^n amplitudes); a controllable quantum system can host the bookkeeping natively.

Q2. What unitary does simulation compile, and from what data?

A. e^{-iHt} , from H 's Pauli-term shopping list — the same list VQE measures, now evolved instead.

Q3. The Trotter recipe and its error?

A. Alternate small doses: $(e^{-iAt/n}e^{-iBt/n})^n$; error $\lesssim t^2\|[A, B]\|/(2n)$ — halved by doubling the slices, priced by the terms' refusal to commute.

Q4. Why do commuting terms cost nothing?

A. $[A, B] = 0$ makes the product exact: $e^{-i(A+B)t} = e^{-iAt}e^{-iBt}$ — one rotation per term, no slicing.

Q5. Digital vs analog simulators?

A. Digital: gate circuits, universal, error-correctable, expensive. Analog: the hardware *is* the model — hundreds of particles today, special-purpose, limited measurement.

Q6. The honest prize list's top entry?

A. Strongly correlated chemistry and materials — FeMoco-class enzymes, exotic magnets — where classical methods fail qualitatively; weakly correlated chemistry stays classical.

Solutions to the Practice Problems

Problem 1 [Basic] When $[A, B] = 0$ the exponentials' power series multiply exactly as for numbers ($e^{a+b} = e^a e^b$ needs commuting exponents). The single-term evolutions are the rotation gates: $e^{-i\theta Z/2} = R_z(\theta)$ and $e^{-i\theta X/2} = R_x(\theta)$ (Chapter 7).

Problem 2 [Basic] $0.4/n \leq 0.01 \Rightarrow n = 40$ slices; $0.001 \Rightarrow n = 400$.

Problem 3 [Basic] All three act on disjoint qubits — $(1, 2)$, $(3, 4)$, $\{5\}$ — so all three share one parallel column. Better: they pairwise *commute* (disjoint supports), so no Trotterization is needed at all: one exact column, e^{-iHt} in a single slice.

Problem 4 [Standard] Error $\propto t^2/n$: doubling t at fixed n multiplies the bound by 4. Restoring the error needs $n \rightarrow 4n$ — and the gate bill (slices $\times$ terms) also quadruples: simulated time is bought at quadratic gate cost in this first-order scheme.

Problem 5 [Standard] Per slice: $99 \times 3 + 100 = 397$ gates; total $200 \times 397 \approx 79,400$ gates, of which $99 \times 2 \times 200 = 39,600$ are CNOTs. Depth: ≈ 7 per slice (odd couplings, even couplings, field column) $\rightarrow \approx 1400$. Survival at $p_L = 10^{-6}$: $e^{-79400 \times 10^{-6}} = e^{-0.079} \approx 92\%$ — comfortably runnable on an error-corrected machine, hopeless raw.

Problem 6 [Standard] The first CNOT writes the two qubits' parity onto the target ($t \rightarrow t \oplus c$); the R_z then phases by an angle whose *sign* depends on that parity bit ($|0\rangle$ vs $|1\rangle$ of the target); the second CNOT uncomputes the parity. Net effect: phase $e^{\mp i c t/n}$ on even/odd parity states — exactly $e^{-i(ct/n)ZZ}$, a two-qubit rotation from one single-qubit dial and two entanglers (kickback bookkeeping, Chapter 10 style).

Problem 7 [Standard] Better from the analog machine: equilibrium and dynamical behavior of *that specific model* at sizes (300 spins) digital machines cannot touch — e.g. watching magnetic domains form in real time. Impossible for it: anything off-model — changing the Hamiltonian's form, applying arbitrary gates, or measuring arbitrary observables (it reads only what its lasers can see). Moral: special-purpose machines trade universality for scale — superb instruments, not computers.

Problem 8 [Standard] Falls come from better algorithms (smarter encodings, qubitization-style methods replacing naive Trotter) and tighter error analysis (proving the old bounds were pessimistic — less overhead needed for the same accuracy). Identical dynamics drove factoring's $2 \times 10^7 \rightarrow < 10^6$ (Chapter 19): requirements are theorems plus engineering, and both improve — which is why resource estimates are dated documents, not constants.

Problem 9 [Challenge] First order: $n = 400 \text{ slices} \times 2 \text{ rotations} = 800 \text{ rotations}$. Second order: $0.4/n^2 \leq 0.001 \Rightarrow n = 20 \text{ slices} \times 1.5 \text{ rotations} = 30 \text{ rotations}$ — roughly $27\times$ cheaper for the same accuracy. Higher-order slicing is why real simulation papers never use the plain first-order formula at tight tolerances.

Problem 10 [Challenge] $[A, B] = 0.4(XZ - ZX) = 0.4(-iY - iY) = -0.8 iY$: $\|[A, B]\| = 0.8$ ✓ Two commuting Z -type terms (e.g. $0.5 Z_1 + 0.8 Z_2$, or even Z and ZZ) have $[A, B] = 0$: the bound is zero, one slice is exact, and the chapter's whole apparatus becomes unnecessary — non-commutativity is the problem being managed.

Problem 11 [Challenge] All the required observables ($Z_1, \dots, Z_{50}$ and $Z_1 Z_2$) are diagonal: **one** standard-basis setting serves everything — read all 50 bits per shot, compute each $\langle Z_i \rangle$ and the pair correlation from the same records. Budget: each $\langle Z_i \rangle$ needs wobble ≤ 0.01 : $n \approx (1/0.01)^2 = 10^4$ shots (the magnetization, an average of 50 such, comes out even sharper). Total: 10^4 shots, one setting — measurement design at its cheapest, because everything commuted.

Problem 12 [Challenge] (Model questions.) (1) *Classical bar*: which classical method (exact statevector reaches ~ 50 spins; tensor networks far more for low entanglement) was compared, tuned by whom? (2) *Fidelity bar*: what circuit depth did the simulation need, and what does the device's error budget $((1 - p_2)^{n_2}$, Chapter 28) say survives — with what measured evidence (echo/benchmark circuits)? (3) *Observable bar*: were the reported quantities verified against exactly solvable limits or small instances? (4) *Model bar*: is the simulated Hamiltonian one whose entanglement actually grows (else tensor networks eat it)? The claim stands only if the classical baseline was strong, the fidelity evidence direct, the checks passed, and the model genuinely hard — most 2020s “beyond classical” simulation claims fell at (1) or (4).

CHAPTER TWENTY-TWO

Encoding Data into Quantum States

Teaching notes, self-test, and complete solutions

TEACHING NOTES

Part Seven opens with its least glamorous, most decisive topic: how data gets in. Drill the three encodings until students can price any dataset on sight (the invoice worked example is the template; Problems 3, 5, 12 are the drills). Two traps deserve classroom theater: the wrap-around/aliasing trap (encode 0 and 2π live, watch them collide — Problems 11, 17) and the loading-cost trap behind amplitude encoding’s “exponential compression” (Problems 19–20 are the audit muscles). The scaling lesson’s leakage rule (ranges from training data only) is professional ML hygiene many students meet here first — enforce it in every later project. The entangling feature-map challenge (Problem 18) bridges to kernels; assign it to your strongest students as presentation material. Stumbles: degrees-vs-radians slips, and forgetting that amplitude encoding requires normalization (scale information dies).

SELF-TEST — QUESTIONS WITH ANSWERS

Q1. The three encodings’ qubit invoices for d features?

A. Basis: one qubit per *bit*; angle: d qubits (one rotation each); amplitude: $\lceil \log_2 d \rceil$ qubits — with a loading circuit that may cost $O(d)$ gates.

Q2. Angle-encode x : which gate, and what state?

A. $R_y(x) |0\rangle = \cos \frac{x}{2} |0\rangle + \sin \frac{x}{2} |1\rangle$ — the feature becomes a latitude.

Q3. Why scale features into $[0, \pi]$?

A. Rotations live on a circle: unscaled features can differ by 2π and encode to the *same* state (aliasing); half a turn keeps every distinct value distinguishable.

Q4. What is data leakage in the scaling step?

A. Computing $x_{\min}, x_{\max}$ (or any preprocessing) on test data — information flowing from test to training-time choices, inflating reported accuracy.

Q5. Amplitude encoding’s fine print?

A. The vector must be normalized (overall scale is lost), and loading an arbitrary vector needs a circuit that grows

like the data size — compression in storage, not necessarily in time.

Q6. What is the “input problem”?

A. Claimed quantum speedups that ignore data-loading cost: if loading is linear (or exponential), a fast quantum core cannot rescue the end-to-end time.

Solutions to the Practice Problems

Problem 1 [Basic] $|1101\rangle$ — four qubits, one per bit.

Problem 2 [Basic] $R_y(120^\circ)|0\rangle = \cos 60^\circ|0\rangle + \sin 60^\circ|1\rangle = 0.5|0\rangle + 0.866|1\rangle$; $P(0) = 0.25$, $P(1) = 0.75$.

Problem 3 [Basic] $256 = 2^8$: 8 qubits. A million values: $\lceil \log_2 10^6 \rceil = 20$ qubits.

Problem 4 [Basic] $\theta(v) = \frac{v}{5,000,000} \pi$. For $v = 2,500,000$: $\theta = \pi/2 = 90^\circ$.

Problem 5 [Basic] Basis: $64 \times 8 = 512$ qubits. Angle: 64 qubits. Amplitude: $\lceil \log_2 64 \rceil = 6$ qubits (plus its loading bill).

Problem 6 [Basic] Norm = 5: state $0.6|0\rangle + 0.8|1\rangle$; probabilities 36%/64%.

Problem 7 [Basic] Range 30: $5 \rightarrow 0^\circ$; $20 \rightarrow \frac{15}{30} \times 180^\circ = 90^\circ$; $35 \rightarrow 180^\circ$.

Problem 8 [Basic] Naive: $\frac{41-5}{30} \times 180^\circ = 216^\circ$ — outside the safe window, where the circle’s geometry starts folding distinct values together (and the model never trained there). Clipping maps it to the window edge 180° , an honest “at or beyond the maximum seen” statement.

Problem 9 [Standard] $\|x\|^2 = 1 + 4 + 4 + 16 = 25$: amplitudes $\frac{1}{5}(1, 2, 2, 4) = (0.2, 0.4, 0.4, 0.8)$; probabilities $0.04 + 0.16 + 0.16 + 0.64 = 1$ ✓

Problem 10 [Standard] Same longitude ($x_2 = 0$), latitudes 30° and 150° : states $\cos 15^\circ|0\rangle + \sin 15^\circ|1\rangle$ and $\cos 75^\circ|0\rangle + \sin 75^\circ|1\rangle$. Overlap: $\cos^2\left(\frac{150^\circ-30^\circ}{2}\right) = \cos^2 60^\circ = 0.25$ ✓ Chapter 24 (quantum kernels) turns exactly this overlap into the similarity score K .

Problem 11 [Standard] $R_y(x + 2\pi) = R_y(x) R_y(2\pi)$, and $R_y(2\pi) = -I$: the states differ by a global -1 — physically identical (Chapter 5). Rule forced: scale each feature so its *entire data range* spans less than 2π (conservatively $[0, \pi]$), making the encoding injective on the data.

Problem 12 [Standard] (a) 8 qubits, depth 1 — trivial loading, the near-term default. (b) 4 qubits, depth 2 — same information, half the width; slightly entangling the two features’ readout. (c) $\lceil \log_2 8 \rceil = 3$ qubits — smallest, but the loader for arbitrary 8-vectors costs a dozen-plus gates and normalization eats the scale: pay only if qubits are the scarce resource.

Problem 13 [Standard] At $x = 60^\circ$: amplitudes $\cos 30^\circ = 0.866$, $\sin 30^\circ = 0.5$: $P(0) = 0.75$, $P(1) = 0.25$, $\langle Z \rangle = 0.75 - 0.25 = 0.5 = \cos 60^\circ$ ✓

Problem 14 [Standard] Angle: 12 qubits, depth-1 single-qubit layer, zero CNOTs — survival essentially set by readout alone ($\approx (0.988)^{12} \approx 87\%$, Chapter 30's numbers). Amplitude: 4 qubits but an arbitrary-16-vector loader costs on the order of 30 CNOT-equivalents: $(0.994)^{30} \approx 84\%$ *before* the model's own circuit, plus normalization loss and design complexity. Verdict: angle encoding — qubits are plentiful (27 available), fidelity is not.

Problem 15 [Standard] Amplitudes must satisfy $\sum |a_i|^2 = 1$ — the Born rule leaves no freedom. Two points differing only by overall scale encode to the *same* state: magnitude information is lost. Harmless: data already meaningful up to scale (word-frequency profiles, direction vectors). Harmful: incomes, sensor magnitudes — anywhere the size is the signal (then append the norm as an extra feature, the standard patch).

Problem 16 [Standard] Angle: qubits bind first — $d = 100$ hits the 100-qubit wall (gates, at depth 1, never bind). Amplitude: qubits bind only at astronomical $d = 2^{100}$; the $O(d)$ loading *depth* binds at roughly $d \approx 200$. So: angle for $d \leq 100$; amplitude buys $100 < d \lesssim 200$ on this machine; beyond that, classical compression first (Problem 21).

Problem 17 [Standard] Time is genuinely periodic — 23:59 and 00:01 are neighbors — so wrap-around is *correct* physics for it: scale $\theta = \frac{v}{24} \times 360^\circ$. Encodings: 23:00 $\rightarrow 345^\circ$; 01:00 $\rightarrow 15^\circ$: angular gap 30° , kernel $\cos^2 15^\circ = 0.93$ — near neighbors, as they should be (linear $[0, \pi]$ scaling would have put them at opposite ends). Periodic features are the one licensed use of the full circle.

Problem 18 [Challenge] After the encodings, the CNOT- $R_z(x_1x_2)$ -CNOT sandwich phases components by $e^{\mp i x_1 x_2 / 2}$ depending on the two qubits' *parity* (the ZZ-rotation mechanics of Chapter 21, Problem 6): at (π, π) a relative phase $e^{i\pi^2/2 \cdot (\pm)}$ appears between even- and odd-parity components. The circuit thus computes a feature proportional to the *product* x_1x_2 — unreachable by any pair of independent single-qubit encodings, whose phases are sums, never products. Entanglement in the feature map = interaction terms in the feature space: the ZZ-feature-map idea of Chapter 24.

Problem 19 [Challenge] Loading: $10^{12}/10^9 = 1000$ s. Quantum core: $\sqrt{10^{12}} = 10^6$ ops $\rightarrow 10^{-3}$ s. End-to-end: ≈ 1000 s, loading-dominated. Classical linear scan: $10^{12}/10^9 = 1000$ s — *a tie*; even $N \log N$ (4×10^4 s) is only $40\times$ slower, not $10^6\times$. Verdict: “The $\sqrt{N}$ advantage exists only downstream of an input the algorithm cannot afford; end-to-end, the quantum pipeline matches a classical scan. Claims must be priced from data-at-rest to answer — the input problem in one line.”

Problem 20 [Challenge] (Model paragraph.) Reading amplitudes back would need $\sim 2^n$ measurements — true, and not the point. Useful operations act on the *whole vector at once*: interfere two encoded vectors and measure one overlap ($|\langle \phi(x) | \phi(y) \rangle|^2$ — a kernel entry, Chapter 24); apply a unitary that concentrates amplitude on a property of the vector and sample it; estimate a handful of expectation values (global summaries) with polynomially many shots. Amplitude encoding is a medium for *holistic* linear algebra — overlaps, projections, evolutions — never a memory you read element by element; algorithms that respect this distinction survive

audits, and those that quietly assume readback do not.

Problem 21 [Challenge] Pipeline: PCA ($O(1024^2)$ classical arithmetic per point, microseconds) $\rightarrow$ 8 angles $\rightarrow$ 8 qubits, depth 1-3 with entanglers: trivially runnable. But the heavy lifting — crushing 1024 dimensions to the 8 most informative — happened classically; the quantum stage sees only what PCA kept, so any claimed advantage must live in the 8-feature residual problem, which small classical models also handle well. Settling experiment: the ablation — same pipeline with the quantum stage swapped for a tuned classical model of equal input; if accuracies tie, the quantum stage added nothing (Chapter 27's verdict template).

Problem 22 [Challenge] (1) “Which split computed the scaling ranges?” — exposes leakage: test-informed ranges inflate accuracy (the definition, verbatim). (2) “Is each feature's span matched to its encoding's period?” — exposes wrap-around aliasing: distinct values silently identified, or periodic features broken by linear scaling. (3) “What happens to test points outside the training range?” — exposes undefined behavior at deployment: no clipping policy means the model's first out-of-range customer produces an unvetted angle. Each question has a one-line honest answer; a paper missing all three has not engineered its first layer.

Variational Quantum Classifiers

Teaching notes, self-test, and complete solutions

TEACHING NOTES

The chapter where a quantum circuit *learns* — and where this book’s rings demo gives every claim a reproducible anchor. Teach the three-stage pipeline (encode, process, read out) as a re-labeled VQE loop; students who traced Chapter 20 will feel at home. The hand-drawable decision-boundary lesson is the conceptual gift: one cosine, one threshold, the whole philosophy of classifiers visible at once — extend it live by sliding and stretching the wave (Problems 6, 14, 15). Give the deployment lessons their due: thresholds and imbalance (the accuracy trap, Problems 7, 8, 17 — assign at least one; it is the most transferable skill in Part Seven) and shot-noise-aware inference (Problems 11, 16, 22). The overfitting diagnosis (Problem 10) previews Chapter 25’s capacity dial. Stumbles: quoting accuracy on imbalanced data, and forgetting that near-boundary scores need shots before their sign means anything.

SELF-TEST — QUESTIONS WITH ANSWERS

Q1. The VQC’s three stages?

A. Encode the features (Chapter 22’s rotations), process with trainable entangling layers, read out a score $s = \langle Z \rangle \in [-1, 1]$; the sign is the label.

Q2. What loss is minimized, and by which gradients?

A. A squared (or similar) loss between scores and $\pm$ targets, with parameter-shift gradients ($\pm 90^\circ$ rule) — the VQE loop chasing labels instead of energies.

Q3. Where is a one-layer model’s decision boundary, structurally?

A. Where its single cosine crosses zero — one threshold per half-period; weights stretch the wave, biases slide it.

Q4. Why is accuracy a trap on rare-event data?

A. The do-nothing model scores 99% on 1% prevalence; quote recall and false-alarm rate (or precision) and set the threshold by costs instead.

Q5. What does re-uploading buy?

A. Each re-encoding layer adds a harmonic: more wiggles, more boundaries, richer shapes — the frequency budget

of Chapter 25.

Q6. Why do borderline predictions cost more shots?

A. The score's sign must beat shot noise $\sim 1/\sqrt{n}$: near-zero scores need hundreds of shots for a statistically honest decision; confident scores need a dozen.

Solutions to the Practice Problems

Problem 1 [Basic] Encoding (Chapter 22), variational processing layers (Chapter 20's ansatz idea), measurement readout as expectation value (Chapter 8).

Problem 2 [Basic] Labels $+, -, -$; confidence order: -0.61 (most), $+0.72$... careful — confidence is $|s|$: ranking $0.72 > 0.61 > 0.05$: the $+0.72$ point first, then -0.61 , then the near-boundary -0.05 (barely a decision at all).

Problem 3 [Basic] Score $+0.5$: $(0.9 - 0.5)^2 = 0.16$. Score -0.5 : $(0.9 + 0.5)^2 = 1.96$ — the wrong-side error costs $12.25\times$ more: the loss punishes misclassification far beyond mere under-confidence.

Problem 4 [Basic] $5 \times 5 = 25$ dials.

Problem 5 [Basic] $s(0.1) = \cos 18^\circ = +0.95$: class $+$, very confident. $s(0.9) = \cos 162^\circ = -0.95$: class $-$, equally confident.

Problem 6 [Basic] Boundary: $180^\circ x - 45^\circ = 90^\circ \Rightarrow x = 0.75$. The -45° slid the boundary *rightward* by $45/180 = 0.25$.

Problem 7 [Basic] Accuracy $= 0.98 \times 0.92 + 0.02 \times 0.90 = 0.9196 \approx 92\%$; do-nothing: 98%. The clinic should quote recall (90%) and false-alarm rate (8%) — the numbers that describe what the model *does* — never bare accuracy.

Problem 8 [Basic] Per 1000 transactions: 10 frauds, 8 caught (recall = 80%); false alarms $= 0.05 \times 990 = 49.5$; total alarms 57.5; precision $= 8/57.5 \approx 14\%$ — six of seven alarms are false, which is normal at 1% prevalence and exactly why alarm-handling capacity, not accuracy, drives deployment.

Problem 9 [Standard] A single encoding gives the score one harmonic of each feature — one wiggle, one boundary per half-period. Re-encoding between layers multiplies phases that add, so L uploads give up to L harmonics: the function family grows from single cosines toward truncated Fourier series. Complex boundaries are higher harmonics; without re-uploading, no amount of dial-tuning can add them.

Problem 10 [Standard] Gap of 27 points: overfitting — the model memorized training accidents. Treatments from this chapter: reduce capacity (fewer layers = fewer harmonics) and get more data (or augment); honorable mention: stronger regularization/early stopping via a validation split (Chapter 25 formalizes the capacity dial).

Problem 11 [Standard] Noise 0.05 against a true score 0.06: a wrong-sign estimate needs only a 1.2σ fluctuation — probability $\approx 11\%$: entirely plausible. For noise $= s/3 = 0.02$: $n = (1/0.02)^2 = 2500$ shots.

Problem 12 [Standard] (Model.) Train a tuned RBF-kernel SVM and a small MLP on the identical 42/18 split, with a hyperparameter search budget matching the quantum model's restart-and-tune budget, scoring by the same test accuracy with seeds and error bars. Honest prediction: the classical models also reach $\approx 100\%$ on rings — the dataset is easy; the defensible quantum claim is parity at 15 parameters, not superiority (the chapter's own audit says so).

Problem 13 [Standard] If the data is centered so that $(0, 0)$ is deep *inside* one class (as with rings), the origin probe only exercises one region — fine for a bias check, blind for the rest. If $(0, 0)$ were a class boundary point or far outside the data range, its score would be uninformative or untrained. Better probe: a training point of known label near the boundary (maximum diagnostic sensitivity), plus one deep-interior point per class.

Problem 14 [Standard] Boundaries where $\cos(360^\circ x) = 0$: $x = 0.25$ and $x = 0.75$. Pattern: $+$ on $[0, 0.25)$, $-$ on $(0.25, 0.75)$, $+$ on $(0.75, 1]$ — the label sequence $+, -, +$ (an enclave), impossible for the single-boundary 180° model. Frequency 2 = two boundaries.

Problem 15 [Standard] Zeros of $\cos(270^\circ x - 60^\circ)$: $270^\circ x - 60^\circ = 90^\circ + k 180^\circ$: $x = \frac{150^\circ}{270^\circ} = 0.56$ (next solution 1.22, outside). One boundary at $x \approx 0.56$; sign pattern: $s(0) = \cos(-60^\circ) = +0.5$: $+$ before, $-$ after.

Problem 16 [Standard] Near a boundary x_0 , $s(x) \approx s'(x_0)(x - x_0)$: confidence grows linearly from zero. A decision is honest only when $|s| \gtrsim 2\sigma = 1/\sqrt{n}$: points close to the boundary (small $|s|$) need $n \gtrsim 1/s^2$ shots — quadratically more as the margin shrinks; far points are decided in dozens.

Problem 17 [Standard] Before: misses $10 \times 20\% = 2 \rightarrow$ cost 200; false alarms $49.5 \rightarrow$ cost 49.5: total ≈ 250 . After: misses $10 \times 8\% = 0.8 \rightarrow 80$; false alarms $0.11 \times 990 = 108.9$: total ≈ 189 . The lower threshold wins at these costs — the cut belongs to the cost model, not to the number zero.

Problem 18 [Challenge] (Model paragraph.) Along a center line, “inside ring” is an interval flanked by “outside” on *both* sides: the label pattern $-, +, -$ needs the score to cross zero twice. A single harmonic $A \cos(wx + c) + B$ on the data's (sub-period) window crosses zero at most... with the offset B it can manage two crossings only by using nearly a full period — which the one-layer frequency budget ($|w| \leq 1$ effective upload) does not reach within the scaled window; within reach it is monotone-then-monotone: one crossing. A second upload doubles the frequency budget, buying the second crossing — the enclave becomes expressible (and the trained three-layer model indeed draws it). Full credit for any argument connecting crossings-needed (2) to harmonics-available (1 vs 2).

Problem 19 [Challenge] Sub-model k trains on labels “ $+1$ if class k , else -1 ” (same data three times, re-labeled). Inference: run all three, predict $\arg \max_k s_k$. Tie failure: two scores equal (often both negative — “none of us” regions); tie-breaker: prefer the sub-model with better validation precision, or fall back to the nearest-training-neighbor's class — and log ties as low-confidence for human review (the Chapter's threshold discipline, promoted to multi-class).

Problem 20 [Challenge] (Model protocol.) Each prediction encodes the point and runs the fixed 15-dial circuit; shots: 500 baseline, escalating to 2500 when $|s| < 0.1$ (Problem 11's arithmetic keeps sign errors under a percent). Readout correction: apply Chapter 28's calibration inversion to each histogram before scoring. Throughput: at 10^4 shots/s, ≈ 70 baseline predictions per minute, ≈ 4000 /hour with escalation mix. Certification: report test accuracy *on-hardware* with error bars over seeds/calibrations, and the escalation rate — the two numbers that let a customer audit both quality and cost.

Problem 21 [Challenge] Zeros of $\cos(w180^\circ x + b)$ occur every half-period $= \frac{180^\circ}{|w|180^\circ} = \frac{1}{|w|}$ in x : on $[0, 1]$ at most $\lceil |w| \rceil \leq L$ crossings. Connection: re-uploading L times raises the reachable frequency to L (Chapter 25's budget), hence up to L boundaries per unit interval — layers literally buy boundary-drawing capacity, one wiggle per upload.

Problem 22 [Challenge] (Model policy.) First look: 50 shots, $\sigma_{50} = 0.071$. Rule: if $|s_{50}| \geq 3\sigma_{50} \approx 0.21$: decide now (sign error $< 0.2\%$). Else escalate to n with $1/\sqrt{n} \leq |s_{50}|/3$, capped at 2500 shots (deciding “low confidence” if still unresolved — route to review). Average cost for uniform scores: $\approx 79\%$ of points decide at 50 shots; the $\approx 21\%$ borderline stratum averages ~ 1500 more: expectation $\approx 50 + 0.21 \times 1500 \approx 370$ shots per transaction — a $7\times$ saving over flat 2500-shot inference at equal error, which is the entire point of adaptive allocation.

CHAPTER TWENTY-FOUR

Quantum Kernels

Teaching notes, self-test, and complete solutions

TEACHING NOTES

The elegant half of QML: no training loop on the quantum side, convex optimization on the classical side, and one honest quantum job — measuring overlaps. Anchor everything on the closed form $K = \cos^2(\Delta/2)$ (Problems 2, 21): a kernel students can compute by hand inoculates against treating the machinery as magic. Give the two economics lessons full weight: the quadratic measurement bill (Problems 4, 10, 15) and the support-vector inference invoice (Problems 8, 17, 22) — the numbers that decide deployments. The health-check table converts theory into a five-minute professional routine; run it on the book's real rings matrix. Concentration (Problem 19) is the kernel-side barren plateau: same disease, same cure (structured, shallow maps). Stumbles: forgetting the dagger in the overlap circuit, and reporting kernel values without shot-noise bars.

SELF-TEST — QUESTIONS WITH ANSWERS

Q1. What is a quantum kernel, in one line?

A. A similarity score $K(x, y) = |\langle \phi(x) | \phi(y) \rangle|^2 \in [0, 1]$ between two data points' encoded states — estimated as a probability on hardware.

Q2. How is one entry measured?

A. The overlap circuit: run $U(x)$ then $U(y)^\dagger$ on $|0 \cdots 0\rangle$; the probability of reading all-zeros is $K(x, y)$.

Q3. Where does learning happen in a kernel method?

A. Classically: an SVM ingests the measured kernel matrix and solves a convex problem — no quantum training loop, no plateaus.

Q4. The kernel bill for N training points?

A. $N(N-1)/2$ entries, each a shot-budgeted circuit — quadratic in dataset size; inference adds (support vectors) $\times$ (shots) per new point.

Q5. Two sick-matrix signatures?

A. All off-diagonals near 1: encoding barely moves states (too weak). All near 0: concentration — over-rich map,

memorization risk; simplify.

Q6. When is a quantum kernel worth trying?

A. When a structured quantum feature map plausibly matches the data (symmetries, interactions) and the dataset is small enough for the quadratic bill — with the tuned classical kernel always run alongside.

Solutions to the Practice Problems

Problem 1 [Basic] Self-similarity is perfect: a state overlaps itself completely ($|\langle\phi|\phi\rangle|^2 = 1$). Similarity has no direction: $|\langle\phi(x)|\phi(y)\rangle|^2 = |\langle\phi(y)|\phi(x)\rangle|^2$ since conjugation preserves modulus (Chapter 3's fidelity symmetry).

Problem 2 [Basic] $K = \cos^2\left(\frac{90^\circ - 30^\circ}{2}\right) = \cos^2 30^\circ = 0.75$.

Problem 3 [Basic] The all-zeros outcome: $P(0 \cdots 0) = |\langle 0 \cdots 0 | U(y)^\dagger U(x) | 0 \cdots 0 \rangle|^2 = |\langle\phi(y)|\phi(x)\rangle|^2$. The dagger makes the second block *un-prepare* $\phi(y)$: only the part of $\phi(x)$ lying along $\phi(y)$ returns to all-zeros.

Problem 4 [Basic] $\binom{20}{2} = 190$ pairs.

Problem 5 [Basic] $K(\pi/2) = \cos^2(\pi/4) = \frac{1}{2}$; $K(\pi) = \cos^2(\pi/2) = 0$. $K = 0$ means the encoded states are *orthogonal* — antipodal Bloch points, perfectly distinguishable.

Problem 6 [Basic] Diagonal must be exactly 1 (a state's overlap with itself); 0.97–0.99 signals estimation bias — on hardware, *readout error* (Chapter 28) taxing the all-zeros probability of every entry. Fix: readout mitigation (Chapter 14), or normalize by the measured diagonal.

Problem 7 [Basic] $f = 0.8(0.9) - 0.5(0.95) - 0.1 = 0.72 - 0.475 - 0.1 = +0.145$: still class +1, but the margin collapsed from 0.42 to 0.15 — the point now resembles the negative support vector too, and shot noise (± 0.02) starts to matter.

Problem 8 [Basic] Far-from-boundary points are classified correctly by their neighbors' influence alone — the optimizer needn't spend weight on them ($\alpha = 0$); only boundary-hugging points constrain the separator. All 200 points as support vectors suggests no margin structure at all — typically a concentrated (near-identity) kernel forcing pure memorization: run the health check.

Problem 9 [Standard] Bloch picture: two equator arrows 90° of longitude apart subtend 90° ; fidelity = $\cos^2(90^\circ/2) = \frac{1}{2}$. Explicit: $|\phi(a)\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ (up to phases), $|\phi(b)\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle)$: inner product $\frac{1+i}{2} \cdot (\text{phase})$, modulus² = $\frac{1}{2}$ ✓

Problem 10 [Standard] Precision 0.02 at worst-case variance: $n \approx (0.5/0.02)^2 = 625$ shots per entry. Total: $190 \times 625 \approx 1.2 \times 10^5$ shots; at 10^4 /s: about 12 seconds — pleasantly cheap at 20 points, and quadratically less pleasant later (Problem 15).

Problem 11 [Standard] Within-class and cross-class similarities nearly coincide (0.55 vs 0.53): the map barely separates the classes — the “no structure” row of the health check, resembling incipient concentration (everything similarly similar). Worth testing: rescale features (wider angular spread), and add an entangling interaction term (ZZ-style, Chapter 22’s Problem 18) to give the map class-relevant geometry.

Problem 12 [Standard] Predict class B; vote margin $0.71 - 0.62 = 0.09$. A trained SVM weights *support vectors* rather than class averages — a few decisive boundary examples (with large α) can outvote a diffuse average, flipping close calls exactly like this one.

Problem 13 [Standard] Kernel method: $\binom{100}{2} = 4950$ distinct overlap circuits, measured once (shot bill up front); no quantum training loop — the SVM trains classically on the fixed matrix. VQC: one circuit template, run millions of times inside the gradient loop (shot bill spread across training); the quantum computer participates in every optimization step and in every inference.

Problem 14 [Standard] $x = 10$ and $x = 16.28$ differ by 2π : identical encoded states, $K = 1$ — two different readings the model constitutionally cannot separate. Recipe: min-max scale the column into $[0, \pi]$ (Chapter 22), reserving full-circle scaling for genuinely periodic features.

Problem 15 [Standard] $\binom{200}{2} = 19,900$ entries $\times 500$ shots $\approx 10^7$ shots — versus the 60-point set’s $1770 \times 500 \approx 9 \times 10^5$: an $11 \times$ jump for a $3.3 \times$ dataset. Scaling law: entries (and shots) grow as N^2 — the kernel method’s structural bill.

Problem 16 [Standard] Healthy: diagonal-adjacent values spread over a broad range (0.996 near, 0.367 far) — similarity decaying with feature distance, exactly what a working ruler does. Uniform $K \approx 0.001$ off-diagonal would be concentration: every point orthogonal to every other, the matrix $\approx$ identity, the model reduced to memorizing labels — simplify the map.

Problem 17 [Standard] Wobble on f bounded by $\sum \alpha_i \times 0.5/\sqrt{n} = 4 \times 0.5/\sqrt{n} \leq 0.02 \Rightarrow \sqrt{n} \geq 100 \Rightarrow n = 10^4$ shots per entry; per decision: $25 \times 10^4 = 2.5 \times 10^5$ circuit runs. (At 10^4 shots/s: 25 s per classification — the invoice that decides whether this deploys.)

Problem 18 [Challenge] The final state is $U(y)^\dagger U(x) |0 \cdots 0\rangle$. By the Born rule in inner-product form, $P(0 \cdots 0) = |\langle 0 \cdots 0 | U(y)^\dagger U(x) |0 \cdots 0\rangle|^2$. Move the dagger: $\langle 0 \cdots 0 | U(y)^\dagger = (U(y) |0 \cdots 0\rangle)^\dagger = \langle \phi(y) |$, so the amplitude is $\langle \phi(y) | \phi(x) \rangle$ and its squared modulus is $K(x, y)$ — for any qubit count, any map ✓

Problem 19 [Challenge] Typical entry $K \sim 1/N = 1/1024 \approx 0.001$. Shot noise at 10^4 shots: $\approx 0.5/\sqrt{10^4} = 0.005$ (even the tighter $\sqrt{K/n}$ estimate gives 3×10^{-4} only if you already know K is tiny). Signal below noise: the measured matrix is statistically indistinguishable from the identity. Conclusion: deep random feature maps destroy their own usefulness — the kernel cousin of barren plateaus, cured the same way (shallow, structured, symmetry-matched maps).

Problem 20 [Challenge] (Model design.) Encode the *radius* only: $r = \sqrt{x_1^2 + x_2^2}$, scaled to $[0, \pi]$, via $R_y(r) |0\rangle$ — longitude never used, so rotating the feature plane changes nothing: K depends on (r_a, r_b) alone. Check:

$(1, 0)$ and $(0, 1)$ have equal r : $K = 1$ (identical states) though the raw points differ — invariance by construction. Why it helps: the hypothesis space shrinks to functions of r — the symmetry the labels actually follow — so few examples pin down the boundary instead of also having to *learn* the symmetry. Building known structure into the map is the kernel designer’s highest-leverage move (and the honest answer to “where could quantum kernels win?”: where the right structure is quantum-natural).

Problem 21 [Challenge] $R_y(-x)R_y(y) = R_y(y-x)$ (same-axis rotations add), and $\langle 0 | R_y(y-x) | 0 \rangle = \cos \frac{y-x}{2}$ (the matrix’s top-left entry): squared, $K = \cos^2 \frac{y-x}{2}$ ✓ Generality: any single-qubit map sends points to Bloch vectors, and fidelity between pure qubit states is $\cos^2(\vartheta/2)$ with ϑ the angle between arrows (Chapter 5) — so the kernel can only ever see that one angle.

Problem 22 [Challenge] At $x = 0.7$: K to both training points is $\cos^2(0.2) = 0.960$ — equidistant, so $b = 0$ makes $f(0.7) = 0$ ✓ At $x = 0.5$: $K_1 = \cos^2(0.1) = 0.990$ (to $x=0.3$), $K_2 = \cos^2(0.3) = 0.913$ (to $x=1.1$): $f = \alpha(0.990 - 0.913) = 0.077\alpha > 0$: class +1 for every $\alpha > 0$. Interpretation: the kernel’s monotone distance-decay guarantees that any point left of the midpoint is more similar to the left (positive) example — the SVM’s geometry reduced to “closer wins,” visible by hand.

Quantum Neural Networks and Training

Teaching notes, self-test, and complete solutions

TEACHING NOTES

The training chapter — where QML’s engine room (parameter-shift gradients) and its two governors (frequency budgets and barren plateaus) live. Prove the shift rule in class (Problem 18’s identity is four lines) — an exact gradient from two circuit runs reliably astonishes students raised on finite differences. The frequency picture (layers = harmonics) unifies three chapters: boundary counting (Chapter 23), capacity/overfitting (this chapter’s second lesson), and initialization strategy. Teach the desert quantitatively: the pricing worked example (24M shots) and the desert table (Problem 14) convert “barren plateaus” from folklore to arithmetic; the pre-flight check (Problem 21) turns it into engineering practice. The quantum-data ranking (Problem 13) sets up Chapter 26 honestly. Stumbles: shift rule applied with $\pm$ arbitrary angles (it is exactly $\pm 90^\circ$ for Pauli dials), and gradient budgets quoted without the $\times 2 \times$ dials multiplier.

SELF-TEST — QUESTIONS WITH ANSWERS

Q1. The parameter-shift rule?

A. $\partial E / \partial \theta = \frac{1}{2}[E(\theta + 90^\circ) - E(\theta - 90^\circ)]$ — exact for Pauli-rotation dials: two runs per parameter, no approximation.

Q2. One gradient’s cost for p dials?

A. $2p$ circuit evaluations, each with its own shot budget — the multiplier that dominates training invoices.

Q3. What do re-uploading layers buy, and what do they risk?

A. Each layer adds a harmonic per feature (capacity: more boundaries); too many buys wiggles that fit noise — overfitting, read from the train–test gap.

Q4. Define the barren plateau quantitatively.

A. Deep scrambled circuits have typical gradients $\sim 2^{-n}$, while shot noise falls only as $1/\sqrt{N}$: resolving them needs $N \sim 4^n$ shots — unpayable past a dozen qubits.

Q5. Three anti-desert practices?

A. Stay shallow and structured; initialize near identity (or grow layerwise); monitor gradient norms from step one and stop early if flat.

Q6. Where does QML have a principled edge?

A. Quantum data — states from simulators, sensors, networks — where coherent processing provably beats measure-first pipelines; classical data remains an uphill, audit-everything field.

Solutions to the Practice Problems

Problem 1 [Basic] Measurement (the Born rule's squared moduli — Chapter 4) and the data re-encoding between layers (cosines of inputs — Chapter 22). The unitary circuit between them is strictly linear.

Problem 2 [Basic] At 0° : $\frac{1}{2}[\cos 90^\circ - \cos(-90^\circ)] = 0$ — the curve's flat top. At 90° : $\frac{1}{2}[\cos 180^\circ - \cos 0^\circ] = -1$ — the steepest descent, matching $-\sin \theta$ at both points.

Problem 3 [Basic] $2 \times 40 = 80$ evaluations.

Problem 4 [Basic] The activation function (ReLU/sigmoid): unitaries cannot bend amplitudes nonlinearly. Stand-ins: measurement's squared moduli and data re-uploading — nonlinearity smuggled in at the boundaries of the circuit.

Problem 5 [Basic] $12 \times 2 \times 100 \times 1000 = 2.4$ million shots. Cheapest cut: shots per evaluation (early in training, coarse gradients steer fine) — steps and parameters are structural, but shot budgets can adapt to training phase.

Problem 6 [Basic] Target noise $= \frac{1}{2} \times 2^{-6} = 0.0078$: $1/\sqrt{N} = 0.0078 \Rightarrow N \approx 16,000$ shots per evaluation.

Problem 7 [Basic] 99/97: healthy (2-point gap — noise-level). 100/60: severe overfitting; remedies: remove re-uploading layers (cut harmonics = smooth the boundary) and enlarge/augment the training set (give the harmonics something real to fit); early stopping on a validation split earns honorable mention.

Problem 8 [Basic] Noisy labels contain errors, and a model matching them *perfectly* has necessarily memorized the errors — fitting the noise is failure wearing a 100% badge.

Problem 9 [Standard] Per gradient: $80 \text{ evaluations} \times 2000 = 160,000$ shots. Noise per evaluation ≈ 0.022 , seven times the 0.003 signal — each shifted pair differs by less than its own error bar: the optimizer walks on static. Seeing 0.003 at $3\times$ margin needs noise 0.001: 10^6 shots per evaluation (8×10^7 per gradient) — the budget explosion that defines the desert.

Problem 10 [Standard] At 30° : $\frac{1}{2}[\cos 120^\circ - \cos(-60^\circ)] = \frac{1}{2}[-0.5 - 0.5] = -0.5 = -\sin 30^\circ$ ✓ For $0.7 \cos \theta + 0.2$: the constant shifts cancel in the difference and the amplitude scales linearly: $\frac{1}{2}\Delta = -0.7 \sin \theta$, again the exact derivative. The proof rides on the curve being a pure $\cos / \sin$ in each dial (period 360°) — guaranteed for Pauli-rotation gates, and the reason the rule is exact rather than approximate.

Problem 11 [Standard] Condition $2^{-n}/2 = 3/\sqrt{N} \Rightarrow N = 36 \cdot 4^n$. At $n = 8$: $N \approx 2.4 \times 10^6$; at $n = 16$:

$N \approx 1.5 \times 10^{11}$. Scaling: 4^n — each added qubit quadruples the bill; sixteen qubits of scrambled circuit cost $65,000\times$ more than eight.

Problem 12 [Standard] Four harmonics per feature needs $L = 4$ re-uploads: 4 layers, $4 \times 5 = 20$ dials. Doubling to 8 layers doubles the frequency budget past the data's actual content — the extra harmonics have nothing true to encode and will happily trace the training sample's accidents: the train-test gap opens (Problem 7's disease, self-inflicted).

Problem 13 [Standard] (a) Classical/uphill: spreadsheets load classically; tuned gradient-boosted trees are brutal competition. (b) Quantum/principled: the simulator's output is a quantum state — process it coherently, no loading problem. (c) Classical/uphill: images are classical data with superb classical baselines. (d) Quantum/principled: network states arrive as qubits; certification is native quantum measurement work (and Chapter 26's proven-advantage territory).

Problem 14 [Standard] Table: $n = 4$: gradient 0.06, shots ~ 256 ; $n = 8$: 0.004, $\sim 65,000$; $n = 12$: 2.4×10^{-4} , $\sim 1.7 \times 10^7$; $n = 16$: 1.5×10^{-5} , $\sim 4.3 \times 10^9$. One gradient *evaluation* at $n = 16$: $4.3 \times 10^9 / 10^4 \approx 4 \times 10^5$ s $\approx$ five days — per parameter, per step: the desert, timetabled.

Problem 15 [Standard] At identity initialization the layers cancel: the circuit's *effective* depth is near zero, so the initial function is low-frequency and smooth — far from the scrambled 2-design regime where gradients vanish. Training then grows frequency content only as the data demands it: capacity is added under supervision instead of pre-scrambled, keeping gradients macroscopic throughout.

Problem 16 [Standard] The statevector simulator computes exact expectations — noise 0 — so even 10^{-5} -sized gradients steer perfectly. Hardware at 4000 shots has noise $1/\sqrt{4000} \approx 0.016$; the 8-qubit, 40-layer scrambled ansatz has gradients $\sim 2^{-8}/2 \approx 0.002$ — eight times *below* the noise: every measured gradient is static. Same model, two noise regimes: the simulator hid the desert the hardware honestly reports.

Problem 17 [Standard] $\sigma = \sqrt{0.9 \times 0.1/20} = 0.067$: two-sigma interval $\approx [77\%, 100\%]$. The competitor's 85% sits well inside it: the 5-point difference is statistically meaningless at $n = 20$ — which is how tiny-test-set QML papers manufacture “wins.” (Chapter 27's error-bar demand, derived.)

Problem 18 [Challenge] $E(\theta \pm 90^\circ) = A \cos \theta \cos 90^\circ \mp A \sin \theta \sin 90^\circ + B \sin \theta \cos 90^\circ \pm B \cos \theta \sin 90^\circ + C = \mp A \sin \theta \pm B \cos \theta + C$. Difference over 2: $\frac{1}{2}[(-A \sin \theta + B \cos \theta + C) - (A \sin \theta - B \cos \theta + C)] = -A \sin \theta + B \cos \theta$ — exactly $dE/d\theta$ ✓ No step size, no truncation error: trigonometry, not approximation.

Problem 19 [Challenge] Global projector: on a scrambled state, $P(0 \cdots 0) \sim 2^{-n}$ — the observable's value, and its gradients, live at the 2^{-n} floor: undetectable. Local average: each “qubit k reads 0” sits near $\frac{1}{2}$ with $O(1)$ sensitivity to that qubit's dials — the average retains gradients suppressed only polynomially. Three sentences: global costs ask one astronomically rare event to move, so nothing measurable changes when a dial turns; local costs ask n ordinary probabilities to move, and they do; hence “train on local observables” is a standard plateau-softener — signal survives where the cost function keeps its feet on the ground.

Problem 20 [Challenge] Fixed Z basis: $|0\rangle$ reads 0 always, $|+\rangle$ reads 0 half the time — outcome 0 is ambiguous: error rate 25% (guessing the likelier source). Fixed X basis: symmetric story. Best fixed basis: the tilted Helstrom basis at 22.5° , success $\cos^2 22.5^\circ = 85.4\%$ — and *no* fixed basis does better, while errors remain on every shot. The scaling: with many states and many shots, measure-first pipelines discard the phase information a coherent (quantum-memory) pipeline retains; Lesson 5’s exponential separations (two-copy joint measurements) are this 15% gap compounded across an experiment — classical notes lose a little every shot, and “a little, exponentially often” is the theorem.

Problem 21 [Challenge] (Model design.) Pick 5 random parameter points; at each, estimate one randomly chosen dial’s shift-rule gradient with 4000 shots per evaluation ($\sigma \approx 0.016$ per E, ≈ 0.011 per gradient): cost $5 \times 2 \times 4000 = 40,000$ shots — under budget with room for a repeat. Decision rule: if the five gradients’ typical magnitude exceeds $3\sigma \approx 0.035$: cleared — training will see signal at production budgets. If all five sit within $\pm 2\sigma$ of zero: grounded — the landscape is flat at your budget; shrink depth or restructure before spending millions. Justification: five samples suffice because plateau gradients are uniformly tiny (one macroscopic gradient disproves the desert); 4000 shots matches the training-time noise you must live with anyway.

Problem 22 [Challenge] (Model design.) Split 60 points: 36 train / 12 validation / 12 test (test locked until the end). Sweep $L = 1, 2, 3, 5, 8$ with equal training budgets (say 2×10^6 shots each, from a 10^7 pool). Expect: training accuracy rising monotonically with L (capacity never hurts the fit); validation rising to a peak near $L = 2$ –3 (the rings need ~ 2 harmonics) then *falling* as overfitting opens the gap — the gap first widens visibly at $L = 5$, badly at 8. Deploy the smallest L within one standard error of the best validation score (expect $L = 3$, the book’s choice); then, and only then, report the locked test number with its Problem-17-style error bar.

Quantum Generative Models and the Question of Advantage

Teaching notes, self-test, and complete solutions

TEACHING NOTES

Two chapters in one: a builder's introduction (Born machines, TV distance, QGANs) and the field's honest advantage map. Run the two-qubit Born machine end to end — untrained $TV = 0.5$, one CNOT, $TV = 0$ — as live theater; the moral (*correlated data needs entangling gates*) is Part Seven's most quotable line. The evaluation lesson (marginals, joint TV, novelty rate) teaches generative-model skepticism transferable far beyond quantum. Give the advantage map a full discussion class: students should leave able to place any headline in one of the four territories and name its catch — with dequantization (Tang's story) told as the fairness principle it is, not as gossip. Problem 14 (quantum data, two-copy measurements) is the deepest in Part Seven; assign it to be written, not just discussed. Stumbles: TV computed without the $\frac{1}{2}$; and "hard to sample" conflated with "useful to sample" (Problem 12 exists to break that reflex).

SELF-TEST — QUESTIONS WITH ANSWERS

Q1. Define the Born machine.

A. A parameterized circuit as a generative model: $P_{\theta}(x) = |\langle x | \psi(\theta) \rangle|^2$ — measurement IS sampling; training reshapes the histogram toward the data's.

Q2. The TV distance, and its teeth?

A. $TV = \frac{1}{2} \sum_x |p - q|$, from 0 to 1; operationally, the best single-sample judge distinguishes the distributions with probability $\frac{1}{2} + TV/2$.

Q3. Why did one CNOT finish the worked example's training?

A. The target's perfect correlation is Bell-state statistics; without entangling gates the generator can only produce independent bits — no dial settings suffice.

Q4. The quantum GAN's division of labor?

A. Quantum generator samples by measurement (parameter-shift updates); classical discriminator judges real-vs-

fake (backpropagation); equilibrium = indistinguishable samples.

Q5. Name the advantage map's four territories with one-word verdicts.

A. Sampling hardness: hard-but-useless (so far). Quantum data: proven. Hard kernels: proven-but-toy. Heuristic QML on classical data: unproven.

Q6. What is dequantization?

A. Granting the classical competitor the quantum algorithm's data-access assumptions and matching its speedup — Tang's fairness rule; advantages must survive equal footing.

Solutions to the Practice Problems

Problem 1 [Basic] $TV = \frac{1}{2}(|0.25 - 1| + 3 \times 0.25) = \frac{1}{2} \times 1.5 = 0.75$. Perfect generator: the empty circuit — $|00\rangle$ measured gives 00 always; doing nothing is sometimes optimal.

Problem 2 [Basic] $\sin^2(\theta/2) = 0.30 \Rightarrow \theta = 2 \arcsin \sqrt{0.30} = 2 \times 33.2^\circ \approx 66.4^\circ$.

Problem 3 [Basic] The Born rule makes every measurement a sample from the circuit's distribution — generation is free. The classical modeler's headache: producing samples requires transforming randomness through the model *and* often normalizing an intractable distribution — sampling is the expensive part, not the cheap one.

Problem 4 [Basic] Four gaps of 0.05: $TV = \frac{1}{2} \times 0.20 = 0.10$.

Problem 5 [Basic] Fail — memorization: the machine reproduces training records rather than the law behind them (worthless for generation, dangerous for privacy); excellent TV is exactly what verbatim copying produces, which is why novelty is checked separately.

Problem 6 [Basic] $TV = 0.10$ (as in Problem 4). Interpretation: the best possible judge, shown one sample, guesses its source correctly with probability $\frac{1}{2} + \frac{0.10}{2} = 55\%$ — barely better than coin-flipping: the generator is close.

Problem 7 [Standard] H on q_0 ; CNOT($q_0 \rightarrow q_1$); CNOT($q_0 \rightarrow q_2$) (or chain through q_1): the GHZ state $\frac{1}{\sqrt{2}}(|000\rangle + |111\rangle)$ — three-way correlation needs the correlation spread by two entanglers.

Problem 8 [Standard] Bell target: two patterns at $p = 0.5$ ($n \approx 0.25/10^{-4} = 2500$ each) and two near 0 (cheaper, but budget 2500 for symmetric bars): $\approx 10^4$ samples per side certifies every entry to ± 0.01 , hence TV to $\pm 0.02 < 0.05$ ✓ At $2^{10} = 1024$ patterns, per-pattern certification needs $\gtrsim$ thousands of samples *per pattern* — millions overall, growing exponentially with qubits: joint certification stops scaling almost immediately (Problem 16 builds the workaround).

Problem 9 [Standard] Generator $\rightarrow$ discriminator: measured fake samples (bit strings). Data $\rightarrow$ discriminator: real samples. Discriminator $\rightarrow$ verdicts: real/fake probabilities. Verdicts $\rightarrow$ both models: training signals —

the discriminator updates by ordinary backpropagation (it is classical); the generator updates by parameter-shift gradients of its fooling score (its dials live in a circuit).

Problem 10 [Standard] TV between $(1, 0, 0, 0)$ and $(0.5, 0, 0, 0.5)$: $\frac{1}{2}(0.5 + 0.5) = 0.5$ — as far from the target as the untrained uniform start. The discriminator soon learns “11 never appears from the generator” and flags every 11-less batch as fake; the generator’s fooling score then collapses, and gradients push it to repopulate 11 — collapse is punished, though the duel may oscillate (the instability tax).

Problem 11 [Standard] Per-pattern wobbles: ≈ 0.016 (the 0.4-ish entries) and ≈ 0.007 (the 0.05 entries), per side. The TV estimate’s bar: roughly $\frac{1}{2} \sum \sigma \approx 0.02$ – 0.03 (both sides contribute). A measured TV of 0.10 sits $\sim 4\sigma$ from zero: yes, distinguishable — but a measured 0.03 would not be; always attach the bar before declaring victory or convergence.

Problem 12 [Challenge] (Model response.) “True premise: sampling from random circuits is believed classically intractable — the supremacy results stand. False step one: intractable-to-imitate is not valuable-to-have; no customer’s data follows a random circuit’s distribution, and the claim names none. False step two: the same randomness that makes the circuits hard makes them *untrainable* — Chapter 25’s barren-plateau results say gradients vanish exactly in the scrambled regime the claim boasts of. So the product occupies the worst quadrant: distributions nobody wants, produced by circuits nobody can steer. Verdict: quantum advantage in difficulty, not in usefulness — please demonstrate training toward a customer-relevant target before claiming either.”

Problem 13 [Challenge] Hidden assumption: data pre-loaded in quantum-accessible form (state preparation / QRAM at negligible cost). The grant: Tang gave classical algorithms an analogous data structure (ℓ^2 -norm sampling access). Fairness rule: compare algorithms under *equal data-access assumptions* — an advantage that evaporates when the classical side gets the same key was never an algorithmic advantage. Applied to amplitude-encoding claims (Chapter 22): any speedup quoted *after* free amplitude encoding must either price the loading honestly (usually $O(N)$, killing the claim) or grant classical algorithms equivalent sampling access and re-compare — Problem 19 of that chapter ran exactly this audit.

Problem 14 [Challenge] (Model paragraph.) A quantum sensor’s output state carries information partly in *incompatible* observables — and Chapter 8 taught that measuring one destroys the others: an immediate-measurement pipeline must choose, every shot, which face of the state to keep and which to erase, so reconstructing joint structure costs statistically many experiments. A quantum memory changes the game: store two copies from consecutive runs and measure them *jointly* — joint observables (like swap tests across copies) read correlation properties that no sequence of single-copy measurements can access with comparable statistics. The 2022 experiments made this concrete: tasks needing $\sim$ exponentially many measure-first experiments were solved with polynomially many memory-assisted ones. For the lab: if your instrument’s output is quantum, a quantum memory beside it is not a luxury — it is an exponential discount on experiment count for the right questions.

Problem 15 [Challenge] (Model audit, canonical DASHB.) D — data access: how were patient records encoded into 12 qubits, at what preprocessing loss and priced loading cost; red flag: heavy classical compression doing

the real modeling. A — ablation: swap the quantum generator for a tuned classical GAN/VAE in the identical pipeline; red flag: no ablation, “entanglement” cited as explanation. S — scaling: trainability at 12 qubits and the certification wall beyond (Problem 16); red flag: “indistinguishable from real” via a metric that cannot be estimated at that dimension. H — hardware honesty: samples from hardware or simulator, error bars over seeds, and a novelty audit (memorized *patient records* are a privacy breach, not a product); red flag: simulator samples marketed as hardware. B — baselines: tuned classical generators on identical data and metrics; red flag: comparison to an absent or stale baseline. Expected outcome: a legitimate engineering demo whose claims outrun its evaluation — the standard 2026 configuration.

Problem 16 [Challenge] Joint certification must estimate 2^n pattern probabilities; even *populating* each pattern once needs $\gtrsim 2^n$ samples (10^3 at $n = 10$, 10^6 at $n = 20$), and precision multiplies that further — exponential, full stop. Polynomial report card: all n single-qubit marginals (± 0.01 : 10^4 samples total); all $\binom{n}{2}$ pairwise correlations $\langle Z_i Z_j \rangle$ (same 10^4 samples reused — they are diagonal); a handful of higher-order parities chosen by domain knowledge; novelty rate on 10^3 samples; and one downstream-utility test — train a classifier on synthetic, test on real, compare with real-trained (the metric a customer actually feels). Certifies: low-order structure, non-memorization, and task usefulness. Cannot certify: full joint fidelity — and the card should say so in print, which is precisely the honesty separating engineering from hype.

CHAPTER TWENTY-SEVEN

Quantum AI in Practice

Teaching notes, self-test, and complete solutions

TEACHING NOTES

Part Seven's graduation exercise: judgment. The honest scoreboard should be *memorized in shape* (mechanisms established, classical-data advantage open, big-data linear algebra withdrawn, quantum data proven, deep training obstructed) — quiz it cold. DASHB — Data access, Ablation, Scaling, Hardware honesty, Baselines — is the chapter's takeaway tool; the press-conference activity (live audits of real claims) is the best assessment I know: grade the audits, not opinions. Tell the dequantization story as a hero story (corrections are gifts), and give the regional road its full due — honest benchmarks and ablation studies are world-class entry points that need laptops, not refrigerators; Problem 13's thesis-proposal exercise regularly produces real proposals. The project template closes the loop to capstones 9–11. Stumbles: letters recalled but questions garbled (drill the five *questions*, not the acronym), and audits written as sneers — require every criticism to name the evidence that would answer it.

SELF-TEST — QUESTIONS WITH ANSWERS

Q1. Recite DASHB as five questions.

A. D: how does data get in, and is loading priced? A: does removing the quantum piece change the result? S: do gradients/shots/depth survive plateau arithmetic at the promised size? H: simulator or hardware, with what noise, shots, and error bars? B: is the classical baseline tuned as hard as the quantum model?

Q2. Scoreboard: what is established, what is open?

A. Established: circuits learn real boundaries; exact hardware gradients; provable advantage on constructed problems and quantum data. Open: advantage on natural classical data. Withdrawn: big-data linear-algebra speedups (dequantization). Obstructed: deep variational training.

Q3. The dequantization fairness rule?

A. Compare like with like: grant classical algorithms comparable data access before claiming quantum speedups — advantages that die on equal footing were bookkeeping, not physics.

Q4. Why are quantum circuits natively generative?

A. Measurement is Born-rule sampling (Chapter 4): one shot, one fresh sample — the Born machine needs no sampling algorithm.

Q5. Where does quantum sit in a realistic 2026 pipeline?

A. A small slot inside a classical stack: kernel provider, feature generator, sampler, or optimizer for a sub-problem — validated by ablation.

Q6. The region's realistic entry points?

A. Reproductions, honest benchmarks, ablation studies, and symmetry-matched designs on laptop simulators and free cloud tiers — publishable-adjacent work with no refrigerator required.

Solutions to the Practice Problems

Problem 1 [Basic] (Sample.) Established: parameterized circuits genuinely learn decision boundaries (this book's rings demo among the evidence). Open: any advantage on natural classical data — tuned baselines still match or win. Withdrawn: exponential speedups for classical big-data linear algebra — dequantized.

Problem 2 [Basic] “What data access did you assume — and what happens if the classical algorithm gets comparable access?” Asked (and answered, devastatingly) by Ewin Tang, then an undergraduate.

Problem 3 [Basic] The Born rule (Chapter 4): measuring a circuit returns pattern x with probability $|a_x|^2$ — every shot is a fresh draw from the circuit's distribution, so sampling is the hardware's native act.

Problem 4 [Basic] D — Data access: how does the data get in, and is loading priced? A — Ablation: does removing the quantum piece change the result? S — Scaling: do gradients, shots, and depth survive plateau arithmetic at the promised size? H — Hardware honesty: simulator or hardware, with what noise and shot budgets? B — Baselines: is the classical opponent tuned as hard as the quantum model?

Problem 5 [Basic] (Sample rewrite.) “On hardware X, with 10^4 shots per prediction and readout mitigation, accuracy was $91 \pm 2\%$ across five seeds and three calibration days.” The three required details: hardware-vs-simulator named; shot budget stated; error bars over seeds/calibrations reported.

Problem 6 [Basic] Ablation: rerun the identical pipeline with the quantum kernel/model replaced by a tuned classical counterpart. Outcomes: accuracy unchanged $\Rightarrow$ the quantum ingredient contributed nothing (claim dissolves); accuracy drops materially $\Rightarrow$ the quantum piece genuinely mattered (claim strengthens and now deserves the other four letters).

Problem 7 [Basic] Order: freeze the splits $\rightarrow$ tune the classical baseline $\rightarrow$ train the quantum model. Traps prevented, in order: leakage (test data influencing choices); “beat classical” with no real opponent; and hindsight-shopping the quantum result against a baseline chosen afterwards.

Problem 8 [Basic] Steps 3 and 4 — start with the smallest circuit (barren-plateau avoidance, Chapter 25) and budget shots in advance with tracked gradients (Chapter 25's pricing; Chapter 24's kernel bills). The rest is

good machine learning everywhere.

Problem 9 [Standard] Ablation run: identical splits, scaling, SVM solver, and tuning budget, with the quantum kernel matrix replaced by a tuned classical kernel (RBF); metric: test accuracy with error bars over seeds. Most-at-risk letters here: **D** (how do field images become few qubits? — classical preprocessing may be the real model) and **B** (agronomy datasets have strong classical baselines that are rarely tuned in QML papers).

Problem 10 [Standard] It nudges only the “mechanisms work” rows: another demonstration that Born machines train at small scale — consistent with the established entries. It does not touch “advantage on natural classical data” (synthetic data, simulator, parity-at-best) nor the quantum-data row (no quantum-born input involved). Scoreboards move on evidence tied to their specific claims — one more nice demo is progress, not news.

Problem 11 [Standard] Referee rule: *state every input-access assumption, then grant the classical competitor the analogous one before comparing*. Applied: “exponentially faster (data assumed amplitude-encoded)” must either price the encoding (typically $O(N)$ — speedup gone) or face classical algorithms with sampling access to the same vectors (Tang’s move — speedup typically gone again). The claim as stated compares a machine with magic memory to one with a filing cabinet.

Problem 12 [Standard] (a) Sampler — quantum proposes candidates, classical screens. (b) Features — measured circuit statistics appended to a classical table. (c) Kernel — quantum similarity matrix feeding a classical SVM. (d) Optimizer — QAOA-style attack on one combinatorial sub-problem inside the logistics stack.

Problem 13 [Standard] (Grading guide — expect one paragraph naming:) a locally plausible dataset (e.g. wheat-disease images from Kurdistan agriculture directorates, Erbil grid-load records, anonymized clinic tabular data); three feature maps (angle; angle+entangler/ZZ; a symmetry-matched design); a *tuned* classical baseline named explicitly (RBF-SVM and gradient-boosted trees); full DASHB reporting with seeds and error bars; and a stated deliverable (benchmark tables + ablation + public code). Full credit requires the baseline and the ablation to be in the plan, not appended.

Problem 14 [Standard] $\sigma \approx \sqrt{0.85 \times 0.15/40} \approx 0.056$ per measurement; the difference of two such accuracies carries $\sigma_\Delta \approx 0.08$. The 7-point gap is $0.07/0.08 \approx 0.9\sigma$: statistically meaningless. Forty test points cannot certify single-digit superiority claims — demand more data or more splits before believing either number.

Problem 15 [Standard] D: pass — two features, angle-encoded, loading trivial and stated. A: needs-info — no ablation reported, though the parity claim makes it low-stakes. S: pass — one qubit, three layers: no plateau exposure. H: pass — hardware, shots, seeds, error bars all present. B: pass — tuned RBF-SVM, same splits, honest parity claim. Verdict: publishable honesty; the single request is the (cheap) ablation for completeness. (This is, of course, this book’s own rings study — students should notice.)

Problem 16 [Standard] PCA or feature selection can capture the class structure classically, handing the quantum stage an almost-solved problem — the quantum model then “achieves” accuracy the preprocessing earned (a

D-question failure in disguise). Control experiment: train a small classical model *directly on the preprocessed features* (the PCA scores); its accuracy measures the preprocessing’s own power, and only the margin above it is attributable to anything quantum.

Problem 17 [Standard] Request: “Please rerun the identical pipeline with the quantum kernel replaced by a tuned classical kernel on the same splits, reporting both accuracies with error bars.” Outcomes: classical matches 94% $\Rightarrow$ the quantum kernel drove nothing — claim withdrawn to parity; classical falls materially short $\Rightarrow$ the contribution is real and the paper strengthens. Either way the field learns — which is the point of asking.

Problem 18 [Challenge] (Model paragraph.) Hardness results certify that classical machines cannot *imitate* certain circuit distributions — they say nothing about steering such circuits toward a *chosen* target: training requires gradients, and Chapter 25’s plateau theorems bite hardest exactly in the scrambled, hardness-bearing regime — the first missing ingredient, guarded by the desert dragons. The second: even a trained hard sampler must sample something someone wants — hardness of imitation is not demand; no natural dataset is known to live in these families. Sampling advantage becomes generative advantage only when trainability and usefulness are both demonstrated — and as of this book, neither has been, which is the scoreboard’s honest “open.”

Problem 19 [Challenge] (Grading guide.) Strengths paragraph: clinically meaningful features (D honest — 12 features load directly), tuned classical baseline present (B partially satisfied), shots stated. Concerns: simulator-only (H — no hardware, no noise story), no ablation (A — is the 3-point gap the feature map’s work?), 3 points on what test size — error bars likely swallow it (the Problem-14 arithmetic), and no scaling statement (S) for the promised clinical deployment. Verdict: major revision. The single decisive request: the ablation with error bars on an adequate test set — it converts the headline from suggestive to science (or dissolves it).

Problem 20 [Challenge] (Grading guide — any territory defended with three named facts passes.) Sample: quantum data — theorem-grade: proven exponential sample savings with quantum memory (2022, scoreboard row); hardware trend: sensors/simulators producing quantum states are maturing faster than fault tolerance (Chapters 21, 30), and the advantage needs *memories*, not million-qubit machines; economics: no shot-hungry kernel bills or plateau budgets — the pipeline replaces *experiments*, whose cost is already enormous. Weakest common answer: classical data “because datasets are big” — big data is the classical stack’s home field.

Problem 21 [Challenge] (Grading guide, one paragraph per letter.) D: demand the loading ledger for “scans” into 12 qubits; evidence: the preprocessing pipeline plus Problem 16’s control. A: demand the swap-in classical ablation; evidence: paired accuracies. S: demand gradient-norm traces or kernel-concentration checks at the working size; evidence: Chapter 25’s pre-flight numbers. H: demand hardware runs with shots, mitigation, seeds; evidence: the three details of Problem 5. B: demand tuned modern baselines; evidence: search protocol and budgets. Close with the constructive list (ablation + tuned baseline + hardware error bars) and the reminder that any claim surviving all five deserves *support* — the checklist cuts both ways.

Problem 22 [Challenge] (Grading guide — expect concrete numbers such as:) splits 60/20/20 with seeds; features scaled per-column into $[0, \pi]$ with ranges logged; architecture 4 qubits, 2–3 re-uploading layers (≤ 30 dials), local readout; shot budget priced via Chapter 25 (e.g. $2 \times 30 \times 2000$ per step, $\times 150$ steps $\approx 2 \times 10^7$, times restarts — state the total); evaluation: test accuracy with binomial error bars, tuned RBF-SVM and boosted-tree baselines, ablation run; audit risks pre-declared: D (records $\rightarrow$ features pipeline), B (medical baselines are strong), H (simulator first, hardware pilot second). Any plan with all six steps numbered meets Project 9's rubric.

CHAPTER TWENTY-EIGHT

Noise, Decoherence, and Errors

Teaching notes, self-test, and complete solutions

TEACHING NOTES

The chapter where the book turns from mathematics to engineering weather. Build everything on three formulas students must own cold: e^{-t/T_1} , the Ramsey envelope e^{-t/T_2} , and the survival chain $(1 - p)^d$ — every hardware discussion for the rest of the course is these three, re-dressed. The measure-it-yourself lesson converts dread into protocols (run both on the app or cloud if at all possible; they are the standard daily calibrations of real labs). Teach mitigation as honest first aid, not resurrection: readout un-flipping and ZNE (with its mush-floor failure, Problem 22) rescue expectation values at a shot-cost, while *correction* (next chapter) is the scalable future. Budgeting problems (10, 19) make superb graded work — they are the datasheet-reading skill in embryo. Stumbles: applying $\sqrt{N}$ wobble to probabilities instead of counts, confusing T_2^* with echo T_2 , and quoting fidelity as if it were survival probability of a whole circuit.

SELF-TEST — QUESTIONS WITH ANSWERS

Q1. Define T_1 and T_2 operationally.

A. T_1 : prepare $|1\rangle$, wait t , measure — survival e^{-t/T_1} . T_2 : H, wait, H, measure — fringe envelope e^{-t/T_2} : the phase memory. Always $T_2 \leq 2T_1$.

Q2. The circuit survival rule?

A. $P \approx (1 - p)^d$: per-step success compounds; trouble begins near $d \approx 1/p$ — the depth wall.

Q3. Which error type usually dominates budgets?

A. Two-qubit gates ($\sim 10\times$ worse than single-qubit) — with readout often the worst single step; hence compilers count CNOTs.

Q4. Mitigation versus correction, one line?

A. Mitigation: statistical repair of *answers* using extra runs (readout inversion, ZNE) — no extra qubits, limited scope. Correction: encode logical qubits to repair *states* mid-flight — scalable, expensive.

Q5. How does ZNE work?

A. Measure at natural and deliberately amplified noise (gate folding), extrapolate the trend to zero noise — trusting

smoothness, paying in runs.

Q6. Why are lifetimes “weather”?

A. They drift with calibration and stray defects hour to hour; serious users check the morning’s T_1/T_2 map before believing yesterday’s budget.

Solutions to the Practice Problems

Problem 1 [Basic] $e^{-1} = 36.8\%$ after one lifetime; $e^{-2} = 13.5\%$ after two.

Problem 2 [Basic] T_2 : interference lives in relative phase, and T_2 is the phase’s lifetime (T_1 merely bounds it via $T_2 \leq 2T_1$).

Problem 3 [Basic] Error rate 0.8%; expected errors reach 1 after $1/0.008 = 125$ uses.

Problem 4 [Basic] $(0.99)^{50} \approx 0.605$: about 60% of runs survive depth 50 at the 1% wall’s halfway mark.

Problem 5 [Basic] $t = T_1$: $e^{-1} = 36.8\%$; $t = 2T_1$: $e^{-2} = 13.5\%$. Rule: each whole T_1 multiplies survival by another $1/e \approx 0.37$.

Problem 6 [Basic] T_1 protocol: (1) X gate; (2) **wait** t ; (3) measure; (4) repeat 1000 $\times$; (5) sweep t . Ramsey T_2 : (1) H; (2) **wait** t ; (3) H; (4) measure; (5) repeat and sweep. The waits sit between preparation and readout — they *are* the experiment.

Problem 7 [Basic] Slope = $(0.28 - 0.40)/2 = -0.06$: $E(0) = 0.40 + 0.06 = 0.46$. Sign check: noise pulls expectations toward 0, so the corrected value moves *away* from 0 — as it does ✓

Problem 8 [Basic] $GG^\dagger G = G$ as a unitary — the ideal circuit is untouched. Physically, three pulses run instead of one: three exposures to per-gate noise — the error budget triples while the logic stands still: a controlled noise dial.

Problem 9 [Standard] $0.1 T_2 = 6 \mu\text{s}$; $6000 \text{ ns}/25 \text{ ns} = 240$ single-qubit gates. Compilers still sweat the two-qubit gates because each costs $\sim 10\times$ the error *and* $\sim 2\text{--}3\times$ the time — ten CNOTs spend the budget of a hundred singles.

Problem 10 [Standard] Survival = $(0.992)^2(0.9995)(0.988)^3 = 0.984 \times 0.9995 \times 0.964 = 0.949$: corrupted fraction $\approx 5\%$. Largest single contributor: the three readouts ($\approx 3.6\%$ combined) — measurement, not the famous CNOTs, taxes this little circuit most.

Problem 11 [Standard] Usable depths $\approx 1/p$: 200 versus 400 steps — B runs algorithms twice as deep. “ $2\times$ better gates” doubles what every qubit can *do*, while “ $10\times$ more qubits” multiplies capacity that the depth wall may forbid anyone to use.

Problem 12 [Standard] False zeros inflate the zero count: 4% of the (many) true 1s are reported as 0, while only 1% leak the other way — the raw $P(0) = 0.30$ *overestimates* the true value. Mitigation inverts the calibration matrix and pushes $P(0)$ down (here to $\approx (0.30 - 0.04)/0.95 \approx 0.27$).

Problem 13 [Standard] Fringes become invisible when the envelope has fallen to roughly the noise floor — $e^{-t/T_2} \sim 0.1$, i.e. $t \approx 2\text{--}2.5 T_2$. From $t \approx 150 \mu\text{s}$: $T_2 \approx 60\text{--}75 \mu\text{s}$ — one significant figure: $\sim 70 \mu\text{s}$.

Problem 14 [Standard] $0.37 = e^{-1}$ at $100 \mu\text{s}$ and $0.135 = e^{-2}$ at $200 \mu\text{s}$: both fit $T_1 = 100 \mu\text{s}$ exactly — two points, one exponential, a consistent clock.

Problem 15 [Standard] $\sigma = 0.5/\sqrt{1000} = 0.016$ per point; the difference 0.06 carries $\sigma_\Delta \approx 0.022$: $0.06/0.022 \approx 2.7$ sigma — borderline-reliable, uncomfortable for calibration decisions. Cheap fix: $4\times$ the shots (halves the wobble) — or fit the whole decay curve instead of trusting one point.

Problem 16 [Standard] T_2^* includes slow drifts of the qubit frequency between shots — each run accumulates a slightly different phase, washing fringes early; the echo's mid-wait X pulse refocuses those slow wanderings, revealing the longer intrinsic T_2 . Long algorithms resemble the *un*-refocused reality unless they embed echo sequences — so T_2^* is the pessimistic-but-honest planning number, echo T_2 the ceiling good pulse design can approach.

Problem 17 [Standard] $\sigma_{E(0)} = \sqrt{(1.5 \times 0.02)^2 + (0.5 \times 0.02)^2} = 0.02\sqrt{2.5} \approx 0.032$ — $1.6\times$ a single measurement's wobble. Mitigation amplifies statistical noise through its lever arms: budget extra shots for every mitigated number, or the “corrected” value is precise-looking guesswork.

Problem 18 [Challenge] $\frac{1}{T_\varphi} = \frac{1}{70} - \frac{1}{200} = 0.014286 - 0.005 = 0.009286 \Rightarrow T_\varphi \approx 108 \mu\text{s}$. Doubling $T_1 \rightarrow 200$: $1/T_2 = 1/400 + 1/107.7 \Rightarrow T_2 \approx 85 \mu\text{s}$. Doubling $T_\varphi \rightarrow 215$: $1/T_2 = 1/200 + 1/215 \Rightarrow T_2 \approx 104 \mu\text{s}$. Dephasing is the binding constraint here — attack T_φ (shielding, dynamical decoupling) before chasing T_1 .

Problem 19 [Challenge] $d \approx 785 \times 40 = 31,400$ steps; survival $\approx e^{-0.001 \times 31400} = e^{-31.4} \approx 2 \times 10^{-14}$, hopeless. Plausibility needs $p_2 d \lesssim 1$: $p_2 \sim 3 \times 10^{-5}$ — order 10^{-5} , a hundredfold beyond the best raw hardware. The technology being begged for: quantum error correction — logical qubits whose *effective* error rate is engineered below 10^{-5} (Chapter 29, on cue).

Problem 20 [Challenge] (Grading guide.) T_1 : prepare $|1\rangle$ (X), insert wait t , measure; repeat 1000 shots per t ; sweep t over several expected lifetimes; plot surviving-1 fraction versus t and read T_1 where it crosses 37% (or fit the exponential). T_2 : H, wait t (optionally with small detuning for visible fringes), H, measure; repeat and sweep; plot $P(0)$ versus t : oscillations under a shrinking envelope; read T_2 where the envelope falls to 37% of full swing. Full credit needs: repetition counts, the sweep, the plotted quantity, and the read-off rule — the four marks of a real protocol.

Problem 21 [Challenge] Repeat a fixed-length sequence many times and *compare statistics to prediction*: decoherence produces a shot-to-shot *random* spread that grows with wait time and never averages away into a clean offset; a miscalibrated over-rotation produces a *deterministic, repeatable* bias — the same wrong probability every run, independent of added idle time. The killer discriminator: run the pulse sequence *reversed*

(or double each pulse): a fixed over-rotation flips sign (or doubles) deterministically; decoherence, indifferent to pulse sign, stays the same. Signature summary: randomness growing with $t \Rightarrow$ decoherence; sign-flipping bias independent of $t \Rightarrow$ calibration — retune the gate, not the fridge.

Problem 22 [Challenge] Both readings sit at ≈ 0.1 , barely apart — consistent with a signal already crushed to the depolarized floor ($E \approx 0$) plus a small bias/noise; extrapolating a flat line manufactures a confident 0.105 from nothing. Exposure experiments: fold to $\lambda = 5$ (a real signal keeps falling on trend; floor-noise stays flat), and run ZNE on a *known-answer* circuit of the same depth (if it returns garbage, the method is dead at this depth). Rule: extrapolation amplifies trends — and when there is no trend, it amplifies noise; never ZNE a signal you cannot first see moving.

CHAPTER TWENTY-NINE

Quantum Error Correction

Teaching notes, self-test, and complete solutions

TEACHING NOTES

The chapter that answers “can this field actually work?” — teach it as a triumph of ideas over three apparent impossibilities (no cloning, no looking, analog errors). The three-qubit code must be traced by hand, including the sacred point that parities are measured *without reading data* (Problem 10; the classroom syndrome game makes it visceral). The crossover computation (Problem 9) is the chapter’s quantitative heart — protection helps only below threshold, and students who compute $q = 60\%$ making things *worse* never forget it. Shor’s nine-qubit lesson shows concatenation and degeneracy; the ledger lesson ($\Lambda^{-(d+1)/2}, 2d^2$) turns headlines into arithmetic — assign at least one ledger problem, since every vendor claim students will ever read is priced by it. Close on the 2023–24 below-threshold milestones: the era where growing codes finally started helping. Stumbles: confusing detecting with locating; forgetting phase errors exist (Problem 11 is the trap); and reading “more qubits” as automatically better *above* threshold.

SELF-TEST — QUESTIONS WITH ANSWERS

Q1. The three impossibilities QEC defeats, and its three answers?

A. No cloning → spread information in entanglement, not copies. Measurement destroys → measure only parities (syndromes), never data. Analog errors → measurement discretizes errors into X/Z events.

Q2. The 3-qubit code’s failure probability, and its lesson?

A. $3q^2(1 - q) + q^3$: beats the bare qubit only for $q < 50\%$ — protection pays only below a threshold.

Q3. What do Shor’s code’s eight checks do?

A. Six ZZ parities locate bit flips to the qubit; two $X^{\otimes 6}$ block comparisons locate phase flips to the block — any single-qubit error, including Y , is repaired.

Q4. State the ledger.

A. $p_L \approx \Lambda^{-(d+1)/2}$ at $\approx 2d^2$ physical qubits per logical: each distance step divides logical error by Λ ; quality (via Λ) acts exponentially, quantity linearly.

Q5. What does “below threshold” mean, and when was it demonstrated?

A. $\Lambda > 1$: bigger codes make better qubits. Demonstrated on hardware in 2023–24 (surface-code scaling; atom-array logical processors).

Q6. Why ~ 1000 physical per logical?

A. At good Λ (~ 10), reaching $p_L \sim 10^{-12}$ needs $d \approx 23$: $2d^2 \approx 10^3$ — the overhead behind every serious machine-size estimate.

Solutions to the Practice Problems

Problem 1 [Basic] $\frac{1}{\sqrt{2}}(|000\rangle + |111\rangle)$ — a GHZ-type state. It is *not* $|+\rangle|+\rangle|+\rangle$ (which expands over all eight patterns): the code entangles one logical amplitude pair across three qubits; it never copies the state (no-cloning is respected, not violated).

Problem 2 [Basic] $\alpha|001\rangle + \beta|110\rangle$. Checks: qubits 1,2 still agree (parity 1-2: agree); qubits 2,3 now differ (disagree). Syndrome (agree, disagree) $\rightarrow$ “flip qubit 3.”

Problem 3 [Basic] (Disagree, agree) fingers the qubit in the first pair but not the second: repair X on qubit 1.

Problem 4 [Basic] $2 \times 25 = 50$ qubits at $d = 5$; $2 \times 81 = 162$ at $d = 9$ (the real $d = 3$ patch’s 17 shows the formula is approximate but honest).

Problem 5 [Basic] X_7 (block 3’s first qubit): $Z_7Z_8 = -1$, $Z_8Z_9 = +1$; all block-1/2 ZZ checks $+1$; both $X^{\otimes 6}$ checks $+1$ (X commutes with X). Repair: X on qubit 7.

Problem 6 [Basic] Either no error occurred, or an undetectable multi-qubit error beyond the code’s distance did (rare at low q). “Do nothing” is right for both: acting without a syndrome would *inject* an error, and the rare invisible case is exactly what larger distance exists to suppress.

Problem 7 [Basic] $(d+1)/2 = 12/\log_{10} 4 = 19.9 \rightarrow d = 39$ (odd). Qubits: $2 \times 39^2 = 3042$ per logical.

Problem 8 [Basic] The ledger reads $p_L \approx \Lambda^{-(d+1)/2}$: at $\Lambda = 1$ every distance gives $p_L \approx 1 \times \text{constant}$ — growing d multiplies cost while dividing error by exactly 1. Scale helps only after quality crosses the line.

Problem 9 [Standard] $q = 10\%$: $3(0.01)(0.9) + 0.001 = 2.8\%$ versus 10% unprotected — $3.6\times$ better. $q = 60\%$: $3(0.36)(0.4) + 0.216 = 64.8\%$ versus 60% — the code now *hurts*. At $q = 50\%$: $37.5\% + 12.5\% = 50\%$ — exact break-even. You have discovered the threshold numerically: redundancy amplifies whichever side of the line the hardware sits on.

Problem 10 [Standard] Branch $|000\rangle$: qubits 1,2 agree; 2,3 agree. Branch $|111\rangle$: agree; agree. *Both* branches give the identical answer (agree, agree), so the parity ancillas learn nothing about α versus β — no which-branch information flows, no collapse of the logical superposition: asking about *differences* is what makes syndrome measurement legal.

Problem 11 [Standard] Z_1 gives $\alpha|000\rangle - \beta|111\rangle$: all bit-values unchanged, so both parities read “agree” — the checks see nothing, yet the logical phase flipped: real damage, invisible to bit-parity eyes. The bit-flip code is blind to phase errors — the gap Shor’s nine-qubit construction (and Problem 18’s phase-flip code) exists to close.

Problem 12 [Standard] $p/p_{th} = 0.5$: $d = 3$: $0.1 \times 0.5^2 = 2.5\%$; $d = 7$: $0.1 \times 0.5^4 = 0.6\%$ — growing the code helps. At $p = 2\%$ (ratio 2): $d = 3$: $0.1 \times 4 = 40\%$; $d = 7$: $0.1 \times 16 \rightarrow$ saturating at total failure — above threshold, bigger codes get worse: more qubits, more places to fail, no net repair.

Problem 13 [Standard] (1) At what code distance, and what is the *measured* logical error rate per operation (not per idle)? (2) What is your Λ — does the logical rate actually fall when the code grows? (3) What logical *gates* run (and the T-gate story), at what logical error and speed? — 48 patches that only idle are memories, not qubits.

Problem 14 [Standard] $Y_4 = iX_4Z_4$: the X part trips $Z_4Z_5 = -1$ ($Z_5Z_6 = +1$) — qubit 4 named; the Z part trips both $X^{\otimes 6}$ comparisons $(-1, -1)$ — block 2 named. Repair: X_4 and Z on any block-2 qubit. The global i multiplies the whole state — a global phase, invisible forever (Chapter 5).

Problem 15 [Standard] Phase flip in block 1: hits comparison 1-2 only: $(-1, +1)$. Block 2: sits in both comparisons: $(-1, -1)$. Block 3: comparison 2-3 only: $(+1, -1)$. Three distinct patterns from two bits — the same two-parities-locate-three logic as the little code, one level up: concatenation reuses arithmetic, not just hardware.

Problem 16 [Standard] Each independent check halves the state space: $9 - 8 = 1$ logical qubit. General rule: n physical qubits with s independent stabilizer checks encode $k = n - s$ logical qubits — the stabilizer bookkeeping identity behind every code in the field.

Problem 17 [Standard] Rival (better gates): $\Lambda \approx p_{th}/p$ doubles yearly from 1.5: within ~ 2 years $\Lambda \approx 6$, and 10^{-9} needs $(d+1)/2 = 9/\log_{10} 6 \approx 11.6 \rightarrow d = 23$: $2d^2 \approx 1058$ qubits — exactly their 1000: goal reached in ≈ 2 years. Vendor (more qubits at $\Lambda = 1.5$): needs $(d+1)/2 = 9/\log_{10} 1.5 \approx 51 \rightarrow d = 101$: $\approx 20,000$ qubits — 4-5 doubling years, with a monstrous machine. Assumptions stated (independent errors, ledger validity): the error-rate racer wins — quality compounds exponentially through the exponent; quantity buys linear real estate.

Problem 18 [Challenge] A single Z_2 , say: $\alpha|+++\rangle + \beta|---\rangle \rightarrow \alpha|+-+\rangle + \beta|-+-\rangle$: one sign flipped in each branch — and $\pm$ -basis parity checks (XX -type: “do neighbors agree in the $\pm$ basis?”) read (disagree, agree)-style syndromes exactly as the bit-flip code did for X errors: located, repaired by one Z. Blind spot: X errors — $X|\pm\rangle = \pm|\pm\rangle$ leaves $\pm$ -labels untouched (only a sign), invisible to these checks. Nesting: Shor’s construction runs bit-protection *inside* blocks and phase-protection *across* them, so each error type meets the code built for it — everything caught.

Problem 19 [Challenge] Solve $(p/p_{th})^{(d+1)/2} = 10^{-8}$. At $p = 0.8\%$ (ratio 0.8): $(d+1)/2 = 8/0.097 \approx 83 \rightarrow d \approx 165$: $2d^2 \approx 54,000$ qubits per logical. At $p = 0.4\%$ (ratio 0.4): $(d+1)/2 = 8/0.398 \approx 20 \rightarrow d \approx$

41: $2d^2 \approx 3400$. One halving of the physical error rate cut the per-logical bill sixteenfold — the ledger’s exponential lever, in one sentence.

Problem 20 [Challenge] Per logical patch: $\sim 2d^2 = 1250$ checks per μs ; 1000 patches: 1.25×10^9 checks per *microsecond*... careful: 1250×10^6 cycles/s = 1.25×10^9 checks/s per patch, $\times 1000 = 1.25 \times 10^{12}$ checks per second machine-wide. A 10^9 -op/s CPU is a thousandfold short *merely to touch each check once* — and decoding costs many operations per check. Conclusion: real-time decoding demands dedicated parallel hardware (FPGAs/ASICs beside the fridge), and decoder engineering is a research frontier co-equal with the qubits themselves.

Problem 21 [Challenge] $Z_1 Z_2 |000\rangle = |000\rangle$; $Z_1 Z_2 |111\rangle = (-1)(-1) |111\rangle = |111\rangle$: identity on both, hence on $|000\rangle \pm |111\rangle$ and on every code word built from such blocks. So if the true error was Z_5 and the repair applied Z_4 , the residue $Z_4 Z_5$ acts as the identity on the code space: the “wrong” repair heals perfectly — degeneracy, the code forgiving our ignorance because the difference between the errors is a stabilizer it cannot feel.

Problem 22 [Challenge] Logical demand: $1500 \times 3 = 4500$ logical qubits (algorithm + factories). At $\Lambda = 10$: $d = 23$, $2d^2 \approx 1060$: ≈ 4.8 million physical qubits. At $\Lambda = 2$: $d = 79$, $2d^2 \approx 12,500$: ≈ 56 million. Bracketing the published trajectory (20M in 2021, $<1\text{M}$ in 2025) — our crude model is the right order but conservative against the 2025 number. Crudest ingredient: the flat “factories triple the count” (real factory overhead depends heavily on T-count scheduling and distillation protocols — the estimate’s true frontier), with the $2d^2$ -and-uniform- Λ idealization a close second.

CHAPTER THIRTY

Quantum Hardware and Our Region

Teaching notes, self-test, and complete solutions

TEACHING NOTES

The chapter that lands the course in the real world — and, deliberately, in *our* world: the regional lessons (cloud access, budgets, the education-first roadmap) are not garnish but the honest answer to “what can we do from here?” Run the datasheet worked example as a live class exercise on a real public dashboard (the scavenger-hunt activity); students who compute one survival budget by hand read every vendor slide differently forever. Keep the two center-of-gravity morals loud: quality beats quantity (the roadmap-audit lesson; Problems 7, 9, 17), and people beat machinery at this stage (Problems 12, 13, 20 — superb essay work). Platform comparisons should stay honest-plural: no winner declared, trade-offs named. Stumbles: comparing machines by qubit count, forgetting SWAP taxes on sparse wiring, and reading median datasheet numbers as guarantees.

SELF-TEST — QUESTIONS WITH ANSWERS

Q1. Four platforms, one signature each?

A. Superconducting: speed and manufacturability. Trapped ions: fidelity and all-to-all connectivity. Neutral atoms: scalable arrays and reconfigurability. Photonics: room temperature and networking affinity.

Q2. The survival budget in one formula?

A. $F \approx (1 - p_1)^{n_1} (1 - p_2)^{n_2} (1 - p_r)^{n_q} e^{-t/T_1}$ — multiply every gate’s, readout’s, and the clock’s survival odds.

Q3. Two rules of thumb from the budget?

A. $n_2 \lesssim 0.1/p_2$ for a $\leq 10\%$ gate toll; $t_{\text{circ}} \lesssim 0.1 T_1$ for a $\leq 10\%$ decay toll.

Q4. Why do medians matter on datasheets?

A. Chips are uneven: hero qubits flatter, duds sabotage; medians describe what a routed circuit actually meets — and good toolchains route around the duds.

Q5. The roadmap audit’s one-line rule?

A. Find the error rate: absent means marketing; falling means serious — count alone cannot buy depth or logical qubits.

Q6. The regional first investment?

A. People and access: cloud credits, curriculum, and honest benchmarking skill — a refrigerator is the last purchase on the list, not the first.

Solutions to the Practice Problems

Problem 1 [Basic] Superconducting: fast gates, chip manufacturability. Trapped ions: highest fidelities, every-ion-talks-to-every-ion. Neutral atoms: large reconfigurable arrays (and 2024's logical-qubit splash). Photonics: room-temperature operation and native fit to communication.

Problem 2 [Basic] Qubit count; coherence times (T_1/T_2); gate fidelities; connectivity; speed. The depth rule $d \approx 1/p$ connects to *gate fidelity* — the metric that sets how much of the count is usable.

Problem 3 [Basic] Millikelvin cold suppresses thermal excitations that would randomly kick the qubits — heat is continuous accidental measurement, i.e. decoherence, and the fridge buys silence.

Problem 4 [Basic] A laptop with internet; a free cloud-provider account; the simulator-first discipline (debug before queueing); and this book's budget arithmetic to interpret the noisy results honestly.

Problem 5 [Basic] Readout: $(0.988)^5 = 0.94$ (a 6-point toll). Twenty CNOTs: $(0.994)^{20} = 0.89$ (11 points): the two-qubit toll dominates — as usual.

Problem 6 [Basic] Two-qubit error rate p_2 (sets n_2 's toll and the depth wall); coherence T_1/T_2 (sets the time toll e^{-t/T_1}); readout error p_r (taxes every qubit once). Qubit count alone fixes none of the exponents.

Problem 7 [Basic] Depth budgets: A: $0.1/0.006 \approx 17$; B: $0.1/0.0015 \approx 67$. A 200-CNOT algorithm: A survives $(0.994)^{200} \approx 30\%$, B $(0.9985)^{200} \approx 74\%$: B — the ten-times-smaller machine — wins decisively.

Problem 8 [Basic] Qubit count; error rates; coherence times; connectivity; calibration stability. The exponential pair through the ledger: *error rates* (via Λ) and, jointly, *coherence* (which feeds effective error rates) — error rates first among equals.

Problem 9 [Standard] Both have enough qubits (60 needed; A's 940 spare are irrelevant). Survival at depth 200: A: $(0.98)^{200} \approx 1.8\%$ — useless; B: $(0.998)^{200} \approx 67\%$ — workable. Machine B, without hesitation: the algorithm fits in both machines' *width* and only B's *depth*.

Problem 10 [Standard] Ions: $1\text{ s}/100\text{ }\mu\text{s} = 10^4$ gates within T_2 . Superconducting: $100\text{ }\mu\text{s}/50\text{ ns} = 2000$. The $10^4\times$ coherence advantage shrinks to $5\times$ in usable depth because gates slow down in proportion — the honest metric is gates-per-coherence-time, and by it the platforms are peers, competing instead on fidelity, connectivity, and scaling paths.

Problem 11 [Standard] Quantum volume 2^n means the machine reliably runs random square circuits of width

and depth n : doubling $512 \rightarrow 1024$ means one more layer of width-10 depth — i.e. 10-qubit circuits one gate-layer deeper now pass the fidelity bar. Either better two-qubit fidelity or better connectivity/compilation (fewer SWAPs) could each have delivered it — QV deliberately rewards the whole stack, not one spec.

Problem 12 [Standard] (Model allocation.) Cloud credits and paid tiers: \$1200 (the actual hardware access); instructor/TA training time: \$800; laptops-lab upkeep or one shared workstation for simulations: \$700; materials and community events (hackathon day, printing): \$300. Fund *cloud credits* first: everything else in the course can be improvised, but real-hardware runs are the irreplaceable experience — and at $\sim \$60/\text{student}/\text{year}$ they are the cheapest transformative line item in any science budget.

Problem 13 [Standard] (Model answer.) First: build people and pilots — a cloud-based curriculum, benchmarking projects, and PQC-migration training cost a few thousand dollars and start this semester. A purchased machine today would enable education and prestige photography, but not computations beyond free cloud tiers — the useful machines are error-corrected ones that do not yet exist to buy. Realistic scale: research-grade systems run \$10–50M plus facilities and staff — an order of magnitude beyond any sensible first step. Recommendation: spend 1% of that on scholarships, cloud credits, and a regional quantum-skills center, and revisit hardware when error-corrected machines and trained local teams both exist.

Problem 14 [Standard] $n = \ln 0.5 / \ln 0.994 = 115$ gates.

Problem 15 [Standard] $12 \times 3 = 36$ extra CNOTs: $(0.994)^{136} = 0.44$, down from 0.55 — routing alone cost 11 points of fidelity. Moral: connectivity is a first-class metric; a “worse” chip whose wiring matches your circuit can beat a “better” one that needs SWAP chains.

Problem 16 [Standard] $e^{-30/60} = 0.61$: the revised budget is $0.43 \times 0.61 \approx 0.26$. Phase-sensitive algorithms must count T_2 exposure on their critical qubits — idle time is not free time.

Problem 17 [Standard] Depth budget: $0.1/0.001 = 100$ versus 20 — $5\times$ deeper. 300-gate survival: $(0.999)^{300} = 74\%$ versus $(0.995)^{300} = 22\%$. Ledger: $\Lambda \approx p_{\text{th}}/p_2$ improves $2 \rightarrow 10$, cutting physical-per-logical roughly $12\times$ (Chapter 29’s worked numbers). The ledger framing conveys it best: this “0.4-point” fidelity gain is a *twelfefold* cut in the cost of every future logical qubit.

Problem 18 [Challenge] Two SWAP-chains of 5 SWAPs = 10 SWAPs = 30 extra CNOTs (amortized against the 20 payload CNOTs — a 150% overhead). Extra error: $(0.995)^{30} = 0.86$: routing alone burns 14 points before the algorithm computes anything. An all-to-all machine (trapped ions) pays zero routing tax — worth roughly a $1.5\times$ better effective gate error in this comparison: connectivity converts directly into fidelity.

Problem 19 [Challenge] (Grading guide — expect approximately:) Task 1: Bell-pair certification with full budget prediction (Chapters 11, 14) — metric: predicted-vs-measured corrupted fraction, error bars mandatory. Task 2: Bernstein–Vazirani at 4–5 qubits (Chapter 17) — metric: success-bar height per gate, compared across two public machines. Task 3: the repetition-code crossover study (Chapters 29, 31’s Project 8) — metric: measured crossover versus theory. Anti-gaming rule: all teams submit code and raw counts, budgets are pre-registered before hardware runs, and one surprise instance per task is issued 48 hours before the deadline —

reproducibility outweighs raw scores. Judging: highest *honesty-adjusted* accuracy of predictions, not prettiest numbers.

Problem 20 [Challenge] (Model position — either side earns credit if fact-anchored.) For: the region's opportunity is precisely that this era's valuable products are people-products — PQC migration is a services industry with immediate demand (Chapter 15's Mosca clock), QML benchmarking runs on laptops (Chapter 27's regional road), and cloud access removes the capital barrier (Chapter 30) — while incumbents' legacy advantage (fabs, cryogenics) matters only for building machines, not for using them; the mobile-banking analogy holds because usage leapfrogs infrastructure. Against (concession): hardware sovereignty will not leapfrog — fabs and helium plants are decades of industrial base. First investment either way: a university cloud-credits + curriculum program producing fifty budget-literate graduates a year — the input every downstream path shares.

Problem 21 [Challenge] Base: $(0.9997)^{90}(0.994)^{40}(0.988)^{12}e^{-6/80} = 0.973 \times 0.786 \times 0.865 \times 0.928 \approx 0.61$. Halve p_2 : $(0.997)^{40} = 0.887$: budget ≈ 0.69 . Halve p_r : $(0.994)^{12} = 0.930$: ≈ 0.66 . Double T_1 : $e^{-6/160} = 0.963$: ≈ 0.64 . Winner: halving p_2 (+8 points) — the two-qubit error rate is the budget's biggest lever, here as everywhere.

Problem 22 [Challenge] (Grading guide.) Paragraph one credits what count-growth genuinely shows: manufacturing yield, packaging, and control-electronics scaling — necessary and non-trivial. Paragraph two demands the missing exponentials: measured p_2 per generation, T_1/T_2 trends, a Δ measurement ($d = 3$ vs $d = 5$ logical error), and connectivity/calibration-stability data; the requested experiment: publish one below-threshold scaling result on the newest chip. Closing sentence (verbatim-worthy): "In quantum hardware, quality compounds and quantity only adds — show me the error rate's trajectory and I will believe the roadmap."

CHAPTER THIRTY-ONE

Capstone Projects

Teaching notes, self-test, and complete solutions

TEACHING NOTES

Twelve projects, one rubric, and the course's true final exam: work that ends in a written deliverable. Practical staging advice: Projects 1–2 suit everyone (weeks, not months); 3–8 make strong pair-work; 9–11 serve students aiming at QML; 12 (the briefing) deserves to be *universal* — every student should leave able to give it. Grade deliverables with the chapter's rubric and insist on the book's signature habits: predictions *before* runs, error bars on everything, honest verdicts even when boring. The self-tests below cover project-support skills; the solutions cover the chapter's own problem set. A logistics note: cloud queues are slow near deadlines — have teams book hardware runs two weeks early, and accept simulator-with-noise-model results when queues fail (with the substitution documented, itself a professional habit).

SELF-TEST — QUESTIONS WITH ANSWERS

Q1. What single element turns a computation into a capstone, per this chapter?

A. The written deliverable — “the calculation that is not communicated does not exist”; every project ends in a report, brief, or card someone else could act on.

Q2. Project 1's statistical heart?

A. Certifying fairness with shot statistics: expected $n/2$ with wobble $\sqrt{n}/2$; pass windows are two-sigma statements, not exact counts.

Q3. Why does Project 7 predict histograms before running?

A. Prediction-then-measurement is what makes it device *qualification* rather than device observation — the error budget becomes a testable claim.

Q4. What does Project 8 measure, conceptually?

A. The crossover (toy threshold): where protection stops paying — the experimental heart of Chapter 29.

Q5. Projects 9–11 share what discipline?

A. The DASHB audit applied to one's own work: baselines tuned, ablations run, shots priced, claims sized to evidence.

Q6. Project 12's three truths to hold at once?

A. The mathematics is solid and spectacular; the engineering is real but young; the timeline is uncertain in both directions.

Solutions to the Practice Problems

Problem 1 [Basic] Expected 5000; count wobble $\sigma = \sqrt{10000}/2 = 50$. Two-sigma pass window: 5000 ± 100 , i.e. **4900–5100 zeros** (results outside trigger investigation, not panic — then more shots).

Problem 2 [Basic] $|++\rangle + |--\rangle = \frac{1}{2}[(|0\rangle + |1\rangle)(|0\rangle + |1\rangle) + (|0\rangle - |1\rangle)(|0\rangle - |1\rangle)]$: cross terms $|01\rangle, |10\rangle$ cancel; $|00\rangle, |11\rangle$ double: $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = |\Phi^+\rangle$ ✓ (The Bell state is agree-in-both bases — the fact Project 2's X-basis check exploits.)

Problem 3 [Basic] Per symbol: $1 - (0.98)^3 = 5.9\%$ error rate. The simulation should show symbol errors climbing $\approx$ linearly with per-gate error at small p (three chances per symbol), matching $\approx 6\%$ at $p = 2\%$ — and the report should note readout errors add on top.

Problem 4 [Standard] $P(3) = \sin^2(7 \times 20.7^\circ) = \sin^2 145^\circ \approx 0.33$ — down from 94.5% at $k = 2$. On noisy hardware each extra round also *adds gate errors*, so stopping early trades a little ideal probability for a lot of survival — often a net win (the overshoot and the noise both punish greed).

Problem 5 [Standard] Protected: $3(0.25)^2(0.75) + (0.25)^3 = 0.141 + 0.016 = 15.6\%$ versus unprotected 25%: protection still helps — below the 50% crossover, though the margin is thinning (at $q = 10\%$ it was $3.6\times$; here $1.6\times$).

Problem 6 [Challenge] (Grading guide — any complete, honest spec passes. Model: the T1/T2 campaign.) **Mission:** measure a public machine's lifetimes and compare with its datasheet. **Uses:** Chapters 4, 14, 28 (and 30's datasheet reading). **Tasks:** (1) implement inversion-recovery and Ramsey sequences; (2) sweep delays over 0.2–3 lifetimes, 1000 shots per point; (3) fit exponentials, extract T_1, T_2 with uncertainty; (4) repeat across three days and three qubits — quantify drift; (5) deliverable: a two-page calibration report comparing fitted values against the published medians, with a weather-report paragraph on drift. **Calculation kernel:** from $P_1(t) = e^{-t/T_1}$, a measured $P_1 = 0.41$ at $t = 70 \mu\text{s}$ gives $T_1 = 70 / \ln(1/0.41) = 78 \mu\text{s}$ — one worked line showing how each data point converts. Rubric emphasis: prediction before measurement (datasheet values), error bars from shot statistics, honest discussion of disagreement.

APPENDIX A

Complex Numbers

Teaching notes and complete solutions

TEACHING NOTES

Teach this appendix (with Appendix B) as the course's opening fortnight for math-from-zero groups, or assign it as guided self-study with the problems as the checkpoint. Priorities: mechanical fluency with conjugates and $|z|^2 = zz^*$ (the amplitude toolkit), Euler's formula as "angle bookkeeping," and the roots-of-unity lesson as a deliberate seed for the QFT — tell students plainly that Chapter 19 will run on these clock numbers. The interference preview (Problem 12) is worth boardtime: it is the double-slit arithmetic wearing algebra clothes. (Self-tests in this Guide cover the main chapters; for the appendices, the solutions below are the assessment resource.)

Solutions to the Practice Problems

Problem 1 [Basic] (a) -1 ; (b) $+1$; (c) $i^{10} = i^8 i^2 = -1$; (d) $5i$.

Problem 2 [Basic] $z + w = 5 + 5i$; $z - w = 3 - i$. (Plot check: z at $(4, 2)$, w at $(1, 3)$, sum at $(5, 5)$, difference at $(3, -1)$ — parallelogram addition visible.)

Problem 3 [Basic] $(2 + i)(3 + 2i) = 6 + 4i + 3i + 2i^2 = 6 + 7i - 2 = 4 + 7i$.

Problem 4 [Basic] (a) $5 - 4i$; (b) $-2 + 7i$; (c) $-6i$; (d) 9 (real numbers are their own conjugates).

Problem 5 [Standard] Parts: $6^2 + (-8)^2 = 100$. Product: $zz^* = (6 - 8i)(6 + 8i) = 36 + 64 = 100$ ✓ So $|z|^2 = 100$, $|z| = 10$.

Problem 6 [Standard] $|0.6i|^2 = 0.36$ — a legal probability. But $z^2 = (0.6i)^2 = -0.36$: negative, impossible as a probability — which is exactly why the Born rule squares the *modulus*, never the number itself.

Problem 7 [Standard] $101 = 4 \times 25 + 1$: $i^{101} = i^1 = i$. Trick: powers of i cycle with period 4, so only the remainder mod 4 matters.

Problem 8 [Standard] (a) 1; (b) -1 ; (c) $\cos 30^\circ + i \sin 30^\circ = 0.866 + 0.5i$.

Problem 9 [Standard] $|z|^2 = \frac{1}{2} + \frac{1}{2} = 1$: on the unit circle ✓ Phase: $\arctan$ of $(-\frac{1}{\sqrt{2}})/(\frac{1}{\sqrt{2}})$ in the fourth quadrant: -45° (equivalently 315°).

Problem 10 [Challenge] $zw = (1+i)(2-i) = 2-i+2i-i^2 = 3+i$. Moduli: $|zw| = \sqrt{10}$; $|z| = \sqrt{2}$, $|w| = \sqrt{5}$; product $\sqrt{2}\sqrt{5} = \sqrt{10}$ ✓ — sizes multiply (and angles add): polar form's two laws.

Problem 11 [Challenge] Angles add: $\omega^3 = e^{i360^\circ} = 1$ ✓ Rectangular form: $\omega = -0.5 + 0.866i$. (Its two companions, $\omega^2 = -0.5 - 0.866i$ and 1, complete the three-hour clock; the eight-hour version powers the QFT.)

Problem 12 [Challenge] $\theta = 0^\circ$: $|1|^2 \dots$ carefully: $z_1 + z_2 = 1$: probability 1 — constructive. $\theta = 90^\circ$: $\frac{1}{2} + \frac{1}{2}i$: $|\cdot|^2 = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$ — the in-between case. $\theta = 180^\circ$: 0 — destructive. One phase dial sweeps an outcome from certain to impossible: interference, in three lines of arithmetic.

Problem 13 [Basic] Roots: 1, i , -1 , $-i$. Sum: $(1-1) + (i-i) = 0$ ✓ — opposite pairs cancel.

Problem 14 [Standard] $\omega^k \omega^m = e^{i(k+m)360^\circ/N} = \omega^{k+m}$ (angles add); and $\omega^{k+N} = \omega^k \cdot e^{i360^\circ} = \omega^k$: after N steps the clock face repeats — addition of exponents is arithmetic mod N .

Problem 15 [Challenge] $\omega S = \omega + \omega^2 + \dots + \omega^N = S$ (the list merely rotated: $\omega^N = 1$ re-enters as the first term). So $(\omega - 1)S = 0$; since $\omega \neq 1$ for $N \geq 2$, $S = 0$ ✓ At $N = 1$: $\omega = e^{i360^\circ} = 1$, the division by $\omega - 1 = 0$ is illegal — and indeed $S = 1 \neq 0$ there.

APPENDIX B

Vectors and Matrices

Teaching notes and complete solutions

TEACHING NOTES

The second math-from-zero fortnight. Non-negotiable fluencies: matrix-vector multiplication (drill daily), the dagger with complex entries, and the unitarity check $U^\dagger U = I$ — the “entrance exam for gates” framing carries the whole course’s physics. The two seeded identities deserve explicit flagging: $HZH = X$ (Problem 12 — returns as the sandwich rule) and the new eigenvector lesson (Problems 13–15 — returns as measurement outcomes, kickback, and phase estimation). The non-unitary machine M (Problem 9) is the best two-minute discussion in the appendix: *why* nature refuses it. (Appendix solutions below serve as the assessment resource; self-tests cover the main chapters.)

Solutions to the Practice Problems

Problem 1 [Basic] $\begin{pmatrix} 1 \\ 8 \end{pmatrix}$ and $\begin{pmatrix} 4 \\ -8 \end{pmatrix}$.

Problem 2 [Basic] $\langle u|u \rangle = 9 + 16 = 25$: length $\sqrt{25} = 5$.

Problem 3 [Basic] $\begin{pmatrix} 0.8 \\ 0.6 \end{pmatrix}$ — the components swap (X ’s whole personality).

Problem 4 [Basic] Transpose, then conjugate: $\begin{pmatrix} 2 & 1+i \\ -3i & 4 \end{pmatrix}$.

Problem 5 [Standard] $\langle u|v \rangle = \frac{1}{2}(1 \cdot 1 + 1 \cdot (-1)) = 0$: **orthogonal** — perfectly distinguishable, a measurement’s two answers (these are $|+\rangle$ and $|-\rangle$).

Problem 6 [Standard] $XZ = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, $ZX = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$: unequal (opposite signs). Meaning: quantum operations in different orders can give different results — circuits read like sentences, and word order matters (Chapter 6 builds the full field guide).

Problem 7 [Standard] $H\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -1 \end{pmatrix}$, versus $H\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix}$: same first component, opposite *sign* in the second — the difference that the whole $|+\rangle$ -vs- $|-\rangle$ story (and every interference effect) hangs on.

Problem 8 [Standard] $Z^\dagger = Z$ (real diagonal), and $Z^\dagger Z = Z^2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$ ✓ unitary.

Problem 9 [Standard] $M \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \end{pmatrix}$ (length 2); $M \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1/2 \end{pmatrix}$ (length $\frac{1}{2}$). It stretches some states and shrinks others — probabilities would total more or less than 100%: nature runs only length-preserving (unitary) machines, and M fails the entrance exam.

Problem 10 [Challenge] $(UV)^\dagger(UV) = V^\dagger U^\dagger UV = V^\dagger IV = V^\dagger V = I$ ✓ Closure means gates compose into circuits that are themselves legal gates — arbitrary-depth circuits stay physical, which is the whole premise of circuit-model computing.

Problem 11 [Challenge] $M^\dagger M = |c|^2 I = I \iff |c| = 1$: every $c = e^{i\theta}$ — the entire unit circle. (These are the global-phase gates: physically invisible, mathematically legal — Chapter 5’s story foreshadowed.)

Problem 12 [Challenge] $ZH = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$; then $H(ZH) = \frac{1}{2} \begin{pmatrix} 1 & -1 & 1 & 1 \\ 1 & 1 & 1 & -1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = X$ ✓ “X is Z dressed in Hadamards” — the sandwich identity, delivered to Chapter 6 as promised.

Problem 13 [Basic] Eigenvectors $\begin{pmatrix} 1 \\ 0 \end{pmatrix} = |0\rangle$ (eigenvalue +1) and $\begin{pmatrix} 0 \\ 1 \end{pmatrix} = |1\rangle$ (eigenvalue −1): the standard measurement’s answer states, with the eigenvalues as the recorded ± 1 outcomes.

Problem 14 [Standard] $H\vec{v} = \frac{1}{\sqrt{2}} \begin{pmatrix} 0.924+0.383 \\ 0.924-0.383 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1.307 \\ 0.541 \end{pmatrix} = \begin{pmatrix} 0.924 \\ 0.383 \end{pmatrix} = \vec{v}$ ✓ eigenvalue +1. On the Bloch sphere this direction is the axis exactly *halfway between* z and x — H’s mirror line, which is why H swaps those two axes.

Problem 15 [Challenge] $U\vec{v} = \lambda\vec{v}$ with $\|\vec{v}\| = 1$: lengths give $\|U\vec{v}\| = |\lambda| \|\vec{v}\| = |\lambda|$, while unitarity demands $\|U\vec{v}\| = 1$: hence $|\lambda| = 1$ — every unitary eigenvalue is a pure phase $e^{i\varphi}$. Consequence: phase estimation’s “answers” are always angles on a circle — which is precisely why the QFT (a machine for reading angles) is the right readout instrument.

APPENDIX C

The Probability You Need

Teaching notes and complete solutions

TEACHING NOTES

The third from-zero fortnight (or a two-lecture refresher). Everything the book's statistics rest on is here: the complement trick, independence chains, expectation/spread, the square-root law, and base-rate reasoning — each solution below names the main-text chapter that reuses it, and telling students so in advance is the best motivation this appendix gets. The worked rare-disease table (Bayes without formalism) prevents the single most common statistical misreading educated adults make; give it classroom time even for strong groups. Problems 7 and 8 are the appendix's graduation: deriving the book's two most-used formulas from scratch.

Solutions to the Practice Problems

Problem 1 [Basic] $P(1) = 0.64$. At least one 1 in three shots: $1 - P(\text{all zeros}) = 1 - 0.36^3 = 1 - 0.047 = 95.3\%$ — the complement trick doing what direct counting would sweat over.

Problem 2 [Basic] $\mathbb{E} = 0.8(+1) + 0.2(-1) = 0.6$; $\mathbb{E}[X^2] = 1$, so $\sigma = \sqrt{1 - 0.36} = 0.8$. These are precisely $\langle Z \rangle = 0.6$ and $\Delta Z = \sqrt{1 - \langle Z \rangle^2} = 0.8$ for a qubit with $P(0) = 0.8$ (Chapter 8's formulas, met early).

Problem 3 [Basic] Exact: $(0.995)^{60} = 0.740$. Approximation: $1 - 60 \times 0.005 = 0.70$. Close but visibly drifting ($pd = 0.3$ is no longer small) — use the approximation for quick reads, the exact form for budgets (Chapter 28 does both).

Problem 4 [Standard] $0.5/\sqrt{n} \leq 0.01 \Rightarrow n = 2500$; $\leq 0.001 \Rightarrow n = 250,000$. Rule: tenfold precision costs a hundredfold shots ($n \propto 1/\text{margin}^2$) — the square-root law's price tag.

Problem 5 [Standard] $\hat{p} = 236/400 = 0.59$; $\sigma = \sqrt{0.59 \times 0.41/400} = 0.025$. Distance from 0.5: $0.09/0.025 = 3.7$ sigma — reject the fair-coin hypothesis at the two-sigma standard with room to spare (Chapter 4's certification logic).

Problem 6 [Standard] Per 10,000: sick 100 $\rightarrow$ 98 true positives; healthy 9900 $\rightarrow$ 198 false positives.

$P(\text{condition} \mid \text{positive}) = 98/296 = 33\%$ — up from 10%. The bigger help came from cutting the false-alarm rate $8\% \rightarrow 2\%$ (false positives fell $792 \rightarrow 198$), not from the catch-rate gain: with rare conditions, specificity rules the posterior.

Problem 7 [Challenge] $p(1-p) = \frac{1}{4} - (p - \frac{1}{2})^2 \leq \frac{1}{4}$, with equality at $p = \frac{1}{2}$ ✓ Hence $\sigma = \sqrt{p(1-p)/n} \leq \sqrt{1/(4n)} = 0.5/\sqrt{n}$ — the worst-case wobble quoted from Chapter 4 to Chapter 30, now proven in one completed square.

Problem 8 [Challenge] Ways to fail: exactly two flips — choose which two of three qubits (3 ways), probability $q^2(1-q)$ each: $3q^2(1-q)$; all three flip: q^3 . Total $P_{\text{fail}} = 3q^2(1-q) + q^3$ ✓ At $q = 0.1$: $0.027 + 0.001 = 2.8\%$ versus the unprotected 10% — a $3.6\times$ improvement, purchased entirely with counting: Chapter 29's central formula, derived before the course ever mentions a qubit.